

Published in final edited form as:

J Pain. 2009 September ; 10(9): 895–926. doi:10.1016/j.jpain.2009.06.012.

Central Sensitization: A Generator of Pain Hypersensitivity by Central Neural Plasticity

Alban Latremoliere and Clifford J. Woolf

Neural Plasticity Research Group, Department of Anesthesia and Critical Care, Massachusetts General Hospital and Harvard Medical School, Charlestown, Massachusetts

Abstract

Central sensitization represents an enhancement in the function of neurons and circuits in nociceptive pathways caused by increases in membrane excitability and synaptic efficacy as well as to reduced inhibition and is a manifestation of the remarkable plasticity of the somatosensory nervous system in response to activity, inflammation, and neural injury. The net effect of central sensitization is to recruit previously subthreshold synaptic inputs to nociceptive neurons, generating an increased or augmented action potential output: a state of facilitation, potentiation, augmentation, or amplification. Central sensitization is responsible for many of the temporal, spatial, and threshold changes in pain sensibility in acute and chronic clinical pain settings and exemplifies the fundamental contribution of the central nervous system to the generation of pain hypersensitivity. Because central sensitization results from changes in the properties of neurons in the central nervous system, the pain is no longer coupled, as acute nociceptive pain is, to the presence, intensity, or duration of noxious peripheral stimuli. Instead, central sensitization produces pain hypersensitivity by changing the sensory response elicited by normal inputs, including those that usually evoke innocuous sensations.

Perspective—In this article, we review the major triggers that initiate and maintain central sensitization in healthy individuals in response to nociceptor input and in patients with inflammatory and neuropathic pain, emphasizing the fundamental contribution and multiple mechanisms of synaptic plasticity caused by changes in the density, nature, and properties of ionotropic and metabotropic glutamate receptors.

Keywords

Central sensitization; inflammatory pain; neuropathic pain; scaffolding protein; heterosynaptic facilitation

Acute nociceptive pain is that physiological sensation of hurt that results from the activation of nociceptive pathways by peripheral stimuli of sufficient intensity to lead to or to threaten tissue damage (noxious stimuli).³⁷⁴ Nociception, the detection of noxious stimuli,²⁸² is a protective process that helps prevent injury by generating both a reflex withdrawal from the stimulus and as a sensation so unpleasant that it results in complex behavioral strategies to avoid further contact with such stimuli. An additional important phenomenon that further enhances this protective function is the sensitization of the nociceptive system that occurs after repeated or particularly intense noxious stimuli, so that the threshold for its activation falls and responses to subsequent inputs are amplified.^{132,376,380} In the absence of ongoing tissue injury,

Address reprint requests to Dr Clifford J. Woolf, Neural Plasticity Research Group, Department of Anesthesia and Critical Care, Massachusetts General Hospital and Harvard Medical School, Charlestown, MA 02129. cwoolf@partners.org.

Editor's Note: This article is 1 in a series of invited Critical Review articles designed to celebrate *The Journal of Pain's* 10th year anniversary of publication.

this state of heightened sensitivity returns over time to the normal baseline, where high-intensity stimuli are again required to initiate nociceptive pain; the phenomenon is long lasting but not permanent. The nociceptor-induced sensitization of the somatosensory system is adaptive in that it makes the system hyperalert in conditions in which a risk of further damage is high, for example, immediately after exposure to an intense or damaging stimulus. This sensitization is the expression of use-dependent synaptic plasticity triggered in the central nervous system (CNS) by the nociceptor input and was the first example of central sensitization, discovered 26 years ago.³⁶⁹ Since then, we have learned that a number of different forms of functional, chemical, and structural plasticity can sensitize the central nociceptive system to produce pain hypersensitivity under both normal and pathological circumstances, some of which are persistent.

In many clinical syndromes, pain is no longer protective. The pain in these situations arises spontaneously, can be elicited by normally innocuous stimuli (allodynia), is exaggerated and prolonged in response to noxious stimuli (hyperalgesia), and spreads beyond the site of injury (secondary hyperalgesia). Central sensitization has provided a mechanistic explanation for many of the temporal, spatial, and threshold changes in pain sensibility in acute and chronic clinical pain settings and has highlighted the fundamental contribution of changes in the CNS to the generation of abnormal pain sensitivity. Although phenomenologically central sensitization may appear to be comparable to peripheral sensitization, it differs substantially, both in terms of the molecular mechanisms responsible and its manifestation. Peripheral sensitization represents a reduction in threshold and an amplification in the responsiveness of nociceptors that occurs when the peripheral terminals of these high-threshold primary sensory neurons are exposed to inflammatory mediators and damaged tissue^{46,105,124,242} and, in consequence, is restricted to the site of tissue injury.¹²⁴ Although peripheral sensitization certainly contributes to the sensitization of the nociceptive system and thereby to inflammatory pain hypersensitivity at inflamed sites (primary hyperalgesia), it nevertheless represents a form of pain elicited by activation of nociceptors, albeit one with a lower threshold due to the increased peripheral transduction sensitivity, and generally requires ongoing peripheral pathology for its maintenance. Peripheral sensitization appears to play a major role in altered heat but not mechanical sensitivity, which is a major feature of central sensitization.

Central sensitization, in contrast to peripheral sensitization, co-opts novel inputs to nociceptive pathways including those that do not normally drive them, such as large low-threshold mechanoreceptor myelinated fibers to produce A β fiber-mediated pain.³⁷⁶ It also produces pain hypersensitivity in noninflamed tissue by changing the sensory response elicited by normal inputs and increases pain sensitivity long after the initiating cause may have disappeared and when no peripheral pathology may be present. Because central sensitization results from changes in the properties of neurons in the CNS, the pain is no longer coupled, as acute nociceptive pain is, to the presence, intensity, or duration of particular peripheral stimuli. Instead, central sensitization represents an abnormal state of responsiveness or increased gain of the nociceptive system. The pain is effectively generated as a consequence of changes within the CNS that then alter how it responds to sensory inputs, rather than reflecting the presence of peripheral noxious stimuli. In this respect, central sensitization represents a major functional shift in the somatosensory system from high-threshold nociception to low-threshold pain hypersensitivity. We all experience pain as arising from “out there,” and, in consequence, imagine that it is actually triggered by noxious stimuli where we feel the pain. Central sensitization reveals, however, that this in many cases is a sensory illusion; specific alterations in the CNS can result in painful sensations occurring in the absence of either peripheral pathology or noxious stimuli, and the target for treatment in these situations must be the CNS not the periphery.

Central sensitization corresponds to an enhancement in the functional status of neurons and circuits in nociceptive pathways throughout the neuraxis caused by increases in membrane excitability, synaptic efficacy, or a reduced inhibition. The net effect is that previously subthreshold synaptic inputs are recruited to generate an increased or augmented action potential output, a state of facilitation, potentiation, or amplification. The reason that these cellular changes alter the system so profoundly is that normally only a small fraction of the synaptic inputs to dorsal horn neurons contribute to their action potential output.³⁷³ Nociceptive-specific neurons, for example, although dominated by large monosynaptic and polysynaptic synaptic potentials from nociceptors in their receptive field, typically also have small-amplitude synaptic inputs from low-threshold afferents and from nociceptor inputs outside their receptive fields, which constitute a subliminal fringe that normally does not drive the output of the cells (Fig 1). Recruiting these subthreshold inputs to the output of a neuron markedly alters its receptive field properties, with profound changes in receptive field threshold, spatial, and temporal properties (Fig 2). This provides an opportunity for rapid functional plasticity that can be revealed experimentally by increasing the excitability of the neuron or by blocking inhibitory transmitters. After administration of GABA or glycine receptor antagonists, for example, $A\beta$ inputs are recruited to neurons in the superficial dorsal horn,¹⁷ and pain-like behavior can be elicited by movement of just a few hairs.²⁸⁹ The receptive field of somatosensory neurons are, therefore, not fixed or hard wired, but are instead highly malleable. This malleability or plasticity is the substrate for the functional effects of central sensitization, and the means is a change in synaptic efficacy.

When neurons in the dorsal horn spinal cord are subject to central sensitization, they exhibit some or all the following: development of or increases in spontaneous activity, a reduction in the threshold for activation by peripheral stimuli, increased responses to suprathreshold stimulation, and an enlargement of their receptive fields (Fig 2). Several features appear particular to central sensitization: conversion of nociceptive-specific neurons to wide-dynamic neurons that now respond to both innocuous and noxious stimuli, progressive increases in the responses elicited by a standard series of repeated innocuous stimuli (temporal windup), an expansion of the spatial extent of their input, and changes that outlast an initiating trigger.^{132,368,369,372,376} These electrophysiological changes correlate remarkably with the development in human experimental subjects after a noxious conditioning input of allodynia (particularly dynamic tactile or brush-evoked allodynia), the temporal summation of repeated low-intensity stimuli from an innocuous sensation to pain, with “afterpain” on cessation of the stimulus, and widespread secondary hyperalgesia. These changes can be elicited in human volunteers by noxious stimulation of the skin (as with topical or intradermal capsaicin or repeated heat stimuli³⁴⁰) and in the gastrointestinal tract by exposure to low pH solutions.²⁷²

Central sensitization contributes to neuropathic³⁷ and inflammatory pain,^{26,274,395} migraine,³⁵ and irritable bowel syndrome.²⁵³ In these patients, it is involved in producing abnormal responsiveness to noxious and innocuous stimuli and a spread of tenderness beyond lesion sites. Central sensitization may also play a fundamental role in the abnormal and widespread pain sensitivity in patients with fibromyalgia.^{6,68,301-303} Given the major role of central sensitization in the generation of clinical pain hypersensitivity, it is essential that we understand the triggers and mechanisms responsible for the induction and maintenance of the switch in the somatosensory system from the physiological state, in which the sensory experiences evoked by low-intensity stimuli (innocuous sensations) and noxious stimuli (pain) are quite distinct and separate, to a dysfunctional hypersensitive system in which this discrimination is lost.

The Discovery of Central Sensitization

The first evidence for a central component to acute pain hypersensitivity was provided in 1983.³⁶⁹ Electrophysiological recordings from single biceps femoris α -motoneuron axons were used to measure the output of the nociceptive system, in this case the flexor reflex withdrawal response elicited by noxious stimuli (Fig 3). These recordings revealed, as expected, that under normal conditions there was no spontaneous activity in the motor neurons and that their activation required a noxious mechanical or thermal stimulus to the skin. These neurons had high-threshold nociceptive-specific receptive fields restricted to the toes or hind paw, in keeping with their activation only as part of the flexion withdrawal reflex. After repeated peripheral noxious heat stimuli sufficient to generate mild inflammation of the hind paw, however, an increased excitability of the motor neurons was detected that lasted for several hours and included a reduction in threshold and enlargement of the cutaneous receptive fields. The flexor motor neurons were now no longer nociceptive-specific but could be activated by low-intensity (innocuous) peripheral inputs such as light touch.³⁶⁹ Three experiments showed that this change in receptive field properties was due to alterations in the CNS and not the periphery. First, electric stimulation of $A\beta$ sensory fibers began to elicit responses in the motor neurons after the conditioning noxious heat stimuli, whereas these inputs elicited no response before. Second, a local anesthetic block of the site of the peripheral injury did not result in collapse of the expanded receptive fields: The change was autonomous once it was triggered by the peripheral input. Finally, the hypersensitivity produced by the noxious heat could be mimicked in extent and duration by a brief 20-second low-frequency electrical stimulation of the sural nerve only at C-fiber strength, which produced changes lasting for tens of minutes. The interpretation of all these data was that noxious heat stimulation, by activating C-fiber nociceptors, had induced a central plasticity of the nociceptive system, which was thereafter capable of responding to stimuli outside of the injury area and to low-threshold afferents that previously did not activate the nociceptive system. This led to the articulation of a more general hypothesis that brief trains of nociceptor C-fiber input could trigger or condition a long-lasting sensitization of the nociceptive system (an effect termed central sensitization) by producing activity-dependent changes in the functional properties of neurons in the dorsal horn of the spinal cord and that this contributed both to postinjury flexor reflex and pain hypersensitivity.

Before the discovery of central sensitization, the receptive field properties of dorsal horn neurons was thought to be fixed by the geometry of their dendrites relative to the central terminals of sensory axons.³³ Although plasticity of the receptive fields of dorsal horn neurons had been shown to occur after peripheral nerve injury, this was thought to be due to a loss of presynaptic inhibition increasing synaptic input from silent or ineffective synapses and not to plasticity in dorsal horn neurons.⁷¹ After the first demonstration of central sensitization in flexor motor neurons, essentially identical changes were soon described in many studies in lamina I and V neurons in the dorsal horn of the spinal cord^{55,79,173,176,192,287,365,372} (Fig 3) as well as in spinal nucleus pars caudalis (Sp5c),³⁶ thalamus⁷⁸ (Fig 3), amygdala,^{219,220} and anterior cingulate cortex.³⁶⁴ More recently, functional magnetic resonance imaging, positron emission tomography, and magnetoencephalography have revealed in human subjects that several other brain structures implicated in pain (parabrachial nucleus, periaqueductal gray [PAG], superior colliculus, prefrontal cortex) also exhibit changes compatible with increases in excitability corresponding to central sensitization^{187,209,212,244,283} (Fig 3).

Activity-Dependent Central Sensitization

The original description of central sensitization referred to an activity- or use-dependent form of functional synaptic plasticity that resulted in pain hypersensitivity after an intense noxious stimulus. This plasticity was triggered by the activity evoked in dorsal horn neurons by input

from C-nociceptors, as after repeated heat stimuli above 49°C,³⁶⁹ electrical stimulation of C-fibers (1 Hz for 10 to 20 seconds),³⁵⁸ and chemical activation of nociceptors by irritant compounds such as allyl isothiocyanate (mustard oil)³⁷² and formalin, which both act through the TRPA1 channel^{135,199} as well as capsaicin, which activates TRPV1 channels.¹⁵⁸ To induce central sensitization, the noxious stimulus must be intense, repeated, and sustained. Input from many fibers is required over tens of seconds; a single stimulus, such as a pinch, is insufficient. Peripheral tissue injury is not necessary, although the degree of noxious stimulation that produces frank tissue injury almost always induces central sensitization, so that the phenomenon is very prominent after post-traumatic or surgical injury. Interestingly, nociceptor afferents innervating muscles or joints produce a longer-lasting central sensitization than those that innervate skin.³⁵⁸

Once the phenomenon had been shown to be robust, easily activated, and detected in both preclinical and human subjects, the issue then was what molecular mechanisms were responsible. The first major mechanistic insight was that the induction and maintenance of acute activity-dependent central sensitization was dependent on NMDA receptors,³⁷⁹ revealing a key involvement of glutamate and its receptors. We now appreciate from 2 decades of investigation by many labs that central sensitization comprises 2 temporal phases, each with specific mechanisms. The early phosphorylation-dependent and transcription-independent phase results mainly from rapid changes in glutamate receptor and ion channel properties.³⁷⁶ The later, longer-lasting, transcription-dependent phase drives synthesis of the new proteins responsible for the longer-lasting form of central sensitization observed in several pathological conditions.³⁷⁶ We will review the current understanding of these mechanisms.

Triggers of Activity-Dependent Central Sensitization

Glutamate, the fast transmitter of primary afferent neurons, binds to several receptors on postsynaptic neurons in the dorsal horn of spinal cord, including ionotropic amino-3-hydroxy-5-methyl-4-isoxazole propionate (AMPA), N-methyl-D-Aspartate (NMDA), and Kainate (KA) receptors and several metabotropic (G-protein coupled) glutamate receptor subtypes (mGluR). In the superficial laminae of the dorsal horn, AMPAR and NMDAR are present in virtually every synapse and are arranged in a mosaic-like manner, whereas mGluRs sit at the extremities of the postsynaptic density zone (PSD).^{10,11,15,247} At the subunit level, NMDAR is a tetramer that contains 2 low-affinity glycine-binding NR1 subunits and 2 subunits from the 6 different NR2_{A-D} or NR3_{A/B} subunits.³⁰⁵ The most common NMDA complexes in the dorsal horn are composed of NR1-NR2_{A/B} subunits.^{202,215,243} AMPAR is also a tetramer, and its most abundant subunits in the dorsal horn of the spinal cord are the calcium (Ca²⁺)-permeable subunits GluR1 and GluR3 and the non-Ca²⁺-permeable GluR2 subunit.²⁵² In basal conditions, inhibitory interneurons appear to preferentially express GluR1, whereas the excitatory neurons appear to express mostly GluR2.^{145,338} The AMPAR complex can also be a GluR1/GluR2 heteromer,¹¹ in which case the receptor displays mainly GluR2 properties.²³³ A subpopulation of lamina I neurons lacking the NK1 receptor and expressing the GluR4 (Ca²⁺-permeable) subunit has been identified.²⁴⁸ The mGluR family is composed of 8 receptors that form 3 groups, based on their sequence similarities and their coupling with specific G α -proteins.⁵⁴ Group I mGluRs (mGluR1 and 5) are coupled with G α_q -proteins (whose activation causes an increase of [Ca²⁺]_i), whereas group II (mGluR 2 and 3) and group III (mGluR4, 6, 7 and 8) are coupled with G α_i/o -proteins. All mGluRs except for mGluR6 and 8 are expressed in the spinal cord,³⁴⁶ whereas only mGluR6 appears not to be expressed by primary afferent neurons.³⁹ In addition, a lamina-specific pattern of expression has been characterized for mGluR1 α (lamina V), mGluR5 (laminae I-II),^{10,247} and mGluR2/3 (lamina II inner),¹² suggesting precise and distinct physiological roles for the different subtypes.

Activation of NMDAR is an essential step in both initiating and maintaining activity-dependent central sensitization as its blockade by noncompetitive (MK801) or competitive (D-CPP) NMDAR antagonists prevent and reverse the hyperexcitability of nociceptive neurons induced by nociceptor conditioning inputs^{184,379} and conditional deletion of NR1 abolishes NMDA synaptic inputs and acute activity-dependent central sensitization.³⁰⁰ NMDAR is both a trigger and effector of central sensitization. Under normal conditions, the NMDAR channel is blocked in a voltage-dependent manner by a magnesium (Mg^{2+}) ion sitting in the receptor pore.¹⁹⁸ Sustained release by nociceptors of glutamate and the neuropeptides substance P and CGRP leads to sufficient membrane depolarization to force Mg^{2+} to leave the NMDAR pore, whereupon glutamate binding to the receptor generates an inward current.¹⁹⁸ Removal of this voltage-dependent block is a major mechanism for rapidly boosting synaptic efficacy and allows entry of Ca^{2+} into the neuron, which then activates numerous intracellular pathways that then contribute to the maintenance of central sensitization. In addition to the critical role of NMDAR in increasing the excitability nociceptive neurons, activation of group I mGluRs by glutamate also appear important for the development of central sensitization. Although these receptors do not participate to basal nociception,^{221,392} their activation is necessary for activity-dependent central sensitization mediated by C-fibers.^{14,67,165,296,392,393} In contrast, activation of group II-III mGluRs is associated with a reduction of capsaicin-induced central sensitization.²⁹⁶

Substance P (SP), which is co-released with glutamate by unmyelinated peptidergic nociceptors, is also involved in the generation of central sensitization.^{8,146,183,192,365} Substance P binds to the neurokinin-1 (NK1) G-protein-coupled receptor, which is expressed by spinothalamic, spinoparabrachial, and spino-PAG neurons¹⁰¹ and causes a long-lasting membrane depolarization,¹¹² and contributes to the temporal summation of C-fiber-evoked synaptic potentials^{80,388,389} as well as to intracellular signaling. Ablation of NK1-positive neurons in the spinal cord leads to a reduction in capsaicin-evoked central sensitization, confirming the importance of projecting neurons expressing the substance P receptor in this phenomenon.^{146,192} Calcitonin gene-related peptide (CGRP), also synthesized by small diameter sensory neurons, potentiates the effects of SP³⁶⁷ and participates in central sensitization through postsynaptic CGRP1 receptors, which activate PKA and PKC.^{307,308} CGRP also enhances release of brain-derived neurotrophic factor (BDNF) from trigeminal nociceptors,³⁴ which may contribute to its involvement in migraine and other primary headaches.^{82,103}

BDNF is a neurotrophic factor and synaptic modulator that is synthesized by nociceptor neurons and released into the spinal cord⁴⁰⁴ in an activity-dependent manner,¹⁹ where it also has a role in the production of central sensitization.^{113,144,330} On binding to its high-affinity trkB receptor, BDNF enhances NMDAR-mediated C-fiber-evoked responses¹⁴⁴ and causes activation of several signaling pathways in spinothalamic track neurons, including ERK^{141,245,292} and PKC.²⁹³

The inflammatory kinin bradykinin is produced in the spinal cord in response to intense peripheral noxious stimuli and acts through its G_q -coupled B2 receptor, which is expressed by dorsal horn neurons^{41,359} and boosts synaptic strength by activating PKC, PKA, and ERK.¹⁵² ERK is also activated by a serotonergic (5-HT) descending input involving the ionotropic 5-HT₃ receptor^{143,310,313,396} and possibly the 5-HT₇ G_s -coupled receptor.²⁹ Fig 4 summarizes the key known synaptic triggers of central sensitization.

Signaling Pathways and Activity-Dependent Central Sensitization

An increase in intracellular Ca^{2+} beyond a certain threshold level appears to be the key trigger for initiating activity-dependent central sensitization. Calcium influx through NMDAR

appears to be particularly prominent in the induction phase but can also occur through calcium-permeable AMPARs,^{159,355} voltage-gated calcium channels,^{52,376} as well as from release from intracellular microsomal stores in response to activation of several metabotropic receptors^{106,182} (Fig 5, A through C). Why is the calcium-induced activation of intracellular kinases so important? The reason is that ionotropic NMDA and AMPA glutamate receptors can be phosphorylated on several key residues located on their C-terminus,^{40,42} and this post-translational modification changes their activity as well as their trafficking to or from the membrane,^{40,161} which with similar post-translational changes in other ion channels, produces the functional changes that manifest as central sensitization (Fig 6, A through C). AMPAR subunit GluR1 residue Ser831 is phosphorylated by protein kinase C (PKC) and calcium-calmodulin-dependent protein kinase II (CaM-KII); Ser845 is the phosphorylation target of PKA, and GluR2 has 1 main site for phosphorylation by PKC, on Ser880.⁴⁰ For the NR1 subunit of NMDAR, PKC phosphorylates Ser896, whereas PKA has 2 potential phosphorylation sites on Ser890 and Ser897.^{168,333} NMDAR conductance properties are modified by phosphorylation of tyrosine residues located on the NR2A and NR2B subunits at 3 potential sites (Tyr1292, 1325, or 1387 for NR2A and Tyr1252, 1336, or 1472 for NR2B), through activation of nonreceptor tyrosine kinases such as Src or Fyn.^{42,106,107,268}

Stimulation of AMPAR and group I mGluRs^{89,106,118,281} participate with NMDAR in the activation of the intracellular pathways that sustain central sensitization. These include the PLC/PKC pathway, through opening of Ca²⁺ channels on the endoplasmic reticulum,^{85,391} the phosphatidylinositol-3-kinase (PI3 K) pathway,²⁴⁶ and the mitogen-activated protein kinase (MAPK) pathway that involves the extracellular signal-regulated kinases (ERK1 and ERK2), which are 44- and 42-kDa Ser/Thr kinases, respectively, with 90% sequence identity, and the cAMP response element binding protein (CREB).^{130,138,141,357,363} One way that ERK and CREB are activated is through an elevation in intracellular Ca²⁺ sufficient to drive a calmodulin-induced stimulation of adenylyl cyclases 1 and 8, whose cAMP production in turn activates PKA and subsequent cascade(s).³⁶³

The activation of ERK by phosphorylation is regulated by a core signaling module that consists of an apical MAPK kinase kinase (MAP3 K), a MAPK kinase (MEK or MKK), and the downstream ERK. Many different signaling pathways can induce ERK activation in addition to its canonical *ras/raf* pathway, and this can be readily detected immunohistochemically in the dorsal horn within minutes of peripheral noxious stimuli¹³⁰ (Fig 7). The presence of phosphorylated ERK reveals the anatomical distribution of those neurons whose intracellular signaling has been activated by the nociceptor input and are presumably undergoing the synaptic changes that constitute central sensitization^{120,131,141,152} (Fig 7). In the spinal cord, ERK is only activated in neurons in response to intense peripheral noxious stimulation, effectively identical to those stimuli that induce central sensitization,^{130,363} suggesting that ERK phosphorylation is a better marker of the neural plasticity that mediates central sensitization than *c-Fos*, which can be activated by low threshold stimuli that do not induce central sensitization.¹²⁶ Because the ERK pathway is composed of successive protein kinases, each of which can be activated by several different signaling pathways,¹³¹ most of the triggers of central sensitization such as NMDAR, group I mGluR, TrkB, NK1, or CGRP1 converge to activate ERK^{120,130,131,171} (Fig 8). Once activated, ERK produces translational and post-translational effects that participate in the maintenance of central sensitization in spinal cord neurons.^{131,387} The post-translational effects include an increase of NMDAR function through phosphorylation of its NR1 subunit^{152,293} (Fig 6, A), recruitment of AMPAR to the membrane^{93,254} (Fig 6, B) leading to an increase in AMPAR, and NMDAR currents boosting synaptic efficacy.¹⁵² Furthermore, ERK produces a decrease in K⁺ currents through phosphorylation of the residue Ser616 of Kv4.2 channels leading to an increase in membrane excitability^{118,119} (Fig 6, A). Transcriptional changes are mediated by activation of CREB as well as other transcription factors, which drives expression of several genes including *c-Fos*,

NK1, TrkB, and Cox-2¹³¹ (Fig 6, C). Inhibition of ERK activation using inhibitors of MEK reduces behavioral measures of activity-dependent central sensitization.^{137,142}

Nitric oxide (NO) synthesized by either neuronal or inducible NO synthases in the dorsal horn³⁸¹ also has a role in central sensitization.³⁸¹⁻³⁸³ Potential mechanisms for NO actions include the cGMP synthesis cascade, nitrosylation of membrane channels, ADP-ribosylation, and production of reactive species.^{64,278} The NO-cGMP pathway involves soluble guanylate cyclase, which is expressed by NK1-positive spinothalamic neurons, as well as inhibitory interneurons.⁷³ Fig 8 summarizes key intracellular pathways whose activation contributes to the generation of central sensitization.

Effectors of Activity-Dependent Central Sensitization

AMPA and NMDAR phosphorylation during central sensitization increases the activity/density of these receptors, leading to postsynaptic hyperexcitability.^{32,86-88,134,178,345,394,407,408} The first phase of central sensitization is a fast augmentation of excitatory glutamatergic synapses in the superficial dorsal horn that strengthens nociceptive transmission and recruits non-nociceptive input to the pathway. This is achieved by phosphorylation of numerous receptor and ion channel targets that lead to changes in threshold, channel kinetics, and voltage dependence, as well as a modification in the trafficking of the receptors to the synapse (Fig 6, A and B). On noxious stimulation, PKA phosphorylates GluR1 subunits,^{87,88,214} leading to an insertion of these receptors into the synapse^{84,93} and thereby an increase in synaptic strength.²⁰ Phosphorylation of GluR1-containing AMPAR by PKC and CaMKII also increases the excitability of nociceptive neurons.^{40,88}

NR1 phosphorylation by PKA⁴⁰⁸ or PKC^{32,407} participates in the development of hypersensitivity^{97,262,345} by increasing the response of NMDARs to glutamate.^{44,259} Phosphorylation of the NR2B subunit of NMDAR, mediated through Src activation, increases the opening of the receptor channel²⁶⁸ and prevents endocytosis of activated receptors by disrupting the binding site of AP-2, a protein involved in clathrin-coated endocytotic vesicle formation.⁴² Decoupling Src interaction with NMDAR blocks NR2B phosphorylation and reduces formalin-induced and inflammatory pain without altering basal nociceptive pain.¹⁷⁸

Activation of PKC contributes to hyperexcitability in nociceptive neurons by several different pathways. First, PKC reduces the Mg²⁺ block of NMDAR and increases the probability of channel opening, facilitating the activated state of NMDAR.⁴⁴ Second, activation of PKC decreases inhibitory transmission at the segmental level by reducing GABA and glycine tonic inhibition¹⁷⁴ and the descending inhibition driven from the PAG.¹⁷⁵ Disinhibition, mediated by whatever means, leaves dorsal horn neurons more susceptible to activation by excitatory inputs including non-nociceptive A-fibers, and is 1 of the major mechanisms triggering and maintaining central sensitization.^{17,289,341,390} Finally, PKC contributes with PKA to the activation of ERK in a manner that requires their coactivation and is triggered by the central release of bradykinin.¹⁵²

Activation of guanylate cyclase seems to be the major way that NO contributes to the induction of sensitization^{275,322,403} through increases in neuronal excitability and a reduction in inhibition,^{173,176,275} although an NO-mediated activation of ADP-ribosyltransferase may participate in the maintenance of central sensitization.⁴⁰³

Global Features of Activity-Dependent Central Sensitization

The key features of acute activity-dependent central sensitization are that it is induced with a short latency (seconds) by intense, repeated, or sustained nociceptor inputs and typically lasts for tens of minutes to several hours in the absence of further nociceptor input. It generally

requires activation of NMDA receptors for its induction, and these receptors contribute to its maintenance. Nevertheless, as reviewed above, multiple different triggers can contribute to the establishment of this form of central sensitization: glutamate acting on NMDAR, but also on AMPAR and mGluR, the neuropeptides substance P and CGRP, the kinin bradykinin, as well as BDNF and NO (Fig 4). The reason so many different transmitters, modulators, and their receptors are involved is that it is not their specific action that is important but rather that they are released directly from or induced in response to nociceptor afferent activity, and each can separately or together initiate the activation of those multiple intracellular signaling pathways that lead to the establishment of hyperexcitability in dorsal horn neurons (Figs 6 and 8). In other words, there are many parallel inputs to dorsal horn neurons that can independently or cooperatively initiate central sensitization. Elevation in intracellular calcium, by whatever means, is 1 major trigger, activating multiple calcium-dependent kinases that act on receptors and ion channels to increase synaptic efficacy (Figs 5 and 6). Many central sensitization-inducing inputs also activate ERK, and this MAPK appears to have a pivotal role, contributing to increases in AMPA and NMDA currents as well as reducing potassium currents (Fig 6, A). However, even this kinase may not be essential. Other kinases such as PKC, PKA, and Src can, independent of ERK, also modulate ionotropic receptors to lead to an increase in synaptic efficacy (Fig 6, A).

What has become clear is that there is no single defining molecular mechanism of central sensitization but rather that it is a general phenomenon, one that produces distinct changes in somatosensory processing but nevertheless can be mediated by several different processes that, in response to nociceptor input, can (1) increase membrane excitability, (2) facilitate synaptic strength, or (3) decrease inhibitory influences in dorsal horn neurons. Similarly, the effectors of this plasticity are multiple: changes in the threshold and activation kinetics of NMDA and AMPA receptors and in their trafficking to the membrane, alterations in ion channels to increase inward currents and reduce outward currents, and reductions in the release or activity of GABA and glycine (Fig 9). These changes produce dramatic alterations in function. However, they are usually relatively short-lasting and reversible. Phosphatases will dephosphorylate receptors and ion channels resetting their activity to baseline levels, trafficking to the membrane will reverse by endocytosis, and, with time, the increased gain of the nociceptive neurons will fade, at least in the absence of any further triggering inputs.^{186,400-402} Different, transcription-dependent changes are required for longer-lasting effects, and these generally do not occur in response only to nociceptor activity but are the consequence of peripheral inflammation and nerve injury (see below). Activity-dependent central sensitization, even though it increases pain sensitivity, is in most situations an adaptive mechanism. Unlike nociceptive pain, which warns of potential damage in response to noxious stimuli, central sensitization creates a situation in which pain is elicited by innocuous stimuli. This change is protective, because it helps healing by limiting use of an injured body part until the injury is fully repaired. Central sensitization becomes pathological, however, if inflammation persists, as with rheumatoid arthritis, in which no healing occurs, and in situations in which central sensitization becomes autonomous and is maintained in the absence of active peripheral pathology. Central sensitization represents not only a state in which pain can be triggered by less intense inputs but in which the central sensitization itself can be maintained by a lower level or different kind of input. Ongoing activity in C-fibers, even at levels that do not elicit central sensitization in basal conditions, is sufficient to maintain central sensitization once it has been induced for prolonged periods (days).¹⁵³ Furthermore, although nociceptor input is required to trigger central sensitization, phenotypic changes in myelinated fibers after inflammation and nerve injury can enable these afferents to acquire the capacity to generate central sensitization (see later).

Activity-Dependent Central Sensitization and Synaptic Plasticity

That the activity-dependent synaptic plasticity in the dorsal horn responsible for central sensitization is reversible differs from the permanent activity-dependent synaptic change in the cortex that leads to long-term memory, long-term potentiation (LTP), in which the efficacy only of activated synapses is changed. Synaptic changes with some resemblance to cortical LTP do occur in the spinal cord, that is, a form of homosynaptic potentiation. However, the major synaptic alteration underlying activity-dependent central sensitization is heterosynaptic potentiation, in which activity in 1 set of synapses enhances activity in nonactivated synapses, typically by “sensitizing” the entire neuron, something that never occurs with cortical LTP.

Homosynaptic potentiation is a type of use-dependent facilitation of a synapse evoked by activation of that same synapse (Fig 10, A). Classic LTP in the CA1 region of the hippocampus is formally defined as input-specific homosynaptic facilitation^{27,164,384} and is dependent on NMDAR activation, Ca^{2+} influx, and activation of Ca^{2+} -dependent intracellular signaling pathways, notably the CaMKII pathway.¹⁶⁴ Although the increased Ca^{2+} is relatively widespread in neurons after tetanic conditioning stimulation of afferents,^{128,182,260} only the stimulated synapse is potentiated.^{164,260} The development of LTP by 2 independent synapses using asynchronous pairing stimulation has been described in the hippocampus, but, once again, only conditioned synapses are potentiated.¹²³

One form of homosynaptic facilitation in spinal cord neurons is windup, in which the action potential discharge elicited by a low-frequency (0.5 to 5 Hz) train of identical C-fiber strength stimuli gets larger on each successive stimulus²⁰⁰ (Fig 11). Windup is the result of the activation of NK1 and CGRP1 receptors after release of substance P and CGRP from peptidergic nociceptors to produce a cumulative membrane depolarization from the temporal summation of slow synaptic potentials.²⁹⁰ This then enables activation of NMDAR by removal of the Mg^{2+} block, further boosting the responses in a nonlinear fashion^{63,72,331,379} (Fig 11). The stimuli that induce windup (repeated C-fiber stimulation) can lead to central sensitization,³⁷⁹ and, although windup is often considered to be an aspect of central sensitization, it is instead the reflection of activity-dependent excitability increases in neurons during a nociceptor conditioning paradigm rather than changes that follow such inputs, which is when central sensitization manifests. Windup disappears within tens of seconds of the end of the stimulus train as the membrane potential returns to its normal resting level (Fig 11).

Another form of homosynaptic facilitation occurs in NK1-positive lamina I neurons in the dorsal horn neuron. This has been termed LTP, although, unlike classic hippocampal LTP, this form of homosynaptic facilitation appears not to be persistent, or at least the functional effects on pain sensitivity are not permanent instead lasting, like central sensitization for a few hours, with no evident change equivalent to long-term memory. Perhaps, therefore, to avoid confusion with cortical plasticity, the term LTP should be avoided for homosynaptic potentiation in the spinal cord because the changes in the dorsal horn are long-lasting (hours) rather than long-term (persistent). The original description of this long-lasting homosynaptic potentiation in the dorsal horn referred to an activity-dependent facilitation of excitatory postsynaptic currents in spinoparabrachial neurons in response to high-frequency (tetanic burst; 100 Hz) stimulation of C-fibers.^{127,177,271} The physiological relevance of this phenomenon was questionable because C-fibers do not fire at such high frequencies. Conditioning C-fiber stimulation at a low frequency (2 Hz) was subsequently shown also to elicit a long-lasting homosynaptic potentiation in lamina I spino-PAG neurons but not in spinoparabrachial neurons.¹²⁸ This low-frequency potentiation is dependent on elevations in Ca^{2+} , which activates PLC, PKC, CaMKII, and NOS.¹²⁸ Capsaicin and formalin injection also evoke a homosynaptic long-lasting potentiation, as manifested by an enhancement of C-fiber-evoked synaptic potentials after the capsaicin/formalin evoked conditioning input.¹²⁸ Capsaicin is, of course, also a potent

inducer of activity-dependent central sensitization,^{294,340,383} characterized by the production of secondary hyperalgesia and tactile allodynia. However, both of these particular forms of pain hypersensitivity reflect heterosynaptic and not homosynaptic facilitation. Indeed, heterosynaptic facilitation characterizes most major changes in the receptive field properties of neurons and in pain sensitivity, in preclinical models, and human subjects^{368,369,376} (Fig 10, B).

Interestingly, healthy human subjects receiving high-frequency stimulation of C-fibers exhibit increased pain in the stimulated region, quite possibly caused by homosynaptic facilitation, but also show evidence of heterosynaptic facilitation, as manifested by dynamic mechanical allodynia in adjacent nonstimulated areas.¹⁴⁹ The combination of the homosynaptic potentiation of conditioning nociceptor inputs and the heterosynaptic facilitation of nonconditioned fibers in the nociceptive pathway constitutes central sensitization.

Heterosynaptic facilitation represents a form of activity-dependent facilitation where activity in 1 set of synapses (the conditioning input) augments subsequent activity in another nonactivated group of synapses (the test input) (Fig 10, B). For homosynaptic potentiation, the test and conditioning inputs are the same; for heterosynaptic facilitation they are different. "LTP"-like phenomena in spinobrachial neurons can only account for the augmentation of the same C-fiber inputs that evoked the facilitation and cannot contribute to either secondary hyperalgesia or tactile allodynia. Repeated nociceptor input, such as that generated by capsaicin, will simultaneously generate both a potentiation of the activated C-fiber synapses (homosynaptic), and, unlike LTP in the hippocampus, also a potentiation of neighboring nonactivated synapses (heterosynaptic). It seems likely, therefore, that long-lasting potentiation in projecting dorsal horn neurons is simply a restricted aspect of the general widespread changes induced in these neurons by nociceptor activity.

Heterosynaptic potentiation appears to dominate the functional sensory manifestations of use-dependent central sensitization. After injection of capsaicin, for example, the thresholds of sensory fibers innervating the area surrounding the injection site are not modified,^{23,157,340,365} but pain hypersensitivity in these areas is prominent and depends on centrally mediated heterosynaptic facilitation. The same argument holds for the activation of pain in response to tactile stimulation or A β fiber inputs during central sensitization. It is no surprise, then, that spinal "LTP" shares major mechanisms with central sensitization (NMDAR, Ca²⁺, kinases, and NO) because it is very likely that the phenomenon of central sensitization includes both homosynaptic and heterosynaptic facilitations triggered by the same process; the major difference is that heterosynaptic potentiation results from the spread of signaling from the conditioning synapse to other synapses in the neuron^{182,270,371} (Fig 10, B). Homosynaptic changes will contribute with peripheral sensitization to primary hyperalgesia,^{128,270} whereas heterosynaptic facilitation alone is responsible for secondary hyperalgesia and allodynia.

Although several different forms of LTP have been characterized in the hippocampus,^{188,224,384} "spreading" or heterosynaptic LTP has not been reported, even though release of Ca²⁺ from intracellular stores can cause a spread of long-term depression (LTD) to neighboring, unstimulated synapses.²²⁷ What, then, is responsible for heterosynaptic facilitation in dorsal horn neurons? Two major candidates are the activation of mGluRs and NO. mGluRs are coupled to the Ca²⁺ channels of the endoplasmic reticulum⁸⁵ and play an important role in central sensitization.⁷ Consequently, the release of intracellular Ca²⁺ in spinal cord neurons on mGluR activation may participate in spreading facilitation from conditioned synapses to neighboring test synapses. NO is also a major effector of spinal cord neuronal plasticity^{211,309} and diffuses rapidly from the site of its production to produce multiple effects at a distance via its downstream signaling pathways, and in this way may contribute to the heterosynaptic facilitation characteristic of central sensitization. It is certainly likely that these

and other “spreading” signals cooperate to produce the widespread synaptic facilitation so characteristic to central sensitization. Scaffolding proteins play a major role in the addressing of specific kinases to the synapse and represent another potential mechanism for widespread synaptic facilitation. A recent study has shown that in the hippocampus, CaMKII activation is restricted to the synaptic bouton of a conditioned synapse, thus only allowing homosynaptic facilitation at that specific site.¹⁶⁴ It is likely in the dorsal horn that CaMKII activation will be much more widespread and indeed the dendrites of dorsal horn neurons lack synaptic boutons.

Central Sensitization in Pathological Settings

In addition to its role in rapidly and reversibly sensitizing the nociceptive system by activity-dependent changes in synaptic strength and excitability, central sensitization also contributes to the longer-lasting and sometimes persistent pain hypersensitivity present in pathological situations involving inflammation and damage to the nervous system. The molecular and cellular mechanisms involved include some that are also responsible for activity-dependent central sensitization and others that are unique to either inflammation or nerve injury. NMDAR,^{47,193,255,280,315} AMPAR,^{180,239} group I mGluR,^{7,65,75,92,102,118,221,288,392,405} group II-III mGluR,^{45,104,194,207,286,398} BDNF,^{144,179,190,230} SP and CGRP,^{2,4,163} NO,^{50,316} and bradykinin²⁴¹ have all been shown to contribute both to the development of central sensitization and to pain hypersensitivity in inflammatory and neuropathic pain models.

Inflammatory Pain

Peripheral inflammation induces a phenotypic switch in primary sensory neurons that comprises a change in their neurochemical character and properties due to alterations in transcription and translation. We will only discuss here those changes that relate specifically to central sensitization by virtue of changes in the synaptic input produced by the afferents and will not review the major changes that also alter peripheral transduction sensitivity and membrane excitability (peripheral sensitization), although of course, anything that increases nociceptor afferent input will also indirectly lead to increased central sensitization. Large DRG neurons begin, unlike in their naive condition, to express SP and BDNF when their peripheral terminals are exposed to inflammatory signals and nerve growth factor (NGF).^{191,223} Consequently, activation of the myelinated fibers by low-intensity innocuous stimuli now releases these neuropeptides in the spinal cord, and conditioning stimulation of the afferents acquires the capacity to generate central sensitization, something they normally cannot do^{185,190,223} (Fig 12). After peripheral inflammation, A β -mediated synaptic input to superficial dorsal horn neurons is substantially increased from the very low levels found in noninflamed animals.¹⁶ TrkA-expressing nociceptors, instead of a phenotypic switch, begin to express higher levels of neuropeptides and other NGF-dependent proteins as a result of exposure to the increased NGF produced by inflammation.^{370,375}

A critical central pathway for the generation of inflammatory pain hypersensitivity involves induction of cyclooxygenase-2 (Cox-2) in dorsal horn neurons, to drive production of prostaglandin E2 (PGE2).^{269,349} PGE2 binds to its EP2 GPCR on dorsal horn neurons to potentiate AM-PAR and NMDAR currents,¹⁵² activate nonselective cation channels,¹⁸ and reduce inhibitory glycinergic neurotransmission by blocking glycinergic receptors with $\alpha 3$ subunits^{9,111,152,213} (Fig 12). PGE2 also acts on EP4 receptors on presynaptic terminals to increase transmitter release.³⁵⁰ The importance of the central neuronal induction of COX-2 to inflammatory hyperalgesia is revealed by conditional deletion of COX-2 only in neurons, which results in the retention of peripheral inflammation and heat hyperalgesia but an almost complete loss of mechanical pain hypersensitivity.³⁵⁰

Under normal conditions, microglia are the only immunocompetent cells of the nervous system^{66,362} and constantly probe or survey the CNS parenchyma to maintain homeostasis.

^{62,225} After peripheral inflammation, some spinal cord microglial cells change their shape, function, and chemical expression.^{115,258,311,312} In particular, p38 MAPK is activated^{311,312} and leads to the synthesis and release of pro-inflammatory cytokines,^{115,258} among which, IL-1 β and TNF- α contribute to the development of central sensitization by enhancing excitatory and reducing inhibitory currents and by activating induction of COX-2^{142,269} (Fig 12).

Neurons in the superficial lamina of the dorsal horn usually display a GluR2 AMPAR phenotype (ie, are Ca²⁺-impermeable)²⁵²; however, peripheral inflammation triggers a shift from GluR2/3 to GluR1-containing AMPARs at the membrane^{159,238,355} (see “PSD Proteins and AMPAR Recycling and Subunit Switch,” below). Under these conditions, activation of AMPAR elicits entry of Ca²⁺, which can then participate in the activation of the signaling pathways that drive central sensitization. Ca²⁺-permeable AMPARs appear to be a major source of the [Ca²⁺]_i increase in inflammatory pain, generating as much Ca²⁺ influx as with NMDAR activation.¹⁸² The functional state of NMDAR is also modified in response to peripheral inflammation, with phosphorylation of NR2B subunits by Src resulting in increased activity of the receptors^{106,107,178} and in their maintenance at the synapse.⁴² Finally, peripheral inflammation also promotes group I mGluR insertion into the membrane (mGluR5) and closer to the synapse (mGluR1), thereby further clustering these receptors at the synapse.²⁴⁷

Neuropathic Pain

After peripheral nerve injury, damaged and nondamaged A- and C-fibers begin to generate spontaneous action potentials. Because these do not arise from the peripheral terminal, it is a form of ectopic input.^{70,74} Such input in C-fibers can initiate and then maintain activity-dependent central sensitization in the dorsal horn.¹⁵⁴ However, because of chemical and structural changes in A fibers,^{56,229,386} input in these afferents can also begin to drive central sensitization.⁶⁹ Injured, and to a much lesser extent, noninjured sensory neurons in the dorsal root ganglion exhibit a massive change in transcription that alter their membrane properties, growth, and transmitter function.^{56,231,232,386} These changes are much greater than those that occur in response to peripheral inflammation, where only a few tens of transcripts are altered in the DRG¹⁹⁵ and involve altered expression of about 1000 transcripts, including ion channels, receptors, transmitters, and the molecular machinery necessary for axon regeneration.^{56,261} Among the many changes, large fibers begin to express new transmitters and neuromodulators including substance P and BDNF and the synthetic enzymes for tetrahydrobiopterin, an essential cofactor for NOS. Stimulation of non-nociceptive fibers now triggers release of factors that can drive central sensitization.^{24,56,91,229,327,386}

Structural changes also contribute to altered synaptic function. Peripheral nerve injury leads to a transganglionic degeneration of C-fiber terminals in lamina II.^{13,136} This loss of presynaptic input, together with the triggering of increases in the intrinsic axonal growth capacity as part of the regenerative response of the injured neurons, provides an opportunity and the molecular means for myelinated A- β fibers to sprout from laminae III-IV into laminae I-II and make contact with nociceptive-specific neurons.^{166,191,285,377,378} The original experiments describing the sprouting phenomenon were conducted using cholera toxin B subunit as a selective tracer for A-fibers as well as single axonal label with HRP. The selectivity of this toxin after peripheral nerve injury is somewhat controversial.^{125,339} Nevertheless, immunostaining for c-Fos activation and electrophysiological recordings have clearly established that peripheral nerve injury causes large myelinated fibers to begin to drive nociceptive neurons in superficial lamina.^{24,151,235,366}

A reduction in the synthesis, release, or activity of inhibitory transmitters leads to a state of disinhibition, whose net functional effects are very similar to that produced by increases in synaptic strength of excitatory synapses and in membrane excitability.^{289,341,390} In

neuropathic pain states, there is substantial disinhibition in the superficial dorsal horn with loss of GABAergic and a reduction in glycinergic inhibitory currents²¹⁰ that can be attributed, at least in part, to apoptosis of inhibitory interneurons.²⁷⁷ This neuronal death appears to be the result of an NMDAR-induced excitotoxicity that develops over time rather than to the large amount of glutamate released centrally at the time of nerve injury.²⁷⁷ One laboratory failed to find significant loss of neurons or of GA-BAergic content in the dorsal horn of neuropathic pain animal models.²⁴⁹⁻²⁵¹ The reasons for this discrepancy are not clear but may reflect technical differences in how the studies were performed. Interestingly, the reduction in glycinergic neurotransmission caused by the activation of EP2 receptors after peripheral inflammation does not appear to operate after nerve injury, further indicating that some inflammatory and neuropathic pain mechanisms differ.¹¹⁷

Another mechanism contributing to the reduction in segmental inhibition in a subpopulation of lamina I neurons in the spinal cord after nerve injury is dependent on BDNF effects on an anion transporter, changing anion gradients across neuronal membrane to alter the inhibitory efficacy of GABA. Under normal conditions, the intracellular concentrations of Cl^- are maintained by the opposed effects of Cl^- -cotransporter K^+ - Cl^- exporter 2 channels (KCC2) and Na^+ - K^+ - Cl^- exporter 1 channels (NKCC1). KCC2 drives Cl^- ions out of the cells (along with K^+) and NKCC1 is responsible for an influx of K^+ , Na^+ , and Cl^- into the cells. The net effect of these 2 co-transporters is a steady-state Cl^- concentration gradient in which opening of Cl^- channels (such as GABA_A receptors) causes entry of Cl^- into the neuron and hyperpolarizes the neurons. After peripheral nerve injury, BDNF released by activated microglial cells results in a reduction of KCC2 expression in a subset of neurons in the superficial lamina of the dorsal horn.^{57,58,203} Consequently, activation of GABA_A receptors by GABA result in a diminution or absence of Cl^- entry into the cell and thus a disinhibition of these nociceptive neurons^{57,58,179,203} (Fig 12). As after peripheral inflammation, there is also an increase in descending excitatory controls from the RVM in the brainstem after peripheral nerve injury, as well as a reduction of descending inhibitory controls.^{60,95,351,356}

Peripheral nerve injury causes a massive activation of, and change in, glial cells in the spinal cord as well as infiltration of peripheral immune-competent cells, notably macrophages and T-cells.^{38,317,360} The extent and duration of the changes in microglia and astrocytes is much greater than in response to peripheral inflammation. Activated microglia produce and release trophic factors, neurotransmitters, cytokines, and reactive oxygen species^{263,361} and appear to play an essential step in the development of pain after nerve injury by triggering central sensitization through their interaction with neurons.^{133,160,162,201,206,256,257,352} Numerous signals trigger microglial activation and recruitment, including ATP and NO,^{62,81,225} cytokines, and chemokines, some of which are released by injured sensory neurons and others by microglial cells themselves or by astrocytes and T-cells.^{1,3,66,76,204,348,361,362} Release of cytokines by microglia increases neuronal excitability through activation of ERK and CREB.^{131,142} Activated microglia also release BDNF and NO,^{58,116} promoting segmental disinhibition.⁵⁷ Finally, microglia can also provoke neuronal death by producing ROS, pro-apoptotic cytokines such as TNF,¹²¹ and by a diminished glutamate uptake.^{48,326,332} T-cells produce specific cytokines such as IFN- γ , which reduce GABAergic currents in the dorsal horn³⁵⁴ through activation of IFN- γ receptors³⁵³ and also activate and recruit microglia. Astrocytes also become activated after peripheral nerve injury,^{98,131,205} with a slower onset and more prolonged time course than microglia, and may play more of a role in the maintenance of neuropathic pain hypersensitivity than microglia.^{94,399,406} What seems clear is that multiple different mechanisms operate after nerve injury to increase excitability and reduce inhibition.

Scaffolding Proteins, Synaptic Plasticity, and Central Sensitization During Inflammation and After Nerve Injury

The proteins that make up the PSD can drive a major functional reorganization of synapses, modifying post-synaptic efficacy by altering receptor density at the membrane and producing switches from Ca^{2+} -impermeable to Ca^{2+} -permeable AMPARs (Fig 13). The PSD is not simply a structural landmark of the synapse but contains elements essential both for the formation of the synapse and for changes in its properties. Absence of scaffolding proteins or specific disruption of their binding sites results in a dramatic reduction in synaptic plasticity because the proteins contribute both to transcriptional and post-translational events. They initiate signaling cascades that lead to the activation of transcription factors, traffic newly synthesized receptors to the PSD, and “address” kinases and phosphatases to specific receptors in a stimulus-dependent manner. Although the involvement of the PSD in synaptic plasticity in the cortex is much better established than in the spinal cord, there is increasing evidence for a major role for the PSD in changing synaptic efficacy in response to peripheral inflammation and nerve injury.

The PSD consists of cytoskeletal proteins, signaling molecules, membrane receptors, and scaffolding proteins.²³⁴ Scaffolding proteins are families of proteins characterized by their ability to interact with numerous partners, and these proteins form the dense molecular structure of the postsynaptic component of the synapse. A particularly abundant component of the PSD are proteins containing a specific peptidergic domain called PDZ, which is named after the protein in which the sequence was first identified (postsynaptic density protein 95 [PSD-95]/discs large/zonula occludens 1). This family of proteins includes, among hundreds of members, the 4 membrane-associated guanylate kinases (MAGUK): PSD-95, PSD-93, synapse associated protein (SAP)-97, SAP-102. The MAGUKs represent the most abundant scaffolding protein family in the PSD²³⁴ and are characterized by 3 PDZ domains, an Src homology region (SH3) domain, and a guanylate kinase-like (GK) domain,¹⁴⁸ making them central elements of the synapse scaffold. The prime binding protein for the MAGUK family is the NMDAR subunit NR2,¹⁵⁵ but it also binds to the transmembrane AMPAR regulatory proteins (TARPs),⁴³ nonreceptor tyrosine kinases,³²⁹ nNOS,^{30,31} GKAP,²¹⁷ and AKAP79/150.⁵³ MAGUKs can be seen as the functional scaffold of the PSD and are essential for the structural integrity of synapses but also modulate the insertion of glutamate receptors into the synapse and physically bring together key enzymes to the PSD.^{59,148,167,197,265} Knock-down of PSD-95 and PSD-93, as well as targeted mutagenesis of the residues required for their protein:protein interaction, both prevent and reduce central sensitization in normal conditions³²¹ as well as in inflammatory^{319,323,397} and neuropathic pain models^{99,320,323,397} but do not alter nociception or locomotor functions.^{319-321,323}

Another member of the PDZ family, stargazin, is a 4-transmembrane domain protein whose putative secondary structure is close to the Ca^{2+} -channel γ subunit, and was named $\gamma 2$.¹⁷⁰ Stargazin however, does not play an important role in neuronal Ca^{2+} channels¹⁷⁰ but is instead highly concentrated in the PSD and co-immunoprecipitates with GluR1, 2, and 4.^{43,335} The protein is a major AMPAR partner, along with the 4 other isoforms, $\gamma 3$, $\gamma 4$, $\gamma 7$, and $\gamma 8$,^{140,335} which form the TARP subgroup.³³⁵ Stargazin traffics AMPAR from the endoplasmic reticulum to the extrasynaptic membrane.^{43,334} Once stargazin and AMPAR are addressed to the extrasynaptic membrane, their recruitment to the synapse requires interaction of the C-terminus segment of stargazin with PSD-95.^{21,276} Activity-dependent phosphorylation of stargazin by PKC and CaMKII³⁴² produces a massive insertion of AMPAR into the membrane,³³⁷ whereas stargazin's dephosphorylation by PP1 or PP2B reduces the number of AMPAR at the synapse.³³⁷ In addition, via the interaction between stargazin's ectodomain and the glutamate binding region of AMPAR, stargazin modulates the activity of AMPAR by slowing channel deactivation and desensitization and increasing the affinity of the receptors for

glutamate,^{49,334,336} thereby potentiating synaptic strength. Disruption of stargazin in the spinal cord inhibits the second phase of formalin-induced pain and reduces the heat hyperalgesia caused by intraplantar CFA injection.³¹⁸ Recently, cornichon homolog 2 (CNIH-2) and cornichon homolog 3 (CNIH-3) have been found to be novel partner proteins for AMPAR in the CNS.²⁷⁹ Cornichon proteins bind to GluR1-4 AMPAR and, as for stargazin, they promote AMPAR surface expression and slow their deactivation kinetics.²⁷⁹ Because their expression in the CNS is estimated to be in 70% of neurons, and because cornichon and stargazin appear to be mutually exclusive,²⁷⁹ the determination of their presence in spinal cord neurons and their role in central sensitization is something that needs to be investigated.

EphrinB-ephBR receptor interactions participate in NMDAR clustering at the PSD.⁹⁰ EphBRs are receptor tyrosine kinases expressed by postsynaptic neurons, whereas EphrinB is anchored to the presynaptic membrane.¹⁵⁶ The kinase activity of EphBR is not required for the initial clustering of NMDAR at the synapse but is essential for their maintenance.⁶¹ Stimulation of EphBR potentiates NMDAR-induced Ca^{2+} influx and the phosphorylation of CREB through the activation of the nonreceptor tyrosine kinase src³¹⁴ and recruits CaM-KII to the synapse²³⁶ to increase NMDAR activity.³¹⁴ Activation of EphBR in the spinal cord induces thermal hyperalgesia (but not allodynia),^{22,299} without modifying nociception.²² Inhibition of EphRB prevents or reverses inflammatory^{22,291} and neuropathic^{150,299} pain and prevents establishment of NMDAR-induced spinal cord "LTP."²⁹⁹ More specifically, targeted disruption of the coupling between EphRB-activated Src and NR2B also prevents the development of central sensitization without altering basal nociceptive transmission.¹⁷⁸ In addition, sustained nociceptive activity leads to an upregulation and reorganization of presynaptic EphB increasing EphB-EphRB interaction.^{22,298,299}

The nonreceptor tyrosine kinase family also includes Fyn, which phosphorylates NR2 subunits²⁶⁸ and binds to PSD-93 to phosphorylate NR2_{A/B}²⁷³ and could play a role in the maintenance of neuropathic pain.⁵

NMDA receptors are structurally connected with group I metabotropic glutamate receptors through a complex composed of PSD-95, GKAP, Shank1, and Homer1b/c. GKAP binds to the GK domain of PSD-95^{147,217} and recruits Shank1 to the PSD via their PDZ domain.^{216,264} Shank1 then binds to the EVH1 domain of homer1b/c.³⁴³ Homer1b/c proteins have a coiled-coil structure in their C-terminus region that enables them to form homotetramers or heterotetramers.²⁸⁴ This assembly of Homers leaves 3 available EVH1 domains that can bind several other targets such as group I mGluRs (but not group 2 or 3 mGluRs),²⁸ inositol triphosphate receptors (IP3R), or the actin cytoskeleton.^{284,344} The interaction between Homer1b and IP3R and between Shank1 and Homer1b/c controls local Ca^{2+} release from the endoplasmic reticulum upon mGluRs activation.^{266,267} Homer1a, a short isoform of Homer1b/c that lacks the coiled-coil structure, is an immediate early gene activated on neuronal activity and participates in remodeling synapses in an activity-dependent manner.¹²⁹ Homer1a is upregulated in dorsal horn neurons after the subcutaneous injection of formalin or CFA³²⁴ as well as transiently after peripheral nerve injury.²⁰⁸ Factors responsible for Homer1a activation include NMDAR, ERK1/2 and Src.^{208,324} Knock-down of Homer1a increases pain-like behaviors specific to central sensitization and not those associated with peripheral sensitization,³²⁴ whereas overexpression reduced inflammatory pain-like behavior without altering basal nociception.³²⁴ Homer1a probably causes a disruption of the clustering properties of Homer1b/c protein and of Ca^{2+} release on NMDAR or mGluR activation.³²⁴ In contrast, homer1b activation leads to an increase of AMPAR activity after the stimulation of mGluR, whereas activation of Homer1a inhibits this.³⁴⁷ In addition, Homer1b/c proteins could play an important role in clustering group 1 mGluRs at the synapse after CFA injection,²⁴⁷ an effect that is reduced by Homer1a overexpression.³⁸⁵ Homer proteins are, therefore, convergent factors that potentially link major glutamate receptors with Ca^{2+} stores in neurons, and the balance between

Homer1b/c and Homer1a may play an important role in the development and maintenance of central sensitization.

PSD Proteins and AMPAR Recycling and Subunit Switch

Stargazin may be important in creating the switch from GluR2- to GluR1-containing AMPAR in response to peripheral inflammation^{159,355} by specifically addressing GluR1-containing AMPARs to the synapse and then strengthening their activity via its ectodomain.³³⁴ Peripheral inflammation leads via PKA to a phosphorylation of Ser831 and Ser845 on GluR1 in the spinal cord,¹⁸⁰ which, in association with CaMKII activation, promotes a transfer of GluR1-containing AMPARs to the membrane,⁸⁴ where they are maintained by synaptic activity.¹⁸⁹

GluR2-containing AMPARs are associated with high affinity to GRIP-1, which clusters the receptors at the synapse,^{77,114} whereas PICK-1 is a critical element for activity-dependent endocytosis of the receptor.^{51,96,226} NMDAR activation leads to a PKC-mediated phosphorylation of GluR2 at Ser880,^{172,238} which decreases GluR2's affinity for GRIP-1, whereas the increase in intracellular Ca^{2+} recruits PICK-1 to the synapse,^{51,108,238} where it reduces the clustering of GluR2-containing AMPAR^{51,172,196,238,304,328} (Fig 14). On endocytosis, vesicles can either be inserted back into the synapse under the action of NSF,^{122,226,228,297} which disrupts the [GluR2-PICK-1] complex,¹⁰⁹ or be maintained out of the synapse through PICK-1.¹⁷² During inflammation, NSF expression is reduced in the spinal cord,¹³⁹ thereby preventing GluR2-containing AMPAR reinsertion into the synapse.

The net effect of these complex changes is an increase in GluR1 and a decrease in GluR2 containing AMPAR at the synapse (Fig 14). Once inserted into the synapse, GluR1-containing AMPAR remain there for as long as there is glutamate release,^{83,189} and their activation is potentially increased by stargazin's ectodomain, thereby further promoting the Ca^{2+} -dependent pathways required for maintenance of central sensitization. In addition, the phosphorylation of stargazin's C-terminus domain by PKC and CaMKII³⁴² increases its affinity for PSD-95, thereby re-enforcing the clustering of AMPAR with NMDAR at the synapse.²⁷⁶

After peripheral nerve injury, GluR2 and GRIP-1 are up-regulated in dorsal horn neurons,^{100,110} whereas PICK-1 expression is not modified and NSF is downregulated.¹⁰⁰ Peptides that disrupt the binding of GRIP-1 or NSF with GluR2 partially decrease neuropathic pain-like behaviors.¹⁰⁰ PICK-1 is also required for $\text{G}_{i/o}$ -coupled mGluR7 trafficking to the membrane.³⁰⁶ Activation of mGluR7 can block P/Q-type Ca^{2+} channels²⁴⁰ and reduces the pain caused by injection of capsaicin²¹⁸ or peripheral nerve injury.²³⁷ The upregulation of GRIP-1 but not of PICK-1 in neuropathic pain models would promote AMPAR maintenance at the synapse but not that of mGluR7, resulting in increased excitability in these cells.

Finally, the A kinase-anchoring protein 79/150 (AKAP79/150) binds to PSD-95⁵³ and is a scaffold for protein kinases and phosphatases,^{53,169} specifically trafficking enzymes within the PSD to increase (kinases) or reduce (phosphatases) synaptic transmission. When AKAP79/150 recruits PKA^{181,295} or PKC³²⁵ in the PSD, it promotes insertion of new AMPAR to the membrane (via PKA) as well as increasing their activity (via PKC). In contrast, recruitment of PP2B^{25,169} triggers AMPAR endocytosis.²⁵ AKAP79/150 may function therefore as a “master switch” of central sensitization by promoting phosphorylation or dephosphorylation of stargazin and AMPAR.

Conclusion

Before central sensitization was discovered, there were 2 major models of pain. The first was that it was a labeled-line system, in which specific “pain pathways” were activated only by particular peripheral “pain stimuli” and that the amplitude and duration of pain was determined

solely by the intensity and timing of these inputs. The second model evoked “gate controls” in the CNS, which, by opening or closing, enabled or prevented pain. Neither model envisaged, however, that pain may arise as a result of changes in the properties of neurons in the CNS: central sensitization. We now appreciate that there are indeed specific nociceptive pathways and that these are subject to complex facilitating and inhibitory controls; both models were in part correct. We also know though, that changes in the functional properties of the neurons in these pathways are sufficient to reduce pain threshold, increase the magnitude and duration of responses to noxious input, and permit normally innocuous inputs to generate pain sensations. Pain is not then simply a reflection of peripheral inputs or pathology but is also a dynamic reflection of central neuronal plasticity. The plasticity profoundly alters sensitivity to an extent that it is a -major contributor to many clinical pain syndromes and represents a major target for therapeutic intervention. The past 26 years have seen enormous advances both in our appreciation of the nature and manifestations of central sensitization and in its underlying molecular mechanisms. We have great insight into what triggers can induce central sensitization, through which signaling pathways and by means of which effectors. The complexity is daunting because the essence of central sensitization is a constantly changing mosaic of alterations in membrane excitability, reductions in inhibitory transmission, and increases in synaptic efficacy, mediated by many converging and diverging molecular players on a background of phenotypic switches and structural alterations. Nevertheless, enormous progress has been made in dissecting out where, when, and how the plasticity occurs, although clearly, more is still waiting to be learned.

Acknowledgments

Supported by the National Institutes of Health and the Fondation pour la Recherche Médicale.

References

1. Abbadie C, Bhangoo S, De Koninck Y, Malcangio M, Melik-Parsadaniantz S, White FA. Chemokines and pain mechanisms. *Brain Res Rev* 2009;60:125–134. [PubMed: 19146875]
2. Abbadie C, Brown JL, Mantyh PW, Basbaum AI. Spinal cord substance P receptor immunoreactivity increases in both inflammatory and nerve injury models of persistent pain. *Neuroscience* 1996;70:201–209. [PubMed: 8848125]
3. Abbadie C, Lindia JA, Cumiskey AM, Peterson LB, Mudgett JS, Bayne EK, DeMartino JA, MacIntyre DE, Forrest MJ. Impaired neuropathic pain responses in mice lacking the chemokine receptor CCR2. *Proc Natl Acad Sci U S A* 2003;100:7947–7952. [PubMed: 12808141]
4. Abbadie C, Trafton J, Liu H, Mantyh PW, Basbaum AI. Inflammation increases the distribution of dorsal horn neurons that internalize the neurokinin-1 receptor in response to noxious and non-noxious stimulation. *J Neurosci* 1997;17:8049–8060. [PubMed: 9315923]
5. Abe T, Matsumura S, Katano T, Mabuchi T, Takagi K, Xu L, Yamamoto A, Hattori K, Yagi T, Watanabe M, Nakazawa T, Yamamoto T, Mishina M, Nakai Y, Ito S. Fyn kinase-mediated phosphorylation of NMDA receptor NR2B subunit at Tyr1472 is essential for maintenance of neuropathic pain. *Eur J Neurosci* 2005;22:1445–1454. [PubMed: 16190898]
6. Ablin J, Neumann L, Buskila D. Pathogenesis of fibromyalgia: A review. *Joint Bone Spine* 2008;75:273–279. [PubMed: 18375167]
7. Adwanikar H, Karim F, Gereau RW. Inflammation persistently enhances nocifensive behaviors mediated by spinal group I mGluRs through sustained ERK activation. *Pain* 2004;111:125–135. [PubMed: 15327816]
8. Afrah AW, Fiska A, Gjerstad J, Gustafsson H, Tjolsen A, Olgart L, Stiller CO, Hole K, Brodin E. Spinal substance P release in vivo during the induction of long-term potentiation in dorsal horn neurons. *Pain* 2002;96:49–55. [PubMed: 11932060]
9. Ahmadi S, Lippross S, Neuhuber WL, Zeilhofer HU. PGE(2) selectively blocks inhibitory glycinergic neurotransmission onto rat superficial dorsal horn neurons. *Nat Neurosci* 2002;5:34–40. [PubMed: 11740501]

10. Alvarez FJ, Villalba RM, Carr PA, Grandes P, Somohano PM. Differential distribution of metabotropic glutamate receptors 1a, 1b, and 5 in the rat spinal cord. *J Comp Neurol* 2000;422:464–487. [PubMed: 10861520]
11. Antal M, Fukazawa Y, Eorogh M, Muszil D, Molnar E, Itakura M, Takahashi M, Shigemoto R. Numbers, densities, and colocalization of AMPA- and NMDA-type glutamate receptors at individual synapses in the superficial spinal dorsal horn of rats. *J Neurosci* 2008;28:9692–9701. [PubMed: 18815255]
12. Aronica E, Catania MV, Geurts J, Yankaya B, Troost D. Immunohistochemical localization of group I and II metabotropic glutamate receptors in control and amyotrophic lateral sclerosis human spinal cord: Upregulation in reactive astrocytes. *Neuroscience* 2001;105:509–520. [PubMed: 11672616]
13. Arvidsson J, Ygge J, Grant G. Cell loss in lumbar dorsal root ganglia and transganglionic degeneration after sciatic nerve resection in the rat. *Brain Res* 1986;373:15–21. [PubMed: 3719303]
14. Azkue JJ, Liu XG, Zimmermann M, Sandkuhler J. Induction of long-term potentiation of C fibre-evoked spinal field potentials requires recruitment of group I, but not group II/III metabotropic glutamate receptors. *Pain* 2003;106:373–379. [PubMed: 14659520]
15. Azkue JJ, Mateos JM, Elezgarai I, Benitez R, Osorio A, Diez J, Bilbao A, Bidaurrezaga A, Grandes P. The metabotropic glutamate receptor subtype mGluR 2/3 is located at extrasynaptic loci in rat spinal dorsal horn synapses. *Neurosci Lett* 2000;287:236–238. [PubMed: 10863038]
16. Baba H, Doubell TP, Woolf CJ. Peripheral inflammation facilitates Abeta fiber-mediated synaptic input to the substantia gelatinosa of the adult rat spinal cord. *J Neurosci* 1999;19:859–867. [PubMed: 9880605]
17. Baba H, Ji RR, Kohno T, Moore KA, Ataka T, Wakai A, Okamoto M, Woolf CJ. Removal of GABAergic inhibition facilitates polysynaptic A fiber-mediated excitatory transmission to the superficial spinal dorsal horn. *Mol Cell Neurosci* 2003;24:818–830. [PubMed: 14664828]
18. Baba H, Kohno T, Moore KA, Woolf CJ. Direct activation of rat spinal dorsal horn neurons by prostaglandin E2. *J Neurosci* 2001;21:1750–1756. [PubMed: 11222664]
19. Balkowiec A, Katz DM. Activity-dependent release of endogenous brain-derived neurotrophic factor from primary sensory neurons detected by ELISA in situ. *J Neurosci* 2000;20:7417–7423. [PubMed: 11007900]
20. Banke TG, Bowie D, Lee H, Huganir RL, Schousboe A, Traynelis SF. Control of GluR1 AMPA receptor function by cAMP-dependent protein kinase. *J Neurosci* 2000;20:89–102. [PubMed: 10627585]
21. Bats C, Groc L, Choquet D. The interaction between Stargazin and PSD-95 regulates AMPA receptor surface trafficking. *Neuron* 2007;53:719–734. [PubMed: 17329211]
22. Battaglia AA, Sehayek K, Grist J, McMahon SB, Gavazzi I. EphB receptors and ephrin-B ligands regulate spinal sensory connectivity and modulate pain processing. *Nat Neurosci* 2003;6:339–340. [PubMed: 12640461]
23. Baumann TK, Simone DA, Shain CN, LaMotte RH. Neurogenic hyperalgesia: The search for the primary cutaneous afferent fibers that contribute to capsaicin-induced pain and hyperalgesia. *J Neurophysiol* 1991;66:212–227. [PubMed: 1919668]
24. Bester H, Beggs S, Woolf CJ. Changes in tactile stimuli-induced behavior and c-Fos expression in the superficial dorsal horn and in parabrachial nuclei after sciatic nerve crush. *J Comp Neurol* 2000;428:45–61. [PubMed: 11058224]
25. Bhattacharyya S, Biou V, Xu W, Schluter O, Malenka RC. A critical role for PSD-95/AKAP interactions in endocytosis of synaptic AMPA receptors. *Nat Neurosci* 2009;12:172–181. [PubMed: 19169250]
26. Bliddal H, Danneskiold-Samsøe B. Chronic widespread pain in the spectrum of rheumatological diseases. *Best Pract Res Clin Rheumatol* 2007;21:391–402. [PubMed: 17602990]
27. Bliss TV, Collingridge GL. A synaptic model of memory: long-term potentiation in the hippocampus. *Nature* 1993;361:31–39. [PubMed: 8421494]
28. Brakeman PR, Lanahan AA, O'Brien R, Roche K, Barnes CA, Huganir RL, Worley PF. Homer: A protein that selectively binds metabotropic glutamate receptors. *Nature* 1997;386:284–288. [PubMed: 9069287]

29. Brenchat A, Romero L, Garcia M, Pujol M, Burgueno J, Torrens A, Hamon M, Baeyens JM, Buschmann H, Zamanillo D, Vela JM. 5-HT₇ receptor activation inhibits mechanical hypersensitivity secondary to capsaicin sensitization in mice. *Pain* 2009;141:239–247. [PubMed: 19118950]
30. Brenman JE, Chao DS, Gee SH, McGee AW, Craven SE, Santillano DR, Wu Z, Huang F, Xia H, Peters MF, Froehner SC, Brecht DS. Interaction of nitric oxide synthase with the postsynaptic density protein PSD-95 and alpha1-syntrophin mediated by PDZ domains. *Cell* 1996;84:757–767. [PubMed: 8625413]
31. Brenman JE, Christopherson KS, Craven SE, McGee AW, Brecht DS. Cloning and characterization of postsynaptic density 93, a nitric oxide synthase interacting protein. *J Neurosci* 1996;16:7407–7415. [PubMed: 8922396]
32. Brenner GJ, Ji RR, Shaffer S, Woolf CJ. Peripheral noxious stimulation induces phosphorylation of the NMDA receptor NR1 subunit at the PKC-dependent site, serine-896, in spinal cord dorsal horn neurons. *Eur J Neurosci* 2004;20:375–384. [PubMed: 15233747]
33. Brown AG. Neuronal organization in the dorsal horn of the spinal cord. *Acta Morphol Hung* 1983;31:87–99. [PubMed: 6312776]
34. Buldyrev I, Tanner NM, Hsieh HY, Dodd EG, Nguyen LT, Balkowiec A. Calcitonin gene-related peptide enhances release of native brain-derived neurotrophic factor from trigeminal ganglion neurons. *J Neurochem* 2006;99:1338–1350. [PubMed: 17064360]
35. Burstein R, Jakubowski M. Analgesic triptan action in an animal model of intracranial pain: A race against the development of central sensitization. *Ann Neurol* 2004;55:27–36. [PubMed: 14705109]
36. Burstein R, Yamamura H, Malick A, Strassman AM. Chemical stimulation of the intracranial dura induces enhanced responses to facial stimulation in brain stem trigeminal neurons. *J Neurophysiol* 1998;79:964–982. [PubMed: 9463456]
37. Campbell JN, Meyer RA. Mechanisms of neuropathic pain. *Neuron* 2006;52:77–92. [PubMed: 17015228]
38. Cao L, DeLeo JA. CNS-infiltrating CD4⁺ T lymphocytes contribute to murine spinal nerve transection-induced neuropathic pain. *Eur J Immunol* 2008;38:448–458. [PubMed: 18196515]
39. Carlton SM, Hargrett GL. Colocalization of metabotropic glutamate receptors in rat dorsal root ganglion cells. *J Comp Neurol* 2007;501:780–789. [PubMed: 17299761]
40. Carvalho AL, Duarte CB, Carvalho AP. Regulation of AMPA receptors by phosphorylation. *Neurochem Res* 2000;25:1245–1255. [PubMed: 11059799]
41. Chapman V, Dickenson AH. The spinal and peripheral roles of bradykinin and prostaglandins in nociceptive processing in the rat. *Eur J Pharmacol* 1992;219:427–433. [PubMed: 1425970]
42. Chen BS, Roche KW. Regulation of NMDA receptors by phosphorylation. *Neuropharmacology* 2007;53:362–368. [PubMed: 17644144]
43. Chen L, Chetkovich DM, Petralia RS, Sweeney NT, Kawasaki Y, Wenthold RJ, Brecht DS, Nicoll RA. Stargazin regulates synaptic targeting of AMPA receptors by two distinct mechanisms. *Nature* 2000;408:936–943. [PubMed: 11140673]
44. Chen L, Huang LY. Protein kinase C reduces Mg²⁺ block of NMDA-receptor channels as a mechanism of modulation. *Nature* 1992;356:521–523. [PubMed: 1373227]
45. Chen SR, Pan HL. Distinct roles of group III metabotropic glutamate receptors in control of nociception and dorsal horn neurons in normal and nerve-injured Rats. *J Pharmacol Exp Ther* 2005;312:120–126. [PubMed: 15375175]
46. Chen X, Tanner K, Levine JD. Mechanical sensitization of cutaneous C-fiber nociceptors by prostaglandin E₂ in the rat. *Neurosci Lett* 1999;267:105–108. [PubMed: 10400223]
47. Cheng HT, Suzuki M, Hegarty DM, Xu Q, Weyerbacher AR, South SM, Ohata M, Inturrisi CE. Inflammatory pain-induced signaling events following a conditional deletion of the N-methyl-D-aspartate receptor in spinal cord dorsal horn. *Neuroscience* 2008;155:948–958. [PubMed: 18621103]
48. Cheret C, Gervais A, Lelli A, Colin C, Amar L, Ravassard P, Mallet J, Cumano A, Krause KH, Mallat M. Neurotoxic activation of microglia is promoted by a nox1-dependent NADPH oxidase. *J Neurosci* 2008;28:12039–12051. [PubMed: 19005069]

49. Cho CH, St Gelais F, Zhang W, Tomita S, Howe JR. Two families of TARP isoforms that have distinct effects on the kinetic properties of AMPA receptors and synaptic currents. *Neuron* 2007;55:890–904. [PubMed: 17880893]
50. Chu YC, Guan Y, Skinner J, Raja SN, Johns RA, Tao YX. Effect of genetic knockout or pharmacologic inhibition of neuronal nitric oxide synthase on complete Freund's adjuvant-induced persistent pain. *Pain* 2005;119:113–123. [PubMed: 16297560]
51. Chung HJ, Xia J, Scannevin RH, Zhang X, Huganir RL. Phosphorylation of the AMPA receptor subunit GluR2 differentially regulates its interaction with PDZ domain-containing proteins. *J Neurosci* 2000;20:7258–7267. [PubMed: 11007883]
- 52.Coderre TJ, Melzack R. The role of NMDA receptor-operated calcium channels in persistent nociception after formalin-induced tissue injury. *J Neurosci* 1992;12:3671–3675. [PubMed: 1326611]
53. Colledge M, Dean RA, Scott GK, Langeberg LK, Huganir RL, Scott JD. Targeting of PKA to glutamate receptors through a MAGUK-AKAP complex. *Neuron* 2000;27:107–119. [PubMed: 10939335]
54. Conn PJ, Pin JP. Pharmacology and functions of metabotropic glutamate receptors. *Annu Rev Pharmacol Toxicol* 1997;37:205–237. [PubMed: 9131252]
55. Cook AJ, Woolf CJ, Wall PD. Prolonged C-fibre mediated facilitation of the flexion reflex in the rat is not due to changes in afferent terminal or motoneurone excitability. *Neurosci Lett* 1986;70:91–96. [PubMed: 3774223]
56. Costigan M, Befort K, Karchewski L, Griffin RS, D'Urso D, Allchorne A, Sitarski J, Mannion JW, Pratt RE, Woolf CJ. Replicate high-density rat genome oligonucleotide microarrays reveal hundreds of regulated genes in the dorsal root ganglion after peripheral nerve injury. *BMC Neurosci* 2002;3:16. [PubMed: 12401135]
57. Coull JA, Beggs S, Boudreau D, Boivin D, Tsuda M, Inoue K, Gravel C, Salter MW, De Koninck Y. BDNF from microglia causes the shift in neuronal anion gradient underlying neuropathic pain. *Nature* 2005;438:1017–1021. [PubMed: 16355225]
58. Coull JA, Boudreau D, Bachand K, Prescott SA, Nault F, Sik A, De Koninck P, De Koninck Y. Trans-synaptic shift in anion gradient in spinal lamina I neurons as a mechanism of neuropathic pain. *Nature* 2003;424:938–942. [PubMed: 12931188]
59. Cuthbert PC, Stanford LE, Coba MP, Ainge JA, Fink AE, Opazo P, Delgado JY, Komiyama NH, O'Dell TJ, Grant SG. Synapse-associated protein 102/dlg3 couples the NMDA receptor to specific plasticity pathways and learning strategies. *J Neurosci* 2007;27:2673–2682. [PubMed: 17344405]
60. D'Mello R, Dickenson AH. Spinal cord mechanisms of pain. *Br J Anaesth* 2008;101:8–16. [PubMed: 18417503]
61. Dalva MB, Takasu MA, Lin MZ, Shamah SM, Hu L, Gale NW, Greenberg ME. EphB receptors interact with NMDA receptors and regulate excitatory synapse formation. *Cell* 2000;103:945–956. [PubMed: 11136979]
62. Davalos D, Grutzendler J, Yang G, Kim JV, Zuo Y, Jung S, Littman DR, Dustin ML, Gan WB. ATP mediates rapid microglial response to local brain injury in vivo. *Nat Neurosci* 2005;8:752–758. [PubMed: 15895084]
63. Davies SN, Lodge D. Evidence for involvement of N-methylaspartate receptors in 'wind-up' of class 2 neurones in the dorsal horn of the rat. *Brain Res* 1987;424:402–406. [PubMed: 2823998]
64. Davis KL, Martin E, Turko IV, Murad F. Novel effects of nitric oxide. *Annu Rev Pharmacol Toxicol* 2001;41:203–236. [PubMed: 11264456]
65. de Novellis V, Siniscalco D, Galderisi U, Fuccio C, Nolano M, Santoro L, Cascino A, Roth KA, Rossi F, Maione S. Blockade of glutamate mGlu5 receptors in a rat model of neuropathic pain prevents early over-expression of pro-apoptotic genes and morphological changes in dorsal horn lamina II. *Neuropharmacology* 2004;46:468–479. [PubMed: 14975670]
66. DeLeo JA, Yezierski RP. The role of neuroinflammation and neuroimmune activation in persistent pain. *Pain* 2001;90:1–6. [PubMed: 11166964]
67. Derjean D, Bertrand S, Le Masson G, Landry M, Morisset V, Nagy F. Dynamic balance of metabotropic inputs causes dorsal horn neurons to switch functional states. *Nat Neurosci* 2003;6:274–281. [PubMed: 12592405]

68. Desmeules JA, Cedraschi C, Rapiti E, Baumgartner E, Finckh A, Cohen P, Dayer P, Vischer TL. Neurophysiologic evidence for a central sensitization in patients with fibromyalgia. *Arthritis Rheum* 2003;48:1420–1429. [PubMed: 12746916]
69. Devor M. Ectopic discharge in Abeta afferents as a source of neuropathic pain. *Exp Brain Res* 2009;196:115–128. [PubMed: 19242687]
70. Devor M.; Seltzer, Z. Pathophysiology of Damaged Nerves in Relation to Chronic Pain. Edinburgh, UK: Churchill Livingstone; 1999.
71. Devor M, Wall PD. Reorganisation of spinal cord sensory map after peripheral nerve injury. *Nature* 1978;276:75–76. [PubMed: 570248]
72. Dickenson AH, Sullivan AF. Evidence for a role of the NMDA receptor in the frequency dependent potentiation of deep rat dorsal horn nociceptive neurones following C fibre stimulation. *Neuropharmacology* 1987;26:1235–1238. [PubMed: 2821443]
73. Ding JD, Weinberg RJ. Localization of soluble guanylyl cyclase in the superficial dorsal horn. *J Comp Neurol* 2006;495:668–678. [PubMed: 16506200]
74. Djouhri L, Koutsikou S, Fang X, McMullan S, Lawson SN. Spontaneous pain, both neuropathic and inflammatory, is related to frequency of spontaneous firing in intact C-fiber nociceptors. *J Neurosci* 2006;26:1281–1292. [PubMed: 16436616]
75. Dogrul A, Ossipov MH, Lai J, Malan TP Jr, Porreca F. Peripheral and spinal antihyperalgesic activity of SIB-1757, a metabotropic glutamate receptor (mGluR[5]) antagonist, in experimental neuropathic pain in rats. *Neurosci Lett* 2000;292:115–118. [PubMed: 10998562]
76. Dominguez E, Rivat C, Pommier B, Mauborgne A, Pohl M. JAK/STAT3 pathway is activated in spinal cord microglia after peripheral nerve injury and contributes to neuropathic pain development in rat. *J Neurochem* 2008;107:50–60. [PubMed: 18636982]
77. Dong H, O'Brien RJ, Fung ET, Lanahan AA, Worley PF, Huganir RL. GRIP: A synaptic PDZ domain-containing protein that interacts with AMPA receptors. *Nature* 1997;386:279–284. [PubMed: 9069286]
78. Dostrovsky JO, Guilbaud G. Nociceptive responses in medial thalamus of the normal and arthritic rat. *Pain* 1990;40:93–104. [PubMed: 2339022]
79. Dougherty PM, Willis WD. Enhanced responses of spinothalamic tract neurons to excitatory amino acids accompany capsaicin-induced sensitization in the monkey. *J Neurosci* 1992;12:883–894. [PubMed: 1545244]
80. Dougherty PM, Willis WD. Enhancement of spinothalamic neuron responses to chemical and mechanical stimuli following combined micro-iontophoretic application of N-methyl-D-aspartic acid and substance P. *Pain* 1991;47:85–93. [PubMed: 1722895]
81. Duan Y, Sahley CL, Muller KJ. ATP and NO dually control migration of microglia to nerve lesions. *Dev Neurobiol* 2009;69:60–72. [PubMed: 19025930]
82. Edvinsson L. CGRP-receptor antagonism in migraine treatment. *Lancet* 2008;372:2089–2090. [PubMed: 19036426]
83. Ehlers MD, Heine M, Groc L, Lee MC, Choquet D. Diffusional trapping of GluR1 AMPA receptors by input-specific synaptic activity. *Neuron* 2007;54:447–460. [PubMed: 17481397]
84. Esteban JA, Shi SH, Wilson C, Nuriya M, Huganir RL, Malinow R. PKA phosphorylation of AMPA receptor subunits controls synaptic trafficking underlying plasticity. *Nat Neurosci* 2003;6:136–143. [PubMed: 12536214]
85. Fagni L, Chavis P, Ango F, Bockaert J. Complex interactions between mGluRs, intracellular Ca²⁺ stores and ion channels in neurons. *Trends Neurosci* 2000;23:80–88. [PubMed: 10652549]
86. Fang L, Wu J, Lin Q, Willis WD. Calcium-calmodulin-dependent protein kinase II contributes to spinal cord central sensitization. *J Neurosci* 2002;22:4196–4204. [PubMed: 12019337]
87. Fang L, Wu J, Lin Q, Willis WD. Protein kinases regulate the phosphorylation of the GluR1 subunit of AMPA receptors of spinal cord in rats following noxious stimulation. *Brain Res Mol Brain Res* 2003;118:160–165. [PubMed: 14559367]
88. Fang L, Wu J, Zhang X, Lin Q, Willis WD. Increased phosphorylation of the GluR1 subunit of spinal cord alpha-amino-3-hydroxy-5-methyl-4-isoxazole propionate receptor in rats following intradermal injection of capsaicin. *Neuro-science* 2003;122:237–245.

89. Ferguson AR, Bolding KA, Huie JR, Hook MA, Santillano DR, Miranda RC, Grau JW. Group I metabotropic glutamate receptors control metaplasticity of spinal cord learning through a protein kinase C-dependent mechanism. *J Neurosci* 2008;28:11939–11949. [PubMed: 19005059]
90. Fox MA, Umemori H. Seeking long-term relationship: Axon and target communicate to organize synaptic differentiation. *J Neurochem* 2006;97:1215–1231. [PubMed: 16638017]
91. Fukuoka T, Kondo E, Dai Y, Hashimoto N, Noguchi K. Brain-derived neurotrophic factor increases in the uninjured dorsal root ganglion neurons in selective spinal nerve ligation model. *J Neurosci* 2001;21:4891–4900. [PubMed: 11425916]
92. Fundytus ME, Osborne MG, Henry JL, Coderre TJ, Dray A. Antisense oligonucleotide knockdown of mGluR1 alleviates hyperalgesia and allodynia associated with chronic inflammation. *Pharmacol Biochem Behav* 2002;73:401–410. [PubMed: 12117595]
93. Galan A, Laird JM, Cervero F. In vivo recruitment by painful stimuli of AMPA receptor subunits to the plasma membrane of spinal cord neurons. *Pain* 2004;112:315–323. [PubMed: 15561387]
94. Gao YJ, Zhang L, Samad OA, Suter MR, Yasuhiko K, Xu ZZ, Park JY, Lind AL, Ma Q, Ji RR. JNK-induced MCP-1 production in spinal cord astrocytes contributes to central sensitization and neuropathic pain. *J Neurosci* 2009;29:4096–4108. [PubMed: 19339605]
95. Gardell LR, Vanderah TW, Gardell SE, Wang R, Ossipov MH, Lai J, Porreca F. Enhanced evoked excitatory transmitter release in experimental neuropathy requires descending facilitation. *J Neurosci* 2003;23:8370–8379. [PubMed: 12967999]
96. Gardner SM, Takamiya K, Xia J, Suh JG, Johnson R, Yu S, Huganir RL. Calcium-permeable AMPA receptor plasticity is mediated by subunit-specific interactions with PICK1 and NSF. *Neuron* 2005;45:903–915. [PubMed: 15797551]
97. Garraway SM, Xu Q, Inturrisi CE. siRNA-mediated knockdown of the NR1 subunit gene of the NMDA receptor attenuates formalin-induced pain behaviors in adult rats. *J Pain* 2009;10:380–390. [PubMed: 19185544]
98. Garrison CJ, Dougherty PM, Carlton SM. GFAP expression in lumbar spinal cord of naive and neuropathic rats treated with MK-801. *Exp Neurol* 1994;129:237–243. [PubMed: 7957738]
99. Garry EM, Moss A, Delaney A, O'Neill F, Blakemore J, Bowen J, Husi H, Mitchell R, Grant SG, Fleetwood-Walker SM. Neuropathic sensitization of behavioral reflexes and spinal NMDA receptor/CaM kinase II interactions are disrupted in PSD-95 mutant mice. *Curr Biol* 2003;13:321–328. [PubMed: 12593798]
100. Garry EM, Moss A, Rosie R, Delaney A, Mitchell R, Fleetwood-Walker SM. Specific involvement in neuropathic pain of AMPA receptors and adapter proteins for the GluR2 subunit. *Mol Cell Neurosci* 2003;24:10–22. [PubMed: 14550765]
101. Gauriau C, Bernard JF. Pain pathways and parabrachial circuits in the rat. *Exp Physiol* 2002;87:251–258. [PubMed: 11856971]
102. Giles PA, Trezise DJ, King AE. Differential activation of protein kinases in the dorsal horn in vitro of normal and inflamed rats by group I metabotropic glutamate receptor subtypes. *Neuropharmacology* 2007;53:58–70. [PubMed: 17543352]
103. Goadsby PJ. Recent advances in understanding migraine mechanisms, molecules and therapeutics. *Trends Mol Med* 2007;13:39–44. [PubMed: 17141570]
104. Goudet C, Chapuy E, Alloui A, Acher F, Pin JP, Eschalier A. Group III metabotropic glutamate receptors inhibit hyperalgesia in animal models of inflammation and neuropathic pain. *Pain* 2008;137:112–124. [PubMed: 17900808]
105. Guenther S, Reeh PW, Kress M. Rises in $[Ca^{2+}]_i$ mediate capsaicin- and proton-induced heat sensitization of rat primary nociceptive neurons. *Eur J Neurosci* 1999;11:3143–3150. [PubMed: 10510178]
106. Guo W, Wei F, Zou S, Robbins MT, Sugiyo S, Ikeda T, Tu JC, Worley PF, Dubner R, Ren K. Group I metabotropic glutamate receptor NMDA receptor coupling and signaling cascade mediate spinal dorsal horn NMDA receptor 2B tyrosine phosphorylation associated with inflammatory hyperalgesia. *J Neurosci* 2004;24:9161–9173. [PubMed: 15483135]
107. Guo W, Zou S, Guan Y, Ikeda T, Tal M, Dubner R, Ren K. Tyrosine phosphorylation of the NR2B subunit of the NMDA receptor in the spinal cord during the development and maintenance of inflammatory hyperalgesia. *J Neurosci* 2002;22:6208–6217. [PubMed: 12122079]

108. Hanley JG, Henley JM. PICK1 is a calcium-sensor for NMDA-induced AMPA receptor trafficking. *EMBO J* 2005;24:3266–3278. [PubMed: 16138078]
109. Hanley JG, Khatri L, Hanson PI, Ziff EB. NSF ATPase and alpha-/beta-SNAPs disassemble the AMPA receptor-PICK1 complex. *Neuron* 2002;34:53–67. [PubMed: 11931741]
110. Harris JA, Corsi M, Quartaroli M, Arban R, Bentivoglio M. Upregulation of spinal glutamate receptors in chronic pain. *Neuroscience* 1996;74:7–12. [PubMed: 8843072]
111. Harvey RJ, Depner UB, Wassle H, Ahmadi S, Heindl C, Reinold H, Smart TG, Harvey K, Schutz B, Abo-Salem OM, Zimmer A, Poisbeau P, Welzl H, Wolfer DP, Betz H, Zeilhofer HU, Muller U. GlyR alpha3: An essential target for spinal PGE2-mediated inflammatory pain sensitization. *Science* 2004;304:884–887. [PubMed: 15131310]
112. Henry JL. Effects of substance P on functionally identified units in cat spinal cord. *Brain Res* 1976;114:439–451. [PubMed: 953765]
113. Heppenstall PA, Lewin GR. BDNF but not NT-4 is required for normal flexion reflex plasticity and function. *Proc Natl Acad Sci U S A* 2001;98:8107–8112. [PubMed: 11438749]
114. Hirai H. Modification of AMPA receptor clustering regulates cerebellar synaptic plasticity. *Neurosci Res* 2001;39:261–267. [PubMed: 11248365]
115. Honore P, Rogers SD, Schwei MJ, Salak-Johnson JL, Luger NM, Sabino MC, Clohisey DR, Mantyh PW. Murine models of inflammatory, neuropathic and cancer pain each generates a unique set of neurochemical changes in the spinal cord and sensory neurons. *Neuroscience* 2000;98:585–598. [PubMed: 10869852]
116. Horvath RJ, Natile-McMenemy N, Alkatis MS, Deleo JA. Differential migration, LPS-induced cytokine, chemokine, and NO expression in immortalized BV-2 and HAPI cell lines and primary microglial cultures. *J Neurochem* 2008;107:557–569. [PubMed: 18717813]
117. Hosl K, Reinold H, Harvey RJ, Muller U, Narumiya S, Zeilhofer HU. Spinal prostaglandin E receptors of the EP2 subtype and the glycine receptor alpha3 subunit, which mediate central inflammatory hyperalgesia, do not contribute to pain after peripheral nerve injury or formalin injection. *Pain* 2006;126:46–53. [PubMed: 16846696]
118. Hu HJ, Alter BJ, Carrasquillo Y, Qiu CS, Gereau RW. Metabotropic glutamate receptor 5 modulates nociceptive plasticity via extracellular signal-regulated kinase-Kv4.2 signaling in spinal cord dorsal horn neurons. *J Neurosci* 2007;27:13181–13191. [PubMed: 18045912]
119. Hu HJ, Carrasquillo Y, Karim F, Jung WE, Nerbonne JM, Schwarz TL, Gereau RW. The kv4.2 potassium channel subunit is required for pain plasticity. *Neuron* 2006;50:89–100. [PubMed: 16600858]
120. Hu HJ, Gereau RW. ERK integrates PKA and PKC signaling in superficial dorsal horn neurons, II: Modulation of neuronal excitability. *J Neurophysiol* 2003;90:1680–1688. [PubMed: 12750418]
121. Huang Y, Erdmann N, Peng H, Zhao Y, Zheng J. The role of TNF related apoptosis-inducing ligand in neurodegenerative diseases. *Cell Mol Immunol* 2005;2:113–122. [PubMed: 16191417]
122. Huang Y, Man HY, Sekine-Aizawa Y, Han Y, Juluri K, Luo H, Cheah J, Lowenstein C, Haganir RL, Snyder SH. S-nitrosylation of N-ethylmaleimide sensitive factor mediates surface expression of AMPA receptors. *Neuron* 2005;46:533–540. [PubMed: 15944123]
123. Huang YY, Pittenger C, Kandel ER. A form of long-lasting, learning-related synaptic plasticity in the hippocampus induced by heterosynaptic low-frequency pairing. *Proc Natl Acad Sci U S A* 2004;101:859–864. [PubMed: 14711997]
124. Hucho T, Levine JD. Signaling pathways in sensitization: Toward a nociceptor cell biology. *Neuron* 2007;55:365–376. [PubMed: 17678851]
125. Hughes DI, Scott DT, Todd AJ, Riddell JS. Lack of evidence for sprouting of Abeta afferents into the superficial laminae of the spinal cord dorsal horn after nerve section. *J Neurosci* 2003;23:9491–9499. [PubMed: 14573528]
126. Hunt SP, Pini A, Evan G. Induction of c-fos-like protein in spinal cord neurons following sensory stimulation. *Nature* 1987;328:632–634. [PubMed: 3112583]
127. Ikeda H, Heinke B, Ruscheweyh R, Sandkuhler J. Synaptic plasticity in spinal lamina I projection neurons that mediate hyperalgesia. *Science* 2003;299:1237–1240. [PubMed: 12595694]
128. Ikeda H, Stark J, Fischer H, Wagner M, Drdla R, Jager T, Sandkuhler J. Synaptic amplifier of inflammatory pain in the spinal dorsal horn. *Science* 2006;312:1659–1662. [PubMed: 16778058]

129. Inoue Y, Udo H, Inokuchi K, Sugiyama H. Homer1a regulates the activity-induced remodeling of synaptic structures in cultured hippocampal neurons. *Neuroscience* 2007;150:841–852. [PubMed: 18006237]
130. Ji RR, Baba H, Brenner GJ, Woolf CJ. Nociceptive-specific activation of ERK in spinal neurons contributes to pain hypersensitivity. *Nat Neurosci* 1999;2:1114–1119. [PubMed: 10570489]
131. Ji RR, Gereau RWt, Malcangio M, Strichartz GR. MAP kinase and pain. *Brain Res Rev* 2009;60:135–148. [PubMed: 19150373]
132. Ji RR, Kohno T, Moore KA, Woolf CJ. Central sensitization and LTP: Do pain and memory share similar mechanisms? *Trends Neurosci* 2003;26:696–705. [PubMed: 14624855]
133. Ji RR, Suter MR. p38 MAPK, microglial signaling, and neuropathic pain. *Mol Pain* 2007;3:33. [PubMed: 17974036]
134. Jones TL, Sorkin LS. Activated PKA and PKC, but not CaMKIIalpha, are required for AMPA/Kainate-mediated pain behavior in the thermal stimulus model. *Pain* 2005;117:259–270. [PubMed: 16150547]
135. Jordt SE, Bautista DM, Chuang HH, McKemy DD, Zygmunt PM, Hogestatt ED, Meng ID, Julius D. Mustard oils and cannabinoids excite sensory nerve fibres through the TRP channel ANKTM1. *Nature* 2004;427:260–265. [PubMed: 14712238]
136. Kapadia SE, LaMotte CC. Deafferentation-induced alterations in the rat dorsal horn, I: Comparison of peripheral nerve injury vs. rhizotomy effects on presynaptic, postsynaptic, and glial processes *J Comp Neurol* 1987;266:183–197.
137. Karim F, Hu HJ, Adwanikar H, Kaplan D, Gereau RW. Impaired inflammatory pain and thermal hyperalgesia in mice expressing neuron-specific dominant negative mitogen activated protein kinase kinase (MEK). *Mol Pain* 2006;2:2. [PubMed: 16412244]
138. Karim F, Wang CC, Gereau RW. Metabotropic glutamate receptor subtypes 1 and 5 are activators of extracellular signal-regulated kinase signaling required for inflammatory pain in mice. *J Neurosci* 2001;21:3771–3779. [PubMed: 11356865]
139. Katano T, Furue H, Okuda-Ashitaka E, Tagaya M, Watanabe M, Yoshimura M, Ito S. N-ethylmaleimide-sensitive fusion protein (NSF) is involved in central sensitization in the spinal cord through GluR2 subunit composition switch after inflammation. *Eur J Neurosci* 2008;27:3161–3170. [PubMed: 18598260]
140. Kato AS, Zhou W, Milstein AD, Knierman MD, Siuda ER, Dotzlaw JE, Yu H, Hale JE, Nisenbaum ES, Nicoll RA, Breder DS. New transmembrane AMPA receptor regulatory protein isoform, gamma-7, differentially regulates AMPA receptors. *J Neurosci* 2007;27:4969–4977. [PubMed: 17475805]
141. Kawasaki Y, Kohno T, Zhuang ZY, Brenner GJ, Wang H, Van Der Meer C, Befort K, Woolf CJ, Ji RR. Ionotropic and metabotropic receptors, protein kinase A, protein kinase C, and Src contribute to C-fiber-induced ERK activation and cAMP response element-binding protein phosphorylation in dorsal horn neurons, leading to central sensitization. *J Neurosci* 2004;24:8310–8321. [PubMed: 15385614]
142. Kawasaki Y, Zhang L, Cheng JK, Ji RR. Cytokine mechanisms of central sensitization: Distinct and overlapping role of interleukin-1beta, interleukin-6, and tumor necrosis factor-alpha in regulating synaptic and neuronal activity in the superficial spinal cord. *J Neurosci* 2008;28:5189–5194. [PubMed: 18480275]
143. Kayser V, Elfassi IE, Aubel B, Melfort M, Julius D, Gingrich JA, Hamon M, Bourgoin S. Mechanical, thermal and formalin-induced nociception is differentially altered in 5-HT1A-/-, 5-HT1B-/-, 5-HT2A-/-, 5-HT3A-/- and 5-HTT-/-knock-out male mice. *Pain* 2007;130:235–248. [PubMed: 17250964]
144. Kerr BJ, Bradbury EJ, Bennett DL, Trivedi PM, Dassan P, French J, Shelton DB, McMahon SB, Thompson SW. Brain-derived neurotrophic factor modulates nociceptive sensory inputs and NMDA-evoked responses in the rat spinal cord. *J Neurosci* 1999;19:5138–5148. [PubMed: 10366647]
145. Kerr RC, Maxwell DJ, Todd AJ. GluR1 and GluR2/3 subunits of the AMPA-type glutamate receptor are associated with particular types of neurone in laminae I-III of the spinal dorsal horn of the rat. *Eur J Neurosci* 1998;10:324–333. [PubMed: 9753141]

146. Khasabov SG, Rogers SD, Ghilardi JR, Peters CM, Mantyh PW, Simone DA. Spinal neurons that possess the substance P receptor are required for the development of central sensitization. *J Neurosci* 2002;22:9086–9098. [PubMed: 12388616]
147. Kim E, Naisbitt S, Hsueh YP, Rao A, Rothschild A, Craig AM, Sheng M. GKAP, a novel synaptic protein that interacts with the guanylate kinase-like domain of the PSD-95/SAP90 family of channel clustering molecules. *J Cell Biol* 1997;136:669–678. [PubMed: 9024696]
148. Kim E, Sheng M. PDZ domain proteins of synapses. *Nat Rev Neurosci* 2004;5:771–781. [PubMed: 15378037]
149. Klein T, Stahn S, Magerl W, Treede RD. The role of heterosynaptic facilitation in long-term potentiation (LTP) of human pain sensation. *Pain*. 2008 July 3;Epub ahead of print
150. Kobayashi H, Kitamura T, Sekiguchi M, Homma MK, Kabuyama Y, Konno S, Kikuchi S, Homma Y. Involvement of EphB1 receptor/EphrinB2 ligand in neuropathic pain. *Spine* 2007;32:1592–1598. [PubMed: 17621205]
151. Kohno T, Moore KA, Baba H, Woolf CJ. Peripheral nerve injury alters excitatory synaptic transmission in lamina II of the rat dorsal horn. *J Physiol* 2003;548:131–138. [PubMed: 12576493]
152. Kohno T, Wang H, Amaya F, Brenner GJ, Cheng JK, Ji RR, Woolf CJ. Bradykinin enhances AMPA and NMDA receptor activity in spinal cord dorsal horn neurons by activating multiple kinases to produce pain hypersensitivity. *J Neurosci* 2008;28:4533–4540. [PubMed: 18434532]
153. Koltzenburg M, Lundberg LE, Torebjork HE. Dynamic and static components of mechanical hyperalgesia in human hairy skin. *Pain* 1992;51:207–219. [PubMed: 1484717]
154. Koltzenburg M, Wahren LK, Torebjork HE. Dynamic changes of mechanical hyperalgesia in neuropathic pain states and healthy subjects depend on the ongoing activity of unmyelinated nociceptive afferents. *Pflugers Arch* 1992;420:R452.
155. Kornau HC, Schenker LT, Kennedy MB, Seeburg PH. Domain interaction between NMDA receptor subunits and the postsynaptic density protein PSD-95. *Science* 1995;269:1737–1740. [PubMed: 7569905]
156. Kullander K, Klein R. Mechanisms and functions of Eph and ephrin signalling. *Nat Rev Mol Cell Biol* 2002;3:475–486. [PubMed: 12094214]
157. LaMotte RH, Lundberg LE, Torebjork HE. Pain, hyperalgesia and activity in nociceptive C units in humans after intradermal injection of capsaicin. *J Physiol* 1992;448:749–764. [PubMed: 1593488]
158. LaMotte RH, Shain CN, Simone DA, Tsai EF. Neurogenic hyperalgesia: psychophysical studies of underlying mechanisms. *J Neurophysiol* 1991;66:190–211. [PubMed: 1919666]
159. Larsson M, Broman J. Translocation of GluR1-containing AMPA receptors to a spinal nociceptive synapse during acute noxious stimulation. *J Neurosci* 2008;28:7084–7090. [PubMed: 18614677]
160. Latremoliere A, Mauborgne A, Masson J, Bourgoin S, Kayser V, Hamon M, Pohl M. Differential implication of proinflammatory cytokine interleukin-6 in the development of cephalic versus extracephalic neuropathic pain in rats. *J Neurosci* 2008;28:8489–8501. [PubMed: 18716207]
161. Lau CG, Zukin RS. NMDA receptor trafficking in synaptic plasticity and neuropsychiatric disorders. *Nat Rev Neurosci* 2007;8:413–426. [PubMed: 17514195]
162. Ledeboer A, Sloane EM, Milligan ED, Frank MG, Mahony JH, Maier SF, Watkins LR. Minocycline attenuates mechanical allodynia and proinflammatory cytokine expression in rat models of pain facilitation. *Pain* 2005;115:71–83. [PubMed: 15836971]
163. Lee SE, Kim JH. Involvement of substance P and calcitonin gene-related peptide in development and maintenance of neuropathic pain from spinal nerve injury model of rat. *Neurosci Res* 2007;58:245–249. [PubMed: 17428562]
164. Lee SJ, Escobedo-Lozoya Y, Szatmari EM, Yasuda R. Activation of CaMKII in single dendritic spines during long-term potentiation. *Nature* 2009;458:299–304. [PubMed: 19295602]
165. Lefebvre C, Fisher K, Cahill CM, Coderre TJ. Evidence that DHPG-induced nociception depends on glutamate release from primary afferent C-fibres. *Neuroreport* 2000;11:1631–1635. [PubMed: 10852214]
166. Lekan HA, Carlton SM, Coggeshall RE. Sprouting of A beta fibers into lamina II of the rat dorsal horn in peripheral neuropathy. *Neurosci Lett* 1996;208:147–150. [PubMed: 8733291]

167. Leonard AS, Davare MA, Horne MC, Garner CC, Hell JW. SAP97 is associated with the alpha-amino-3-hydroxy-5-methylisoxazole-4-propionic acid receptor GluR1 subunit. *J Biol Chem* 1998;273:19518–19524. [PubMed: 9677374]
168. Leonard AS, Hell JW. Cyclic AMP-dependent protein kinase and protein kinase C phosphorylate N-methyl-D-aspartate receptors at different sites. *J Biol Chem* 1997;272:12107–12115. [PubMed: 9115280]
169. Lester LB, Faux MC, Nauert JB, Scott JD. Targeted protein kinase A and PP-2B regulate insulin secretion through reversible phosphorylation. *Endocrinology* 2001;142:1218–1227. [PubMed: 11181538]
170. Letts VA, Felix R, Biddlecome GH, Arikath J, Mahaffey CL, Valenzuela A, Bartlett FS 2nd, Mori Y, Campbell KP, Frankel WN. The mouse stargazer gene encodes a neuronal Ca²⁺-channel gamma subunit. *Nat Genet* 1998;19:340–347. [PubMed: 9697694]
171. Lever IJ, Pezet S, McMahon SB, Malcangio M. The signaling components of sensory fiber transmission involved in the activation of ERK MAP kinase in the mouse dorsal horn. *Mol Cell Neurosci* 2003;24:259–270. [PubMed: 14572451]
172. Lin DT, Haganir RL. PICK1 and phosphorylation of the glutamate receptor 2 (GluR2) AMPA receptor subunit regulates GluR2 recycling after NMDA receptor-induced internalization. *J Neurosci* 2007;27:13903–13908. [PubMed: 18077702]
173. Lin Q, Palecek J, Paleckova V, Peng YB, Wu J, Cui M, Willis WD. Nitric oxide mediates the central sensitization of primate spinothalamic tract neurons. *J Neurophysiol* 1999;81:1075–1085. [PubMed: 10085334]
174. Lin Q, Peng YB, Willis WD. Inhibition of primate spinothalamic tract neurons by spinal glycine and GABA is reduced during central sensitization. *J Neurophysiol* 1996;76:1005–1014. [PubMed: 8871215]
175. Lin Q, Peng YB, Willis WD. Possible role of protein kinase C in the sensitization of primate spinothalamic tract neurons. *J Neurosci* 1996;16:3026–3034. [PubMed: 8622132]
176. Lin Q, Peng YB, Wu J, Willis WD. Involvement of cGMP in nociceptive processing by and sensitization of spinothalamic neurons in primates. *J Neurosci* 1997;17:3293–3302. [PubMed: 9096162]
177. Liu XG, Sandkuhler J. Long-term potentiation of C-fiber-evoked potentials in the rat spinal dorsal horn is prevented by spinal N-methyl-D-aspartic acid receptor blockage. *Neurosci Lett* 1995;191:43–46. [PubMed: 7659287]
178. Liu XJ, Gingrich JR, Vargas-Caballero M, Dong YN, Sengar A, Beggs S, Wang SH, Ding HK, Frankland PW, Salter MW. Treatment of inflammatory and neuropathic pain by uncoupling Src from the NMDA receptor complex. *Nat Med* 2008;14:1325–1332. [PubMed: 19011637]
179. Lu VB, Biggs JE, Stebbing MJ, Balasubramanian S, Todd KG, Lai AY, Colmers WF, Dawbarn D, Ballanyi K, Smith PA. Brain-derived neurotrophic factor drives the changes in excitatory synaptic transmission in the rat superficial dorsal horn that follow sciatic nerve injury. *J Physiol* 2009;587(Pt 5):1013–1032. [PubMed: 19124536]
180. Lu Y, Sun YN, Wu X, Sun Q, Liu FY, Xing GG, Wan Y. Role of alpha-amino-3-hydroxy-5-methyl-4-isoxazolepropionate (AMPA) receptor subunit GluR1 in spinal dorsal horn in inflammatory nociception and neuropathic nociception in rat. *Brain Res* 2008;1200:19–26. [PubMed: 18289517]
181. Lu Y, Zhang M, Lim IA, Hall DD, Allen M, Medvedeva Y, McKnight GS, Usachev YM, Hell JW. AKAP150-anchored PKA activity is important for LTD during its induction phase. *J Physiol* 2008;586:4155–4164. [PubMed: 18617570]
182. Luo C, Seeburg PH, Sprengel R, Kuner R. Activity-dependent potentiation of calcium signals in spinal sensory networks in inflammatory pain states. *Pain* 2008;140:358–367. [PubMed: 18926636]
183. Ma QP, Woolf CJ. Involvement of neurokinin receptors in the induction but not the maintenance of mechanical allodynia in rat flexor motoneurons. *J Physiol* 1995;486(Pt 3):769–777. [PubMed: 7473237]
184. Ma QP, Woolf CJ. Noxious stimuli induce an N-methyl-D-aspartate receptor-dependent hypersensitivity of the flexion withdrawal reflex to touch: Implications for the treatment of mechanical allodynia. *Pain* 1995;61:383–390. [PubMed: 7478681]

185. Ma QP, Woolf CJ. Progressive tactile hypersensitivity: An inflammation-induced incremental increase in the excitability of the spinal cord. *Pain* 1996;67:97–106. [PubMed: 8895236]
186. Maeda T, Hamabe W, Gao Y, Fukazawa Y, Kumamoto K, Ozaki M, Kishioka S. Morphine has an antinociceptive effect through activation of the okadaic-acid-sensitive Ser/Thr protein phosphatases PP 2 A and PP5 estimated by tail-pinch test in mice. *Brain Res* 2005;1056:191–199. [PubMed: 16102737]
187. Maihofner C, Jesberger F, Seifert F, Kaltenhauser M. Cortical processing of mechanical hyperalgesia: A MEG study. *Eur J Pain*. 2009 April 3;Epub ahead of print
188. Malenka RC, Bear MF. LTP and LTD: An embarrassment of riches. *Neuron* 2004;44:5–21. [PubMed: 15450156]
189. Man HY, Sekine-Aizawa Y, Huganir RL. Regulation of {alpha}-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor trafficking through PKA phosphorylation of the Glu receptor 1 subunit. *Proc Natl Acad Sci U S A* 2007;104:3579–3584. [PubMed: 17360685]
190. Mannion RJ, Costigan M, Decosterd I, Amaya F, Ma QP, Holstege JC, Ji RR, Acheson A, Lindsay RM, Wilkinson GA, Woolf CJ. Neurotrophins: Peripherally and centrally acting modulators of tactile stimulus-induced inflammatory pain hypersensitivity. *Proc Natl Acad Sci U S A* 1999;96:9385–9390. [PubMed: 10430952]
191. Mannion RJ, Doubell TP, Coggeshall RE, Woolf CJ. Collateral sprouting of uninjured primary afferent A-fibers into the superficial dorsal horn of the adult rat spinal cord after topical capsaicin treatment to the sciatic nerve. *J Neurosci* 1996;16:5189–5195. [PubMed: 8756447]
192. Mantyh PW, Rogers SD, Honore P, Allen BJ, Ghilardi JR, Li J, Daughters RS, Lappi DA, Wiley RG, Simone DA. Inhibition of hyperalgesia by ablation of lamina I spinal neurons expressing the substance P receptor. *Science* 1997;278:275–279. [PubMed: 9323204]
193. Mao J, Price DD, Hayes RL, Lu J, Mayer DJ, Frenk H. Intrathecal treatment with dextrophan or ketamine potentially reduces pain-related behaviors in a rat model of peripheral mononeuropathy. *Brain Res* 1993;605:164–168. [PubMed: 8385540]
194. Marabese I, de Novellis V, Palazzo E, Scafuro MA, Vita D, Rossi F, Maione S. Effects of (S)-3,4-DCPG, an mGlu8 receptor agonist, on inflammatory and neuropathic pain in mice. *Neuropharmacology* 2007;52:253–262. [PubMed: 17113112]
195. Maratou K, Wallace VC, Hasnie FS, Okuse K, Hosseini R, Jina N, Blackbeard J, Pheby T, Orengo C, Dickenson AH, McMahon SB, Rice AS. Comparison of dorsal root ganglion gene expression in rat models of traumatic and HIV-associated neuropathic pain. *Eur J Pain* 2009;13:387–398. [PubMed: 18606552]
196. Matsuda S, Mikawa S, Hirai H. Phosphorylation of serine-880 in GluR2 by protein kinase C prevents its C terminus from binding with glutamate receptor-interacting protein. *J Neurochem* 1999;73:1765–1768. [PubMed: 10501226]
197. Mauceri D, Gardoni F, Marcello E, Di Luca M. Dual role of CaMKII-dependent SAP97 phosphorylation in mediating trafficking and insertion of NMDA receptor subunit NR2A. *J Neurochem* 2007;100:1032–1046. [PubMed: 17156128]
198. Mayer ML, Westbrook GL, Guthrie PB. Voltage-dependent block by Mg²⁺ of NMDA responses in spinal cord neurones. *Nature* 1984;309:261–263. [PubMed: 6325946]
199. McNamara CR, Mandel-Brehm J, Bautista DM, Siemens J, Deranian KL, Zhao M, Hayward NJ, Chong JA, Julius D, Moran MM, Fanger CM. TRPA1 mediates formalin-induced pain. *Proc Natl Acad Sci U S A* 2007;104:13525–13530. [PubMed: 17686976]
200. Mendell LM, Wall PD. Responses of single dorsal cord cells to peripheral cutaneous unmyelinated fibres. *Nature* 1965;206:97–99. [PubMed: 14334366]
201. Meunier A, Latremoliere A, Dominguez E, Mauborgne A, Philippe S, Hamon M, Mallet J, Benoliel JJ, Pohl M. Lentiviral-mediated targeted NF-kappaB blockade in dorsal spinal cord glia attenuates sciatic nerve injury-induced neuropathic pain in the rat. *Mol Ther* 2007;15:687–697. [PubMed: 17299402]
202. Mi R, Sia GM, Rosen K, Tang X, Moghekar A, Black JL, McEnery M, Huganir RL, O'Brien RJ. AMPA receptor-dependent clustering of synaptic NMDA receptors is mediated by Stargazin and NR2A/B in spinal neurons and hippocampal interneurons. *Neuron* 2004;44:335–349. [PubMed: 15473971]

203. Miletic G, Miletic V. Loose ligation of the sciatic nerve is associated with TrkB receptor-dependent decreases in KCC2 protein levels in the ipsilateral spinal dorsal horn. *Pain* 2008;137:532–539. [PubMed: 18063479]
204. Milligan ED, Sloane EM, Watkins LR. Glia in pathological pain: A role for fractalkine. *J Neuroimmunol* 2008;198:113–120. [PubMed: 18547654]
205. Milligan ED, Watkins LR. Pathological and protective roles of glia in chronic pain. *Nat Rev Neurosci* 2009;10:23–36. [PubMed: 19096368]
206. Milligan ED, Zapata V, Chacur M, Schoeniger D, Biedenkapp J, O'Connor KA, Verge GM, Chapman G, Green P, Foster AC, Naeve GS, Maier SF, Watkins LR. Evidence that exogenous and endogenous fractalkine can induce spinal nociceptive facilitation in rats. *Eur J Neurosci* 2004;20:2294–2302. [PubMed: 15525271]
207. Minami T, Nishihara I, Uda R, Ito S, Hyodo M, Hayaishi O. Involvement of glutamate receptors in allodynia induced by prostaglandins E2 and F2 alpha injected into conscious mice. *Pain* 1994;57:225–231. [PubMed: 7916453]
208. Miyabe T, Miletic G, Miletic V. Loose ligation of the sciatic nerve in rats elicits transient up-regulation of Homer1a gene expression in the spinal dorsal horn. *Neurosci Lett* 2006;398:296–299. [PubMed: 16448751]
209. Mohr C, Leyendecker S, Mangels I, Machner B, Sander T, Helmchen C. Central representation of cold-evoked pain relief in capsaicin induced pain: An event-related fMRI study. *Pain* 2008;139:416–430. [PubMed: 18614290]
210. Moore KA, Kohno T, Karchewski LA, Scholz J, Baba H, Woolf CJ. Partial peripheral nerve injury promotes a selective loss of GABAergic inhibition in the superficial dorsal horn of the spinal cord. *J Neurosci* 2002;22:6724–6731. [PubMed: 12151551]
211. Moreno-Lopez B, Gonzalez-Forero D. Nitric oxide and synaptic dynamics in the adult brain: Physiopathological aspects. *Rev Neurosci* 2006;17:309–357. [PubMed: 16878402]
212. Moylan Governo RJ, Morris PG, Prior MJ, Marsden CA, Chapman V. Capsaicin-evoked brain activation and central sensitization in anaesthetised rats: A functional magnetic resonance imaging study. *Pain* 2006;126:35–45. [PubMed: 16843597]
213. Muller F, Heinke B, Sandkuhler J. Reduction of glycine receptor-mediated miniature inhibitory postsynaptic currents in rat spinal lamina I neurons after peripheral inflammation. *Neuroscience* 2003;122:799–805. [PubMed: 14622922]
214. Nagy GG, Al-Ayyan M, Andrew D, Fukaya M, Watanabe M, Todd AJ. Widespread expression of the AMPA receptor GluR2 subunit at glutamatergic synapses in the rat spinal cord and phosphorylation of GluR1 in response to noxious stimulation revealed with an antigen-unmasking method. *J Neurosci* 2004;24:5766–5777. [PubMed: 15215299]
215. Nagy GG, Watanabe M, Fukaya M, Todd AJ. Synaptic distribution of the NR1, NR2A and NR2B subunits of the N-methyl-D-aspartate receptor in the rat lumbar spinal cord revealed with an antigen-unmasking technique. *Eur J Neurosci* 2004;20:3301–3312. [PubMed: 15610162]
216. Naisbitt S, Kim E, Tu JC, Xiao B, Sala C, Valtschanoff J, Weinberg RJ, Worley PF, Sheng M. Shank, a novel family of postsynaptic density proteins that binds to the NMDA receptor/PSD-95/GKAP complex and cortactin. *Neuron* 1999;23:569–582. [PubMed: 10433268]
217. Naisbitt S, Kim E, Weinberg RJ, Rao A, Yang FC, Craig AM, Sheng M. Characterization of guanylate kinase-associated protein, a postsynaptic density protein at excitatory synapses that interacts directly with postsynaptic density-95/synapse-associated protein 90. *J Neurosci* 1997;17:5687–5696. [PubMed: 9221768]
218. Neugebauer V, Chen PS, Willis WD. Groups II and III metabotropic glutamate receptors differentially modulate brief and prolonged nociception in primate STT cells. *J Neurophysiol* 2000;84:2998–3009. [PubMed: 11110827]
219. Neugebauer V, Li W. Differential sensitization of amygdala neurons to afferent inputs in a model of arthritic pain. *J Neurophysiol* 2003;89:716–727. [PubMed: 12574449]
220. Neugebauer V, Li W, Bird GC, Bhawe G, Gereau RW. Synaptic plasticity in the amygdala in a model of arthritic pain: Differential roles of metabotropic glutamate receptors 1 and 5. *J Neurosci* 2003;23:52–63. [PubMed: 12514201]

221. Neugebauer V, Lucke T, Schaible HG. Requirement of metabotropic glutamate receptors for the generation of inflammation-evoked hyperexcitability in rat spinal cord neurons. *Eur J Neurosci* 1994;6:1179–1186. [PubMed: 7524965]
222. Neumann S, Braz JM, Skinner K, Llewellyn-Smith IJ, Basbaum AI. Innocuous, not noxious, input activates PKC gamma interneurons of the spinal dorsal horn via myelinated afferent fibers. *J Neurosci* 2008;28:7936–7944. [PubMed: 18685019]
223. Neumann S, Doubell TP, Leslie T, Woolf CJ. Inflammatory pain hypersensitivity mediated by phenotypic switch in myelinated primary sensory neurons. *Nature* 1996;384:360–364. [PubMed: 8934522]
224. Nicoll RA, Schmitz D. Synaptic plasticity at hippocampal mossy fibre synapses. *Nat Rev Neurosci* 2005;6:863–876. [PubMed: 16261180]
225. Nimmerjahn A, Kirchhoff F, Helmchen F. Resting microglial cells are highly dynamic surveillants of brain parenchyma in vivo. *Science* 2005;308:1314–1318. [PubMed: 15831717]
226. Nishimune A, Isaac JT, Molnar E, Noel J, Nash SR, Tagaya M, Collingridge GL, Nakanishi S, Henley JM. NSF binding to GluR2 regulates synaptic transmission. *Neuron* 1998;21:87–97. [PubMed: 9697854]
227. Nishiyama M, Hong K, Mikoshiba K, Poo MM, Kato K. Calcium stores regulate the polarity and input specificity of synaptic modification. *Nature* 2000;408:584–588. [PubMed: 11117745]
228. Noel J, Ralph GS, Pickard L, Williams J, Molnar E, Uney JB, Collingridge GL, Henley JM. Surface expression of AMPA receptors in hippocampal neurons is regulated by an NSF-dependent mechanism. *Neuron* 1999;23:365–376. [PubMed: 10399941]
229. Noguchi K, Kawai Y, Fukuoka T, Senba E, Miki K. Substance P induced by peripheral nerve injury in primary afferent sensory neurons and its effect on dorsal column nucleus neurons. *J Neurosci* 1995;15:7633–7643. [PubMed: 7472514]
230. Obata K, Noguchi K. BDNF in sensory neurons and chronic pain. *Neurosci Res* 2006;55:1–10. [PubMed: 16516994]
231. Obata K, Yamanaka H, Dai Y, Mizushima T, Fukuoka T, Tokunaga A, Noguchi K. Differential activation of MAPK in injured and uninjured DRG neurons following chronic constriction injury of the sciatic nerve in rats. *Eur J Neurosci* 2004;20:2881–2895. [PubMed: 15579142]
232. Obata K, Yamanaka H, Dai Y, Tachibana T, Fukuoka T, Tokunaga A, Yoshikawa H, Noguchi K. Differential activation of extracellular signal-regulated protein kinase in primary afferent neurons regulates brain-derived neurotrophic factor expression after peripheral inflammation and nerve injury. *J Neurosci* 2003;23:4117–4126. [PubMed: 12764099]
233. Oh MC, Derkach VA. Dominant role of the GluR2 subunit in regulation of AMPA receptors by CaMKII. *Nat Neurosci* 2005;8:853–854. [PubMed: 15924137]
234. Okabe S. Molecular anatomy of the postsynaptic density. *Mol Cell Neurosci* 2007;34:503–518. [PubMed: 17321751]
235. Okamoto M, Baba H, Goldstein PA, Higashi H, Shimoji K, Yoshimura M. Functional reorganization of sensory pathways in the rat spinal dorsal horn following peripheral nerve injury. *J Physiol* 2001;532:241–250. [PubMed: 11283238]
236. Omkumar RV, Kiely MJ, Rosenstein AJ, Min KT, Kennedy MB. Identification of a phosphorylation site for calcium/calmodulin-dependent protein kinase II in the NR2B subunit of the N-methyl-D-aspartate receptor. *J Biol Chem* 1996;271:31670–31678. [PubMed: 8940188]
237. Osikowicz M, Mika J, Makuch W, Przewlocka B. Glutamate receptor ligands attenuate allodynia and hyperalgesia and potentiate morphine effects in a mouse model of neuropathic pain. *Pain* 2008;139:117–126. [PubMed: 18442882]
238. Park JS, Voitenko N, Petralia RS, Guan X, Xu JT, Steinberg JP, Takamiya K, Sotnik A, Kopach O, Haganir RL, Tao YX. Persistent inflammation induces GluR2 internalization via NMDA receptor-triggered PKC activation in dorsal horn neurons. *J Neurosci* 2009;29:3206–3219. [PubMed: 19279258]
239. Park JS, Yaster M, Guan X, Xu JT, Shih MH, Guan Y, Raja SN, Tao YX. Role of spinal cord alpha-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptors in complete Freund's adjuvant-induced inflammatory pain. *Mol Pain* 2008;4:67. [PubMed: 19116032]

240. Perroy J, Gutierrez GJ, Coulon V, Bockaert J, Pin JP, Fagni L. The C terminus of the metabotropic glutamate receptor subtypes 2 and 7 specifies the receptor signaling pathways. *J Biol Chem* 2001;276:45800–45805. [PubMed: 11584003]
241. Petcu M, Dias JP, Ongali B, Thibault G, Neugebauer W, Couture R. Role of kinin B1 and B2 receptors in a rat model of neuropathic pain. *Int Immunopharmacol* 2008;8:188–196. [PubMed: 18182225]
242. Petho G, Derow A, Reeh PW. Bradykinin-induced nociceptor sensitization to heat is mediated by cyclooxygenase products in isolated rat skin. *Eur J Neurosci* 2001;14:210–218. [PubMed: 11553274]
243. Petralia RS, Wang YX, Wenthold RJ. The NMDA receptor subunits NR2A and NR2B show histological and ultra-structural localization patterns similar to those of NR1. *J Neurosci* 1994;14:6102–6120. [PubMed: 7931566]
244. Peyron R, Laurent B, Garcia-Larrea L. Functional imaging of brain responses to pain: A review and meta-analysis (2000). *Neurophysiol Clin* 2000;30:263–288. [PubMed: 11126640]
245. Pezet S, Malcangio M, Lever IJ, Perkinson MS, Thompson SW, Williams RJ, McMahon SB. Noxious stimulation induces Trk receptor and downstream ERK phosphorylation in spinal dorsal horn. *Mol Cell Neurosci* 2002;21:684–695. [PubMed: 12504600]
246. Pezet S, Marchand F, D'Mello R, Grist J, Clark AK, Malcangio M, Dickenson AH, Williams RJ, McMahon SB. Phosphatidylinositol 3-kinase is a key mediator of central sensitization in painful inflammatory conditions. *J Neurosci* 2008;28:4261–4270. [PubMed: 18417706]
247. Pitcher MH, Ribeiro-da-Silva A, Coderre TJ. Effects of inflammation on the ultrastructural localization of spinal cord dorsal horn group I metabotropic glutamate receptors. *J Comp Neurol* 2007;505:412–423. [PubMed: 17912745]
248. Polgar E, Al-Khater KM, Shehab S, Watanabe M, Todd AJ. Large projection neurons in lamina I of the rat spinal cord that lack the neurokinin 1 receptor are densely innervated by VGLUT2-containing axons and possess GluR4-containing AMPA receptors. *J Neurosci* 2008;28:13150–13160. [PubMed: 19052206]
249. Polgar E, Gray S, Riddell JS, Todd AJ. Lack of evidence for significant neuronal loss in laminae I–III of the spinal dorsal horn of the rat in the chronic constriction injury model. *Pain* 2004;111:144–150. [PubMed: 15327818]
250. Polgar E, Hughes DI, Riddell JS, Maxwell DJ, Puskar Z, Todd AJ. Selective loss of spinal GABAergic or glycinergic neurons is not necessary for development of thermal hyperalgesia in the chronic constriction injury model of neuropathic pain. *Pain* 2003;104:229–239. [PubMed: 12855333]
251. Polgar E, Todd AJ. Tactile allodynia can occur in the spared nerve injury model in the rat without selective loss of GABA or GABA(A) receptors from synapses in laminae I–II of the ipsilateral spinal dorsal horn. *Neuroscience* 2008;156:193–202. [PubMed: 18675320]
252. Polgar E, Watanabe M, Hartmann B, Grant SG, Todd AJ. Expression of AMPA receptor subunits at synapses in laminae I–III of the rodent spinal dorsal horn. *Mol Pain* 2008;4:5. [PubMed: 18215271]
253. Price DD, Zhou Q, Moshiree B, Robinson ME, Verne GN. Peripheral and central contributions to hyperalgesia in irritable bowel syndrome. *J Pain* 2006;7:529–535. [PubMed: 16885007]
254. Qin Y, Zhu Y, Baumgart JP, Stormetta RL, Seidenman K, Mack V, van Aelst L, Zhu JJ. State-dependent Ras signaling and AMPA receptor trafficking. *Genes Dev* 2005;19:2000–2015. [PubMed: 16107614]
255. Qu XX, Cai J, Li MJ, Chi YN, Liao FF, Liu FY, Wan Y, Han JS, Xing GG. Role of the spinal cord NR2B-containing NMDA receptors in the development of neuropathic pain. *Exp Neurol* 2009;215:298–307. [PubMed: 19046970]
256. Raghavendra V, Tanga F, DeLeo JA. Inhibition of microglial activation attenuates the development but not existing hypersensitivity in a rat model of neuropathy. *J Pharmacol Exp Ther* 2003;306:624–630. [PubMed: 12734393]
257. Raghavendra V, Tanga F, Rutkowski MD, DeLeo JA. Anti-hyperalgesic and morphine-sparing actions of propentofylline following peripheral nerve injury in rats: Mechanistic implications of spinal glia and proinflammatory cytokines. *Pain* 2003;104:655–664. [PubMed: 12927638]

258. Raghavendra V, Tanga FY, DeLeo JA. Complete Freund's adjuvant-induced peripheral inflammation evokes glial activation and proinflammatory cytokine expression in the CNS. *Eur J Neurosci* 2004;20:467–473. [PubMed: 15233755]
259. Raman IM, Tong G, Jahr CE. Beta-adrenergic regulation of synaptic NMDA receptors by cAMP-dependent protein kinase. *Neuron* 1996;16:415–421. [PubMed: 8789956]
260. Regehr WG, Tank DW. Postsynaptic NMDA receptor-mediated calcium accumulation in hippocampal CA1 pyramidal cell dendrites. *Nature* 1990;345:807–810. [PubMed: 1972782]
261. Rodriguez Parkitna J, Korostynski M, Kaminska-Chowanec D, Obara I, Mika J, Przewlocka B, Przewlocki R: Comparison of gene expression profiles in neuropathic and inflammatory pain. *J Physiol Pharmacol* 2006;57:401–414. [PubMed: 17033093]
262. Roh DH, Kim HW, Yoon SY, Seo HS, Kwon YB, Han HJ, Beitz AJ, Lee JH. Depletion of capsaicin-sensitive afferents prevents lamina-dependent increases in spinal N-methyl-D-aspartate receptor subunit 1 expression and phosphorylation associated with thermal hyperalgesia in neuropathic rats. *Eur J Pain* 2008;12:552–563. [PubMed: 17933570]
263. Romero-Sandoval EA, Horvath RJ, DeLeo JA. Neuroimmune interactions and pain: Focus on glial-modulating targets. *Curr Opin Investig Drugs* 2008;9:726–734.
264. Romorini S, Piccoli G, Jiang M, Grossano P, Tonna N, Passafaro M, Zhang M, Sala C. A functional role of postsynaptic density-95-guanlylate kinase-associated protein complex in regulating Shank assembly and stability to synapses. *J Neurosci* 2004;24:9391–9404. [PubMed: 15496675]
265. Rumbaugh G, Sia GM, Garner CC, Hagan RL. Synapse-associated protein-97 isoform-specific regulation of surface AMPA receptors and synaptic function in cultured neurons. *J Neurosci* 2003;23:4567–4576. [PubMed: 12805297]
266. Sala C, Piech V, Wilson NR, Passafaro M, Liu G, Sheng M. Regulation of dendritic spine morphology and synaptic function by Shank and Homer. *Neuron* 2001;31:115–130. [PubMed: 11498055]
267. Sala C, Roussignol G, Meldolesi J, Fagni L. Key role of the postsynaptic density scaffold proteins Shank and Homer in the functional architecture of Ca²⁺ homeostasis at dendritic spines in hippocampal neurons. *J Neurosci* 2005;25:4587–4592. [PubMed: 15872106]
268. Salter MW, Kalia LV. Src kinases: A hub for NMDA receptor regulation. *Nat Rev Neurosci* 2004;5:317–328. [PubMed: 15034556]
269. Samad TA, Moore KA, Sapirstein A, Billet S, Allchorne A, Poole S, Bonventre JV, Woolf CJ. Interleukin-1 β -mediated induction of Cox-2 in the CNS contributes to inflammatory pain hypersensitivity. *Nature* 2001;410:471–475. [PubMed: 11260714]
270. Sandkuhler J. Understanding LTP in pain pathways. *Mol Pain* 2007;3:9. [PubMed: 17407590]
271. Sandkuhler J, Liu X. Induction of long-term potentiation at spinal synapses by noxious stimulation or nerve injury. *Eur J Neurosci* 1998;10:2476–2480. [PubMed: 9749775]
272. Sarkar S, Aziz Q, Woolf CJ, Hobson AR, Thompson DG. Contribution of central sensitisation to the development of non-cardiac chest pain. *Lancet* 2000;356:1154–1159. [PubMed: 11030295]
273. Sato Y, Tao YX, Su Q, Johns RA. Post-synaptic density-93 mediates tyrosine-phosphorylation of the N-methyl-D-aspartate receptors. *Neuroscience* 2008;153:700–708. [PubMed: 18423999]
274. Schaible HG, Schmelz M, Tegeder I. Pathophysiology and treatment of pain in joint disease. *Adv Drug Deliv Rev* 2006;58:323–342. [PubMed: 16626837]
275. Schmidt A, Gao W, Konig P, Heine S, Motterlini R, Ruth P, Schlossmann J, Koesling D, Niederberger E, Tegeder I, Friebe A, Geisslinger G. cGMP produced by NO-sensitive guanylyl cyclase essentially contributes to inflammatory and neuropathic pain by using targets different from cGMP-dependent protein kinase I. *J Neurosci* 2008;28:8568–8576. [PubMed: 18716216]
276. Schnell E, Sizemore M, Karimzadegan S, Chen L, Brecht DS, Nicoll RA. Direct interactions between PSD-95 and stargazin control synaptic AMPA receptor number. *Proc Natl Acad Sci U S A* 2002;99:13902–13907. [PubMed: 12359873]
277. Scholz J, Broom DC, Youn DH, Mills CD, Kohno T, Suter MR, Moore KA, Decosterd I, Coggeshall RE, Woolf CJ. Blocking caspase activity prevents transsynaptic neuronal apoptosis and the loss of inhibition in lamina II of the dorsal horn after peripheral nerve injury. *J Neurosci* 2005;25:7317–7323. [PubMed: 16093381]

278. Schwartz ES, Lee I, Chung K, Chung JM. Oxidative stress in the spinal cord is an important contributor in capsaicin-induced mechanical secondary hyperalgesia in mice. *Pain* 2008;138:514–524. [PubMed: 18375065]
279. Schwenk J, Harmel N, Zolles G, Bildl W, Kulik A, Heimrich B, Chisaka O, Jonas P, Schulte U, Fakler B, Klocker N. Functional proteomics identify cornichon proteins as auxiliary subunits of AMPA receptors. *Science* 2009;323:1313–1319. [PubMed: 19265014]
280. Seltzer Z, Cohn S, Ginzburg R, Beilin B. Modulation of neuropathic pain behavior in rats by spinal disinhibition and NMDA receptor blockade of injury discharge. *Pain* 1991;45:69–75. [PubMed: 1677750]
281. Sevostianova N, Danysz W. Analgesic effects of mGlu1 and mGlu5 receptor antagonists in the rat formalin test. *Neuropharmacology* 2006;51:623–630. [PubMed: 16793067]
282. Sherrington CS. Observations on the scratch-reflex in the spinal dog. *J Physiol* 1906;34:1–50. [PubMed: 16992835]
283. Shih YY, Chiang YC, Chen JC, Huang CH, Chen YY, Liu RS, Chang C, Jaw FS. Brain nociceptive imaging in rats using (18)f-fluorodeoxyglucose small-animal positron emission tomography. *Neuroscience* 2008;155:1221–1226. [PubMed: 18675887]
284. Shiraishi-Yamaguchi Y, Furuichi T. The Homer family proteins. *Genome Biol* 2007;8:206. [PubMed: 17316461]
285. Shortland P, Kinman E, Molander C. Sprouting of A-fibre primary afferents into lamina II in two rat models of neuropathic pain. *Eur J Pain* 1997;1:215–227. [PubMed: 15102403]
286. Simmons RM, Webster AA, Kalra AB, Iyengar S. Group II mGluR receptor agonists are effective in persistent and neuropathic pain models in rats. *Pharmacol Biochem Behav* 2002;73:419–427. [PubMed: 12117597]
287. Simone DA, Sorkin LS, Oh U, Chung JM, Owens C, LaMotte RH, Willis WD. Neurogenic hyperalgesia: Central neural correlates in responses of spinothalamic tract neurons. *J Neurophysiol* 1991;66:228–246. [PubMed: 1919669]
288. Siniscalco D, Giordano C, Fuccio C, Luongo L, Ferraraccio F, Rossi F, de Novellis V, Roth KA, Maione S. Involvement of subtype 1 metabotropic glutamate receptors in apoptosis and caspase-7 over-expression in spinal cord of neuropathic rats. *Pharmacol Res* 2008;57:223–233. [PubMed: 18325779]
289. Sivilotti L, Woolf CJ. The contribution of GABAA and glycine receptors to central sensitization: Disinhibition and touch-evoked allodynia in the spinal cord. *J Neurophysiol* 1994;72:169–179. [PubMed: 7965003]
290. Sivilotti LG, Thompson SW, Woolf CJ. Rate of rise of the cumulative depolarization evoked by repetitive stimulation of small-caliber afferents is a predictor of action potential windup in rat spinal neurons in vitro. *J Neurophysiol* 1993;69:1621–1631. [PubMed: 8389833]
291. Slack S, Battaglia A, Cibert-Goton V, Gavazzi I. EphrinB2 induces tyrosine phosphorylation of NR2B via Src-family kinases during inflammatory hyperalgesia. *Neuroscience* 2008;156:175–183. [PubMed: 18694808]
292. Slack SE, Grist J, Mac Q, McMahon SB, Pezet S. TrkB expression and phospho-ERK activation by brain-derived neurotrophic factor in rat spinothalamic tract neurons. *J Comp Neurol* 2005;489:59–68. [PubMed: 15977164]
293. Slack SE, Pezet S, McMahon SB, Thompson SW, Malcangio M. Brain-derived neurotrophic factor induces NMDA receptor subunit one phosphorylation via ERK and PKC in the rat spinal cord. *Eur J Neurosci* 2004;20:1769–1778. [PubMed: 15379998]
294. Sluka KA, Rees H, Chen PS, Tsuruoka M, Willis WD. Inhibitors of G-proteins and protein kinases reduce the sensitization to mechanical stimulation and the desensitization to heat of spinothalamic tract neurons induced by intradermal injection of capsaicin in the primate. *Exp Brain Res* 1997;115:15–24. [PubMed: 9224830]
295. Snyder EM, Colledge M, Crozier RA, Chen WS, Scott JD, Bear MF. Role for A kinase-anchoring proteins (AKAPS) in glutamate receptor trafficking and long term synaptic depression. *J Biol Chem* 2005;280:16962–16968. [PubMed: 15718245]

296. Soliman AC, Yu JS, Coderre TJ. mGlu and NMDA receptor contributions to capsaicin-induced thermal and mechanical hypersensitivity. *Neuropharmacology* 2005;48:325–332. [PubMed: 15721164]
297. Song I, Kamboj S, Xia J, Dong H, Liao D, Huganir RL. Interaction of the N-ethylmaleimide-sensitive factor with AMPA receptors. *Neuron* 1998;21:393–400. [PubMed: 9728920]
298. Song XJ, Cao JL, Li HC, Zheng JH, Song XS, Xiong LZ. Up-regulation and redistribution of ephrinB and EphB receptor in dorsal root ganglion and spinal dorsal horn neurons after peripheral nerve injury and dorsal rhizotomy. *Eur J Pain* 2008;12:1031–1039. [PubMed: 18321739]
299. Song XJ, Zheng JH, Cao JL, Liu WT, Song XS, Huang ZJ. EphrinB-EphB receptor signaling contributes to neuropathic pain by regulating neural excitability and spinal synaptic plasticity in rats. *Pain* 2008;139:168–180. [PubMed: 18448254]
300. South SM, Kohno T, Kaspar BK, Hegarty D, Vissel B, Drake CT, Ohata M, Jenab S, Sailer AW, Malkmus S, Masuyama T, Horner P, Bogulavsky J, Gage FH, Yaksh TL, Woolf CJ, Heinemann SF, Inturrisi CE. A conditional deletion of the NR1 subunit of the NMDA receptor in adult spinal cord dorsal horn reduces NMDA currents and injury-induced pain. *J Neurosci* 2003;23:5031–5040. [PubMed: 12832526]
301. Staud R. Evidence of involvement of central neural mechanisms in generating fibromyalgia pain. *Curr Rheumatol Rep* 2002;4:299–305. [PubMed: 12126581]
302. Staud R, Bovee CE, Robinson ME, Price DD. Cutaneous C-fiber pain abnormalities of fibromyalgia patients are specifically related to temporal summation. *Pain* 2008;139:315–323. [PubMed: 18538477]
303. Staud R, Robinson ME, Price DD. Temporal summation of second pain and its maintenance are useful for characterizing widespread central sensitization of fibromyalgia patients. *J Pain* 2007;8:893–901. [PubMed: 17681887]
304. Steinberg JP, Takamiya K, Shen Y, Xia J, Rubio ME, Yu S, Jin W, Thomas GM, Linden DJ, Huganir RL. Targeted in vivo mutations of the AMPA receptor subunit GluR2 and its interacting protein PICK1 eliminate cerebellar long-term depression. *Neuron* 2006;49:845–860. [PubMed: 16543133]
305. Stephenson FA, Cousins SL, Kenny AV. Assembly and forward trafficking of NMDA receptors (Review). *Mol Membr Biol* 2008;25:311–320. [PubMed: 18446617]
306. Suh YH, Pelkey KA, Lavezzari G, Roche PA, Huganir RL, McBain CJ, Roche KW. Corequirement of PICK1 binding and PKC phosphorylation for stable surface expression of the metabotropic glutamate receptor mGluR7. *Neuron* 2008;58:736–748. [PubMed: 18549785]
307. Sun RQ, Lawand NB, Willis WD. The role of calcitonin gene-related peptide (CGRP) in the generation and maintenance of mechanical allodynia and hyperalgesia in rats after intradermal injection of capsaicin. *Pain* 2003;104:201–208. [PubMed: 12855330]
308. Sun RQ, Tu YJ, Lawand NB, Yan JY, Lin Q, Willis WD. Calcitonin gene-related peptide receptor activation produces PKA- and PKC-dependent mechanical hyperalgesia and central sensitization. *J Neurophysiol* 2004;92:2859–2866. [PubMed: 15486424]
309. Susswein AJ, Katzoff A, Miller N, Hurwitz I. Nitric oxide and memory. *Neuroscientist* 2004;10:153–162. [PubMed: 15070489]
310. Suzuki R, Morcuende S, Webber M, Hunt SP, Dickenson AH. Superficial NK1-expressing neurons control spinal excitability through activation of descending pathways. *Nat Neurosci* 2002;5:1319–1326. [PubMed: 12402039]
311. Svensson CI, Hua XY, Protter AA, Powell HC, Yaksh TL. Spinal p38 MAP kinase is necessary for NMDA-induced spinal PGE(2) release and thermal hyperalgesia. *Neuroreport* 2003;14:1153–1157. [PubMed: 12821799]
312. Svensson CI, Marsala M, Westerlund A, Calcutt NA, Campana WM, Freshwater JD, Catalano R, Feng Y, Protter AA, Scott B, Yaksh TL. Activation of p38 mitogen-activated protein kinase in spinal microglia is a critical link in inflammation-induced spinal pain processing. *J Neurochem* 2003;86:1534–1544. [PubMed: 12950462]
313. Svensson CI, Tran TK, Fitzsimmons B, Yaksh TL, Hua XY. Descending serotonergic facilitation of spinal ERK activation and pain behavior. *FEBS Lett* 2006;580:6629–6634. [PubMed: 17113581]

314. Takasu MA, Dalva MB, Zigmond RE, Greenberg ME. Modulation of NMDA receptor-dependent calcium influx and gene expression through EphB receptors. *Science* 2002;295:491–495. [PubMed: 11799243]
315. Tal M, Bennett GJ. Neuropathic pain sensations are differentially sensitive to dextrorphan. *Neuroreport* 1994;5:1438–1440. [PubMed: 7948833]
316. Tanabe M, Nagatani Y, Saitoh K, Takasu K, Ono H. Pharmacological assessments of nitric oxide synthase isoforms and downstream diversity of NO signaling in the maintenance of thermal and mechanical hypersensitivity after peripheral nerve injury in mice. *Neuropharmacology* 2009;56:702–708. [PubMed: 19111753]
317. Tanga FY, Nutille-McMenemy N, DeLeo JA. The CNS role of Toll-like receptor 4 in innate neuroimmunity and painful neuropathy. *Proc Natl Acad Sci U S A* 2005;102:5856–5861. [PubMed: 15809417]
318. Tao F, Skinner J, Su Q, Johns RA. New role for spinal Stargazin in alpha-amino-3-hydroxy-5-methyl-4-isoxazole-propionic acid receptor-mediated pain sensitization after inflammation. *J Neurosci Res* 2006;84:867–873. [PubMed: 16791853]
319. Tao F, Su Q, Johns RA. Cell-permeable peptide Tat-PSD-95 PDZ2 inhibits chronic inflammatory pain behaviors in mice. *Mol Ther* 2008;16:1776–1782. [PubMed: 18781143]
320. Tao F, Tao YX, Gonzalez JA, Fang M, Mao P, Johns RA. Knockdown of PSD-95/SAP90 delays the development of neuropathic pain in rats. *Neuroreport* 2001;12:3251–3255. [PubMed: 11711866]
321. Tao YX, Huang YZ, Mei L, Johns RA. Expression of PSD-95/SAP90 is critical for N-methyl-D-aspartate receptor-mediated thermal hyperalgesia in the spinal cord. *Neuroscience* 2000;98:201–206. [PubMed: 10854750]
322. Tao YX, Johns RA. Activation and up-regulation of spinal cord nitric oxide receptor, soluble guanylate cyclase, after formalin injection into the rat hind paw. *Neuroscience* 2002;112:439–446. [PubMed: 12044461]
323. Tao YX, Rumbaugh G, Wang GD, Petralia RS, Zhao C, Kauer FW, Tao F, Zhuo M, Wenthold RJ, Raja SN, Huganir RL, Brecht DS, Johns RA. Impaired NMDA receptor-mediated postsynaptic function and blunted NMDA receptor-dependent persistent pain in mice lacking postsynaptic density-93 protein. *J Neurosci* 2003;23:6703–6712. [PubMed: 12890763]
324. Tappe A, Klugmann M, Luo C, Hirlinger D, Agarwal N, Benrath J, Ehrenguber MU, Düring MJ, Kuner R. Synaptic scaffolding protein Homer1a protects against chronic inflammatory pain. *Nat Med* 2006;12:677–681. [PubMed: 16715092]
325. Tavalin SJ. AKAP79 selectively enhances protein kinase C regulation of GluR1 at a Ca²⁺-calmodulin-dependent protein kinase II/protein kinase C site. *J Biol Chem* 2008;283:11445–11452. [PubMed: 18305116]
326. Tawfik VL, Regan MR, Haenggeli C, Lacroix-Fralish ML, Nutille-McMenemy N, Perez N, Rothstein JD, DeLeo JA. Propentofylline-induced astrocyte modulation leads to alterations in glial glutamate promoter activation following spinal nerve transection. *Neuroscience* 2008;152:1086–1092. [PubMed: 18358622]
327. Tegeder I, Costigan M, Griffin RS, Abele A, Belfer I, Schmidt H, Ehnert C, Nejim J, Marian C, Scholz J, Wu T, Allchorne A, Diatchenko L, Binshtok AM, Goldman D, Adolph J, Sama S, Atlas SJ, Carlezon WA, Parsegian A, Lotsch J, Fillingim RB, Maixner W, Geisslinger G, Max MB, Woolf CJ. GTP cyclohydrolase and tetrahydrobiopterin regulate pain sensitivity and persistence. *Nat Med* 2006;12:1269–1277. [PubMed: 17057711]
328. Terashima A, Pelkey KA, Rah JC, Suh YH, Roche KW, Collingridge GL, McBain CJ, Isaac JT. An essential role for PICK1 in NMDA receptor-dependent bidirectional synaptic plasticity. *Neuron* 2008;57:872–882. [PubMed: 18367088]
329. Tezuka T, Umemori H, Akiyama T, Nakanishi S, Yamamoto T. PSD-95 promotes Fyn-mediated tyrosine phosphorylation of the N-methyl-D-aspartate receptor subunit NR2A. *Proc Natl Acad Sci U S A* 1999;96:435–440. [PubMed: 9892651]
330. Thompson SW, Bennett DL, Kerr BJ, Bradbury EJ, McMahon SB. Brain-derived neurotrophic factor is an endogenous modulator of nociceptive responses in the spinal cord. *Proc Natl Acad Sci U S A* 1999;96:7714–7718. [PubMed: 10393886]

331. Thompson SW, King AE, Woolf CJ. Activity-dependent changes in rat ventral horn neurons in vitro: Summation of prolonged afferent evoked postsynaptic depolarizations produce a d-2-amino-5-phosphonovaleric acid sensitive windup. *Eur J Neurosci* 1990;2:638–649. [PubMed: 12106298]
332. Tilleux S, Hermans E. Neuroinflammation and regulation of glial glutamate uptake in neurological disorders. *J Neurosci Res* 2007;85:2059–2070. [PubMed: 17497670]
333. Tingley WG, Ehlers MD, Kameyama K, Doherty C, Ptak JB, Riley CT, Huganir RL. Characterization of protein kinase A and protein kinase C phosphorylation of the N-methyl-D-aspartate receptor NR1 subunit using phosphorylation site-specific antibodies. *J Biol Chem* 1997;272:5157–5166. [PubMed: 9030583]
334. Tomita S, Adesnik H, Sekiguchi M, Zhang W, Wada K, Howe JR, Nicoll RA, Brecht DS. Stargazin modulates AMPA receptor gating and trafficking by distinct domains. *Nature* 2005;435:1052–1058. [PubMed: 15858532]
335. Tomita S, Chen L, Kawasaki Y, Petralia RS, Wenthold RJ, Nicoll RA, Brecht DS. Functional studies and distribution define a family of transmembrane AMPA receptor regulatory proteins. *J Cell Biol* 2003;161:805–816. [PubMed: 12771129]
336. Tomita S, Shenoy A, Fukata Y, Nicoll RA, Brecht DS. Stargazin interacts functionally with the AMPA receptor glutamate-binding module. *Neuropharmacology* 2007;52:87–91. [PubMed: 16919685]
337. Tomita S, Stein V, Stocker TJ, Nicoll RA, Brecht DS. Bidirectional synaptic plasticity regulated by phosphorylation of stargazin-like TARPs. *Neuron* 2005;45:269–277. [PubMed: 15664178]
338. Tong CK, MacDermott AB. Both Ca²⁺-permeable and -impermeable AMPA receptors contribute to primary synaptic drive onto rat dorsal horn neurons. *J Physiol* 2006;575:133–144. [PubMed: 16763002]
339. Tong YG, Wang HF, Ju G, Grant G, Hokfelt T, Zhang X. Increased uptake and transport of cholera toxin B-subunit in dorsal root ganglion neurons after peripheral axotomy: possible implications for sensory sprouting. *J Comp Neurol* 1999;404:143–158. [PubMed: 9934990]
340. Torebjork HE, Lundberg LE, LaMotte RH. Central changes in processing of mechanoreceptive input in capsaicin-induced secondary hyperalgesia in humans. *J Physiol* 1992;448:765–780. [PubMed: 1593489]
341. Torsney C, MacDermott AB. Disinhibition opens the gate to pathological pain signaling in superficial neurokinin 1 receptor-expressing neurons in rat spinal cord. *J Neurosci* 2006;26:1833–1843. [PubMed: 16467532]
342. Tsui J, Malenka RC. Substrate localization creates specificity in calcium/calmodulin-dependent protein kinase II signaling at synapses. 2006;281:13794–13804.
343. Tu JC, Xiao B, Naisbitt S, Yuan JP, Petralia RS, Brakeman P, Doan A, Aakalu VK, Lanahan AA, Sheng M, Worley PF. Coupling of mGluR/Homer and PSD-95 complexes by the Shank family of postsynaptic density proteins. *Neuron* 1999;23:583–592. [PubMed: 10433269]
344. Tu JC, Xiao B, Yuan JP, Lanahan AA, Leoffert K, Li M, Linden DJ, Worley PF. Homer binds a novel proline-rich motif and links group 1 metabotropic glutamate receptors with IP3 receptors. *Neuron* 1998;21:717–726. [PubMed: 9808459]
345. Ultenius C, Linderöth B, Meyerson BA, Wallin J. Spinal NMDA receptor phosphorylation correlates with the presence of neuropathic signs following peripheral nerve injury in the rat. *Neurosci Lett* 2006;399:85–90. [PubMed: 16469445]
346. Valerio A, Paterlini M, Boifava M, Memo M, Spano P. Metabotropic glutamate receptor mRNA expression in rat spinal cord. *Neuroreport* 1997;8:2695–2699. [PubMed: 9295103]
347. Van Keuren-Jensen K, Cline HT. Visual experience regulates metabotropic glutamate receptor-mediated plasticity of AMPA receptor synaptic transmission by homer1a induction. *J Neurosci* 2006;26:7575–7580. [PubMed: 16855085]
348. van Rossum D, Hanisch UK. Microglia. *Metab Brain Dis* 2004;19:393–411. [PubMed: 15554430]
349. Vardeh D, Wang D, Costigan M, Lazarus M, Saper CB, Woolf CJ, Fitzgerald GA, Samad TA. COX2 in CNS neural cells mediates mechanical inflammatory pain hypersensitivity in mice. *J Clin Invest* 2009;119:287–294. [PubMed: 19127021]
350. Vasko MR, Campbell WB, Waite KJ. Prostaglandin E2 enhances bradykinin-stimulated release of neuropeptides from rat sensory neurons in culture. *J Neurosci* 1994;14:4987–4997. [PubMed: 7519258]

351. Vera-Portocarrero LP, Zhang ET, Ossipov MH, Xie JY, King T, Lai J, Porreca F. Descending facilitation from the rostral ventromedial medulla maintains nerve injury-induced central sensitization. *Neuroscience* 2006;140:1311–1320. [PubMed: 16650614]
352. Verge GM, Milligan ED, Maier SF, Watkins LR, Naeve GS, Foster AC. Fractalkine (CX3CL1) and fractalkine receptor (CX3CR1) distribution in spinal cord and dorsal root ganglia under basal and neuropathic pain conditions. *Eur J Neurosci* 2004;20:1150–1160. [PubMed: 15341587]
353. Vikman K, Robertson B, Grant G, Liljeborg A, Kristensson K. Interferon-gamma receptors are expressed at synapses in the rat superficial dorsal horn and lateral spinal nucleus. *J Neurocytol* 1998;27:749–759. [PubMed: 10640190]
354. Vikman KS, Hill RH, Backstrom E, Robertson B, Kristensson K. Interferon-gamma induces characteristics of central sensitization in spinal dorsal horn neurons in vitro. *Pain* 2003;106:241–251. [PubMed: 14659507]
355. Vikman KS, Rycroft BK, Christie MJ. Switch to Ca²⁺-permeable AMPA and reduced NR2B NMDA receptor-mediated neurotransmission at dorsal horn nociceptive synapses during inflammatory pain in the rat. *J Physiol* 2008;586:515–527. [PubMed: 18033811]
356. Vogel C, Mossner R, Gerlach M, Heinemann T, Murphy DL, Riederer P, Lesch KP, Sommer C. Absence of thermal hyperalgesia in serotonin transporter-deficient mice. *J Neurosci* 2003;23:708–715. [PubMed: 12533631]
357. Walker SM, Meredith-Middleton J, Lickiss T, Moss A, Fitzgerald M. Primary and secondary hyperalgesia can be differentiated by postnatal age and ERK activation in the spinal dorsal horn of the rat pup. *Pain* 2007;128:157–168. [PubMed: 17056180]
358. Wall PD, Woolf CJ. Muscle but not cutaneous C-afferent input produces prolonged increases in the excitability of the flexion reflex in the rat. *J Physiol* 1984;356:443–458. [PubMed: 6520794]
359. Wang H, Kohno T, Amaya F, Brenner GJ, Ito N, Allchorne A, Ji RR, Woolf CJ. Bradykinin produces pain hypersensitivity by potentiating spinal cord glutamatergic synaptic transmission. *J Neurosci* 2005;25:7986–7992. [PubMed: 16135755]
360. Watkins LR, Hutchinson MR, Milligan ED, Maier SF. “Listening” and “talking” to neurons: Implications of immune activation for pain control and increasing the efficacy of opioids”. *Brain Res Rev* 2007;56:148–169. [PubMed: 17706291]
361. Watkins LR, Maier SF. Beyond neurons: Evidence that immune and glial cells contribute to pathological pain states. *Physiol Rev* 2002;82:981–1011. [PubMed: 12270950]
362. Watkins LR, Milligan ED, Maier SF. Glial activation: A driving force for pathological pain. *Trends Neurosci* 2001;24:450–455. [PubMed: 11476884]
363. Wei F, Vadakkan KI, Toyoda H, Wu LJ, Zhao MG, Xu H, Shum FW, Jia YH, Zhuo M. Calcium calmodulin-stimulated adenylyl cyclases contribute to activation of extracellular signal-regulated kinase in spinal dorsal horn neurons in adult rats and mice. *J Neurosci* 2006;26:851–861. [PubMed: 16421305]
364. Wei F, Zhuo M. Potentiation of sensory responses in the anterior cingulate cortex following digit amputation in the anaesthetised rat. *J Physiol* 2001;532:823–833. [PubMed: 11313449]
365. Willis WD. Long-term potentiation in spinothalamic neurons. *Brain Res Brain Res Rev* 2002;40:202–214. [PubMed: 12589918]
366. Woodbury CJ, Kullmann FA, McIlwrath SL, Koerber HR. Identity of myelinated cutaneous sensory neurons projecting to nociceptive laminae following nerve injury in adult mice. *J Comp Neurol* 2008;508:500–509. [PubMed: 18335545]
367. Woolf C, Wiesenfeld-Hallin Z. Substance P and calcitonin gene-related peptide synergistically modulate the gain of the nociceptive flexor withdrawal reflex in the rat. *Neurosci Lett* 1986;66:226–230. [PubMed: 2425287]
368. Woolf CJ. Central sensitization: Uncovering the relation between pain and plasticity. *Anesthesiology* 2007;106:864–867. [PubMed: 17413924]
369. Woolf CJ. Evidence for a central component of post-injury pain hypersensitivity. *Nature* 1983;306:686–688. [PubMed: 6656869]
370. Woolf CJ. Phenotypic modification of primary sensory neurons: The role of nerve growth factor in the production of persistent pain. *Philos Trans R Soc Lond B Biol Sci* 1996;351:441–448. [PubMed: 8730783]

371. Woolf CJ. Windup and central sensitization are not equivalent. *Pain* 1996;66:105–108. [PubMed: 8880830]
372. Woolf CJ, King AE. Dynamic alterations in the cutaneous mechanoreceptive fields of dorsal horn neurons in the rat spinal cord. *J Neurosci* 1990;10:2717–2726. [PubMed: 2388084]
373. Woolf CJ, King AE. Subthreshold components of the cutaneous mechanoreceptive fields of dorsal horn neurons in the rat lumbar spinal cord. *J Neurophysiol* 1989;62:907–916. [PubMed: 2809709]
374. Woolf CJ, Ma Q. Nociceptors: Noxious stimulus detectors. *Neuron* 2007;55:353–364. [PubMed: 17678850]
375. Woolf CJ, Ma QP, Allchorne A, Poole S. Peripheral cell types contributing to the hyperalgesic action of nerve growth factor in inflammation. *J Neurosci* 1996;16:2716–2723. [PubMed: 8786447]
376. Woolf CJ, Salter MW. Neuronal plasticity: Increasing the gain in pain. *Science* 2000;288:1765–1769. [PubMed: 10846153]
377. Woolf CJ, Shortland P, Coggeshall RE. Peripheral nerve injury triggers central sprouting of myelinated afferents. *Nature* 1992;355:75–78. [PubMed: 1370574]
378. Woolf CJ, Shortland P, Reynolds M, Ridings J, Doubell T, Coggeshall RE. Reorganization of central terminals of myelinated primary afferents in the rat dorsal horn following peripheral axotomy. *J Comp Neurol* 1995;360:121–134. [PubMed: 7499558]
379. Woolf CJ, Thompson SW. The induction and maintenance of central sensitization is dependent on N-methyl-D-aspartic acid receptor activation: Implications for the treatment of post-injury pain hypersensitivity states. *Pain* 1991;44:293–299. [PubMed: 1828878]
380. Woolf CJ, Walters ET. Common patterns of plasticity contributing to nociceptive sensitization in mammals and Aplysia. *Trends Neurosci* 1991;14:74–78. [PubMed: 1708540]
381. Wu J, Fang L, Lin Q, Willis WD. Nitric oxide synthase in spinal cord central sensitization following intradermal injection of capsaicin. *Pain* 2001;94:47–58. [PubMed: 11576744]
382. Wu J, Lin Q, Lu Y, Willis WD, Westlund KN. Changes in nitric oxide synthase isoforms in the spinal cord of rat following induction of chronic arthritis. *Exp Brain Res* 1998;118:457–465. [PubMed: 9504841]
383. Wu J, Lin Q, McAdoo DJ, Willis WD. Nitric oxide contributes to central sensitization following intradermal injection of capsaicin. *Neuroreport* 1998;9:589–592. [PubMed: 9559921]
384. Xia Z, Storm DR. The role of calmodulin as a signal integrator for synaptic plasticity. *Nat Rev Neurosci* 2005;6:267–276. [PubMed: 15803158]
385. Xiao B, Tu JC, Petralia RS, Yuan JP, Doan A, Breder CD, Ruggiero A, Lanahan AA, Wenthold RJ, Worley PF. Homer regulates the association of group 1 metabotropic glutamate receptors with multivalent complexes of homer-related, synaptic proteins. *Neuron* 1998;21:707–716. [PubMed: 9808458]
386. Xiao HS, Huang QH, Zhang FX, Bao L, Lu YJ, Guo C, Yang L, Huang WJ, Fu G, Xu SH, Cheng XP, Yan Q, Zhu ZD, Zhang X, Chen Z, Han ZG, Zhang X. Identification of gene expression profile of dorsal root ganglion in the rat peripheral axotomy model of neuropathic pain. *Proc Natl Acad Sci U S A* 2002;99:8360–8365. [PubMed: 12060780]
387. Xu Q, Garraway SM, Weyerbacher AR, Shin SJ, Inturrisi CE. Activation of the neuronal extracellular signal-regulated kinase 2 in the spinal cord dorsal horn is required for complete Freund's adjuvant-induced pain hypersensitivity. *J Neurosci* 2008;28:14087–14096. [PubMed: 19109491]
388. Xu XJ, Dalsgaard CJ, Wiesenfeld-Hallin Z. Intrathecal CP-96,345 blocks reflex facilitation induced in rats by substance P and C-fiber-conditioning stimulation. *Eur J Pharmacol* 1992;216:337–344. [PubMed: 1385160]
389. Xu XJ, Dalsgaard CJ, Wiesenfeld-Hallin Z. Spinal substance P and N-methyl-D-aspartate receptors are coactivated in the induction of central sensitization of the nociceptive flexor reflex. *Neuroscience* 1992;51:641–648. [PubMed: 1283208]
390. Yaksh TL. Behavioral and autonomic correlates of the tactile evoked allodynia produced by spinal glycine inhibition: Effects of modulatory receptor systems and excitatory amino acid antagonists. *Pain* 1989;37:111–123. [PubMed: 2542867]

391. Yashpal K, Fisher K, Chabot JG,Coderre TJ. Differential effects of NMDA and group I mGluR antagonists on both nociception and spinal cord protein kinase C translocation in the formalin test and a model of neuropathic pain in rats. *Pain* 2001;94:17–29. [PubMed: 11576741]
392. Young MR, Fleetwood-Walker SM, Dickinson T, Black-burn-Munro G, Sparrow H, Birch PJ, Bountra C. Behavioural and electrophysiological evidence supporting a role for group I metabotropic glutamate receptors in the mediation of nociceptive inputs to the rat spinal cord. *Brain Res* 1997;777:161–169. [PubMed: 9449425]
393. Young MR, Fleetwood-Walker SM, Mitchell R, Munro FE. Evidence for a role of metabotropic glutamate receptors in sustained nociceptive inputs to rat dorsal horn neurons. *Neuropharmacology* 1994;33:141–144. [PubMed: 8183438]
394. Yu XM, Salter MW. Src, a molecular switch governing gain control of synaptic transmission mediated by N-methyl-D-aspartate receptors. *Proc Natl Acad Sci U S A* 1999;96:7697–7704. [PubMed: 10393883]
395. Yunus MB. Role of central sensitization in symptoms beyond muscle pain, and the evaluation of a patient with widespread pain. *Best Pract Res Clin Rheumatol* 2007;21:481–497. [PubMed: 17602995]
396. Zeitz KP, Guy N, Malmberg AB, Dirajlal S, Martin WJ, Sun L, Bonhaus DW, Stucky CL, Julius D, Basbaum AI. The 5-HT₃ subtype of serotonin receptor contributes to nociceptive processing via a novel subset of myelinated and unmyelinated nociceptors. *J Neurosci* 2002;22:1010–1019. [PubMed: 11826129]
397. Zhang B, Tao F, Liaw WJ, Bredt DS, Johns RA, Tao YX. Effect of knock down of spinal cord PSD-93/chapsin-110 on persistent pain induced by complete Freund's adjuvant and peripheral nerve injury. *Pain* 2003;106:187–196. [PubMed: 14581127]
398. Zhang HM, Chen SR, Pan HL. Effects of activation of group III metabotropic glutamate receptors on spinal synaptic transmission in a rat model of neuropathic pain. *Neuroscience* 2009;158:875–884. [PubMed: 19017536]
399. Zhang J, De Koninck Y. Spatial and temporal relationship between monocyte chemoattractant protein-1 expression and spinal glial activation following peripheral nerve injury. *J Neurochem* 2006;97:772–783. [PubMed: 16524371]
400. Zhang X, Wu J, Fang L, Willis WD. The effects of protein phosphatase inhibitors on the duration of central sensitization of rat dorsal horn neurons following injection of capsaicin. *Mol Pain* 2006;2:23. [PubMed: 16846502]
401. Zhang X, Wu J, Lei Y, Fang L, Willis WD. Protein phosphatase 2A regulates central sensitization in the spinal cord of rats following intradermal injection of capsaicin. *Mol Pain* 2006;2:9. [PubMed: 16549018]
402. Zhang X, Wu J, Lei Y, Fang L, Willis WD. Protein phosphatase modulates the phosphorylation of spinal cord NMDA receptors in rats following intradermal injection of capsaicin. *Brain Res Mol Brain Res* 2005;138:264–272. [PubMed: 15919130]
403. Zhang XC, Zhang YQ, Zhao ZQ. Different roles of two nitric oxide activated pathways in spinal long-term potentiation of C-fiber-evoked field potentials. *Neuropharmacology* 2006;50:748–754. [PubMed: 16427664]
404. Zhou XF, Rush RA. Endogenous brain-derived neurotrophic factor is anterogradely transported in primary sensory neurons. *Neuroscience* 1996;74:945–953. [PubMed: 8895863]
405. Zhu CZ, Wilson SG, Mikusa JP, Wismer CT, Gauvin DM, Lynch JJ 3rd, Wade CL, Decker MW, Honore P. Assessing the role of metabotropic glutamate receptor 5 in multiple nociceptive modalities. *Eur J Pharmacol* 2004;506:107–118. [PubMed: 15588730]
406. Zhuang ZY, Gerner P, Woolf CJ, Ji RR. ERK is sequentially activated in neurons, microglia, and astrocytes by spinal nerve ligation and contributes to mechanical allodynia in this neuropathic pain model. *Pain* 2005;114:149–159. [PubMed: 15733640]
407. Zou X, Lin Q, Willis WD. Effect of protein kinase C blockade on phosphorylation of NR1 in dorsal horn and spinothalamic tract cells caused by intradermal capsaicin injection in rats. *Brain Res* 2004;1020:95–105. [PubMed: 15312791]

408. Zou X, Lin Q, Willis WD. Role of protein kinase A in phosphorylation of NMDA receptor 1 subunits in dorsal horn and spinothalamic tract neurons after intradermal injection of capsaicin in rats. *Neuroscience* 2002;115:775–786. [PubMed: 12435416]

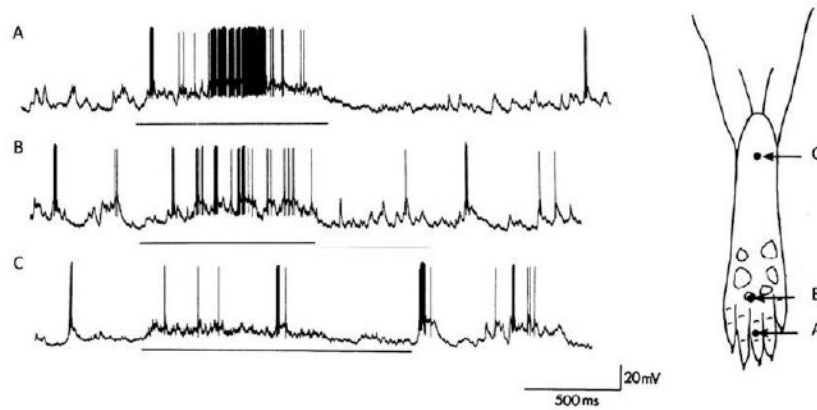


Figure 1.

Subthreshold synaptic inputs. The substrate for receptive field plasticity. Intracellular in vivo recordings from a nociceptive-specific rat dorsal horn neuron revealing subthreshold synaptic inputs. The output of somatosensory neurons is determined by those peripheral sensory inputs that produce sufficiently large-amplitude monosynaptic and polysynaptic potentials to evoke an action potential discharge (**A** and **B**). This constitutes the receptive field or firing zone of the neuron. However, stimuli outside the receptive field can evoke synaptic inputs that are too small normally to produce action potential outputs (**C**), and this constitutes a subliminal fringe or low-probability firing fringe, which can be recruited if synaptic efficacy is increased, to expand and change the receptive field. In this particular neuron, a standard pinch stimulus applied to points A, B, and C evoked only action potentials at points A and B but clear subthreshold synaptic inputs at C. Modified from ^{Reference 365}.

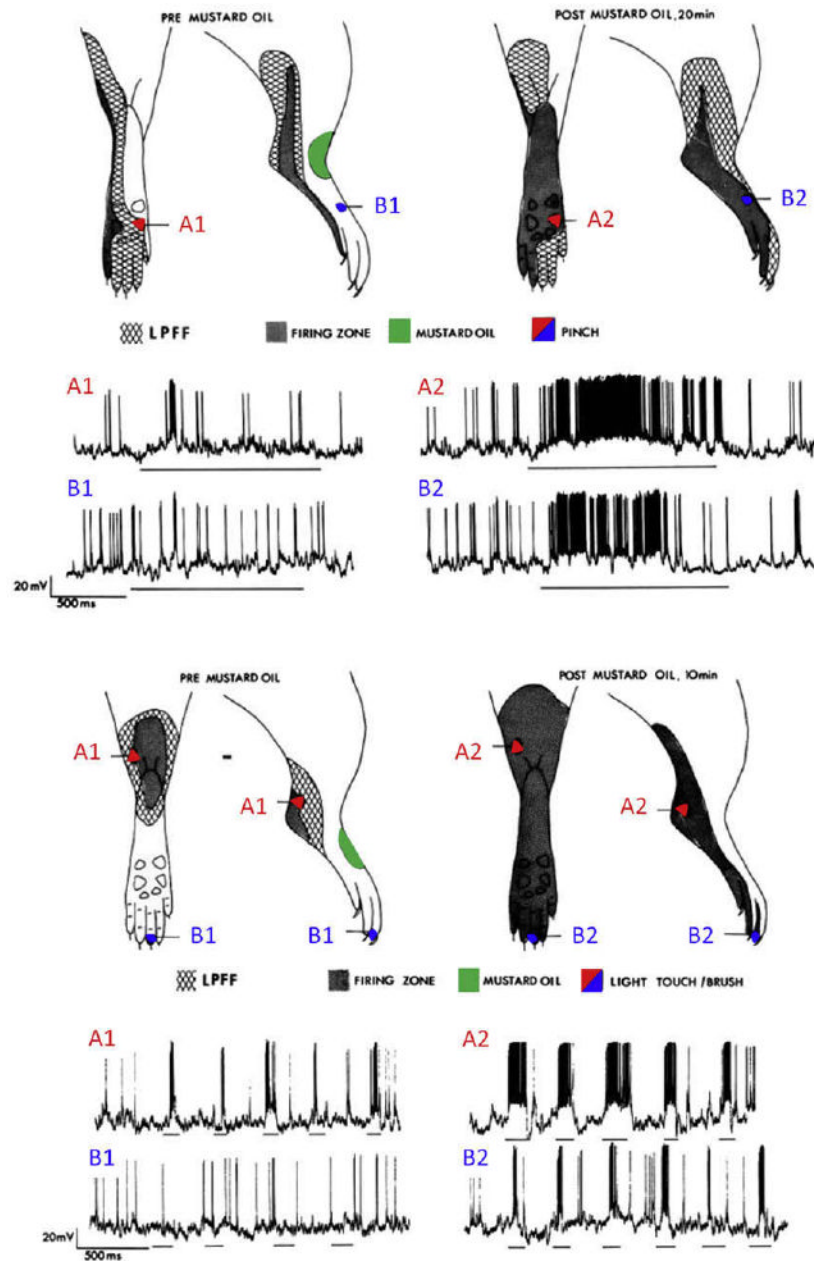


Figure 2.

Expansion of receptor fields during central sensitization. Recruiting subthreshold synaptic inputs to the output of a nociceptive-specific neuron can markedly alter its receptive field properties, producing changes in receptive field threshold and spatial extent. When neurons in the dorsal horn spinal cord are subject to activity-dependent central sensitization, they exhibit some or all the following: development of or increases in spontaneous activity, a reduction in threshold for activation by peripheral stimuli, increased responses to suprathreshold stimulation, and enlargement of their receptive fields. The examples in this figure of intracellular recordings of rat dorsal horn neurons show the cutaneous receptive fields before central sensitization (pre mustard oil) and after the induction of central sensitization (post mustard oil) and indicate how subthreshold nociceptive (pinch, top) and low-threshold (brush, bottom) inputs in the low-probability firing fringe (LPFF) are recruited by central sensitization.

Central sensitization was produced by topical application of mustard oil, which generates a brief burst of activity in TRPA1-expressing nociceptors and resulted in the neural equivalent of secondary hyperalgesia (top) and tactile allodynia (bottom). Note that the mustard oil (conditioning input) was applied to a different area (in red) from the test pinch or brush inputs (in blue), so that the changes observed are due to hetero-synaptic facilitation. Modified from Reference 364.

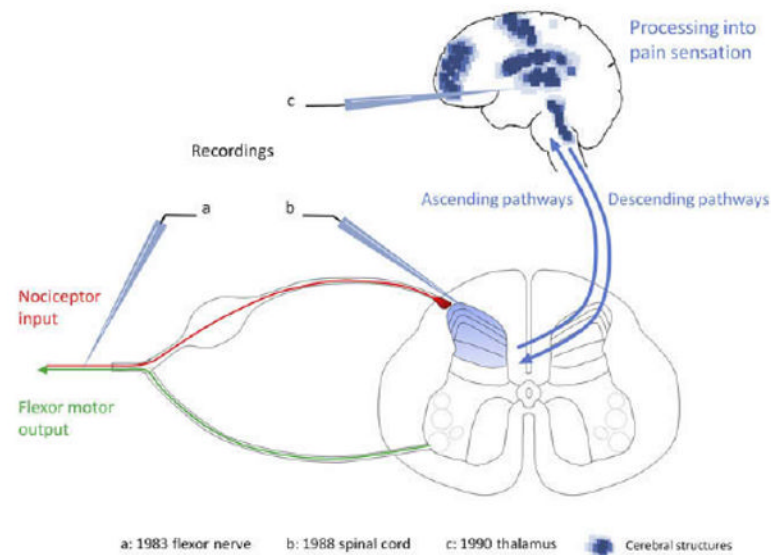
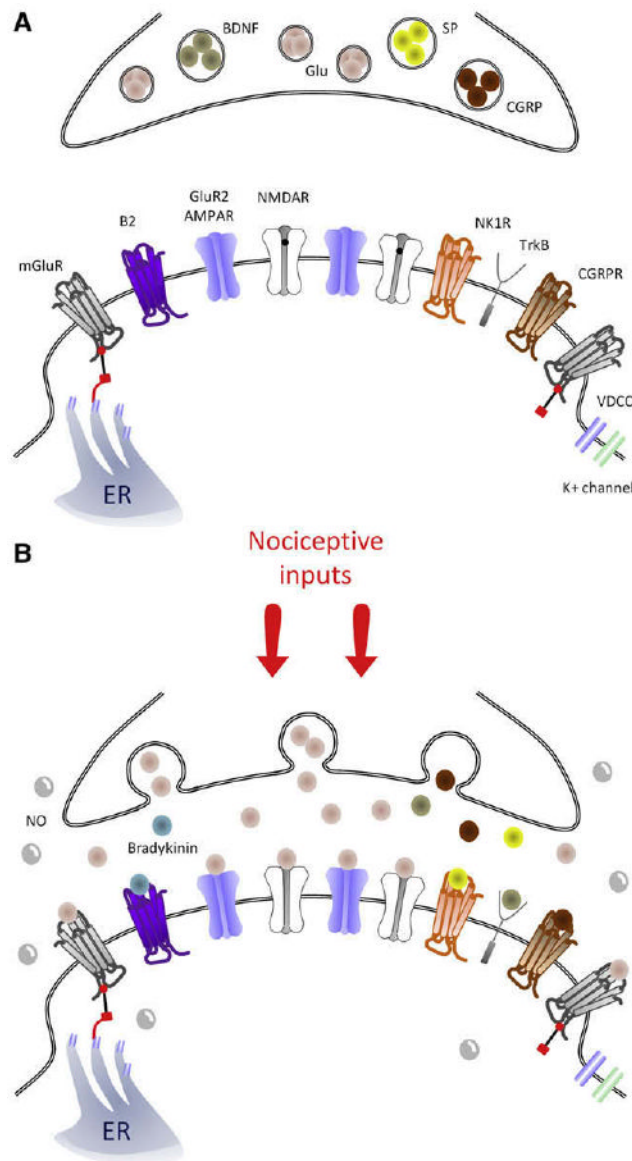


Figure 3.

Schematic representation of the structures exhibiting central sensitization. The first evidence for central sensitization was generated in 1983 by revealing injury-induced changes in the cutaneous receptive field properties of flexor motor neurons as an integrated measure of the functional plasticity in the spinal cord. A conditioning noxious stimulus resulted in long-lasting reductions in the threshold and an expansion of the receptive field of the motor neurons that was shown to be centrally generated. Essentially identical changes were then described in lamina I and V neurons in the dorsal horn of the spinal cord (b) as well as in spinal nucleus pars caudalis (Sp5c), thalamus (c), amygdala, and anterior cingulate cortex. Imaging techniques have revealed several brain structures in human subjects that exhibit changes compatible with central sensitization (blue dots).

**Figure 4.**

Central sensitization triggers: Schematic representation of key synaptic triggers of central sensitization. **(A)**, Model of the synapse between the central terminal of a nociceptor and a lamina I neuron under control, basal conditions. mGluR receptors sit at the extremities of the synapse and are linked to the endoplasmic reticulum (ER). Note that NMDAR channels are blocked by Mg^{2+} in the pore (black dot). After a barrage of activity in the nociceptor **(B)**, the primary afferent presynaptic terminal releases glutamate that binds to AMPAR, NMDAR (now without Mg^{2+}), and mGluR, as well as substance P, CGRP, and BDNF, which bind to NK1, CGRP1, and TrkB receptors, respectively. B2 receptors are also activated by spinally produced bradykinin. NO is produced by several cell types in the spinal cord and can act presynaptically and postsynaptically.

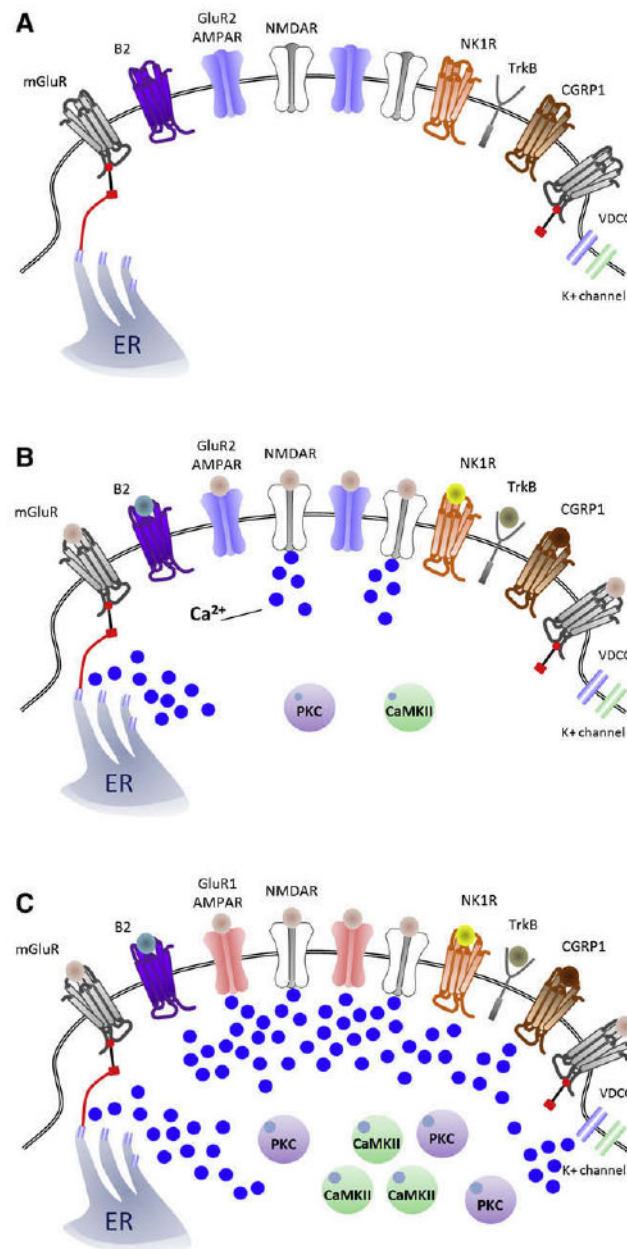


Figure 5. Sources of Ca^{2+} in the synapse of nociceptive neurons for inducing central sensitization. **(A)**, Model of a nociceptor–dorsal horn neuron synapse under control, nonactivated conditions. After nociceptor input **(B)**, activation of NMDAR and mGluR result in a rapid increase of $[\text{Ca}^{2+}]_i$ that activates PKC and CaMKII, 2 major effectors of central sensitization. **(C)**, Representation of a synapse during peripheral inflammation-induced central sensitization, where there is a shift from GluR2/3 to GluR1-containing AMPARs that enables, along with voltage-dependent calcium channels and NMDAR, entry of Ca^{2+} and which, together with activation of the G-coupled mGluR, NK1, B2, and CGRP1 receptors, which release intracellular Ca^{2+} stores, recruits PKC and CaMKII, strengthening the excitatory synapse.

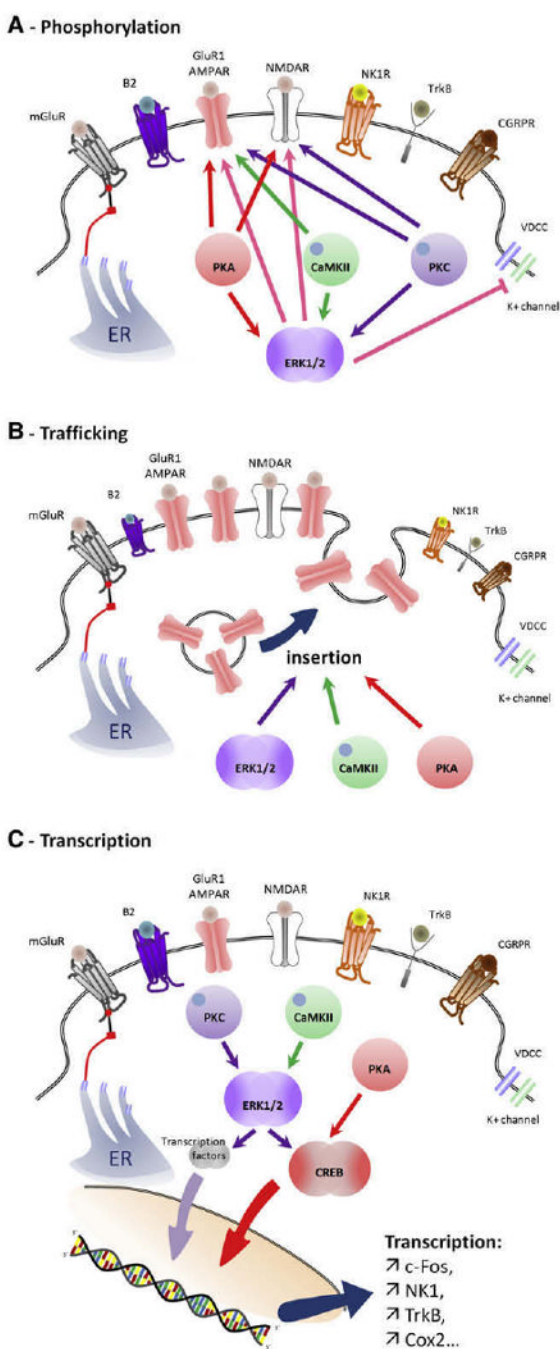


Figure 6. Contribution of PKC, CaMKII, PKA, and ERK activation to central sensitization. **(A)**, Phosphorylation by PKC, CaM-KII, PKA, and ERK cause changes in the threshold and activation kinetics of NMDA and AMPA receptors, boosting synaptic efficacy. ERK also produces a decrease in K^+ currents through phosphorylation of Kv4.2 channels, increasing membrane excitability. **(B)**, PKA, CaMKII, and ERK promote recruitment of GluR1-containing AMPAR to the membrane from vesicles stored under the synapse. **(C)**, Transcriptional changes mediated by activation of CREB and other transcription factors driving expression of genes including c-Fos, NK1, TrkB, and Cox-2, to produce a long-lasting strengthening of the synapse.

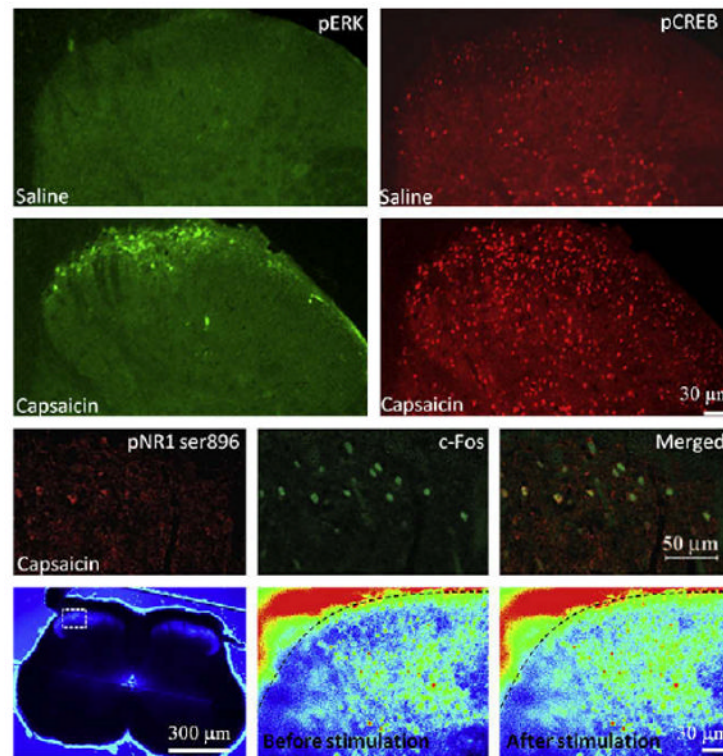


Figure 7.

Key effectors of central sensitization in the dorsal horn. Subcutaneous injection of capsaicin, a potent inducer of central sensitization, causes rapid activation of ERK and CREB (upper panels) as well as a PKC-induced phosphorylation of the NR1 subunit of NMDAR in *c-Fos*-positive neurons of the superficial laminae of the spinal cord (middle panels). ERK activation participates in transcriptional and post-translational changes, CREB activation promotes transcription of several genes involved in central sensitization. NR1 phosphorylation by PKC increases NMDAR activity. ERK and PKC are activated through the increase of intracellular calcium that occurs during stimulation of nociceptive fibers (lower panels). Signals are shown in pseudocolor from blue (weak intensity) to red (strong intensity). ERK and CREB staining are from ^{Reference 141}; NR1 phosphorylation staining from ^{Reference 32}, and calcium imaging from ^{Reference 179}.

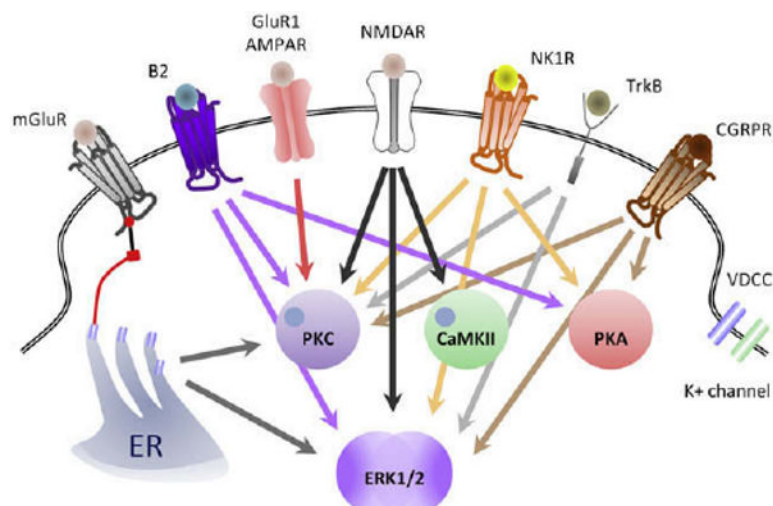
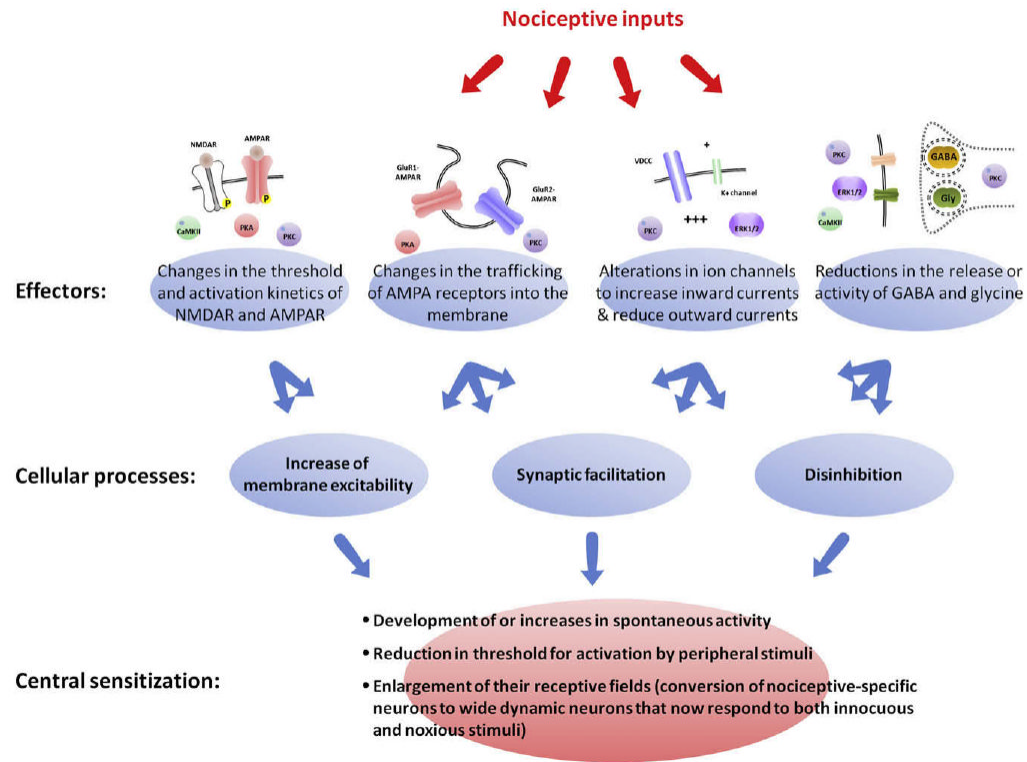


Figure 8.

Key intracellular pathways contributing to the generation of central sensitization. NMDAR activation causes activation of PKC, CaMKII, and ERK (black arrows); GluR1-containing AMPAR activate PKC (red arrow); NK1 and CGRP1 receptors activate PKC, PKA, and ERK (orange and brown arrows, respectively); TrkB s activates of PKC and ERK (purple arrows); and mGluR, via release of Ca^{2+} from microsomal stores, activates PKC and ERK (gray arrows). Note that most of the triggers of central sensitization: Activation of NMDAR, mGluR, TrkB, NK1, CGRP1, or B2 converge to activate ERK.

**Figure 9.**

Multiple cellular processes lead to central sensitization. Central sensitization is not defined by activation of a single molecular pathway but rather represents the altered functional status of nociceptive neurons. During central sensitization, these neurons display 1 or all of the following: i, development of or an increase in spontaneous activity; ii, reduction in threshold for activation; and iii, enlargement of nociceptive neuron receptive fields. These characteristics can be produced by several different cellular processes including increases in membrane excitability, a facilitation of synaptic strength, and decreases in inhibitory transmission (disinhibition). Similarly, these mechanisms can be driven by different molecular effectors including PKA, PKC, CaMKII, and ERK1/2. These kinases participate in changes in the threshold and activation kinetics of NMDA and AMPA receptors and in their trafficking to the membrane, cause alterations in ion channels that increase inward currents and reduce outward currents, and reduce the release or activity of GABA and glycine.

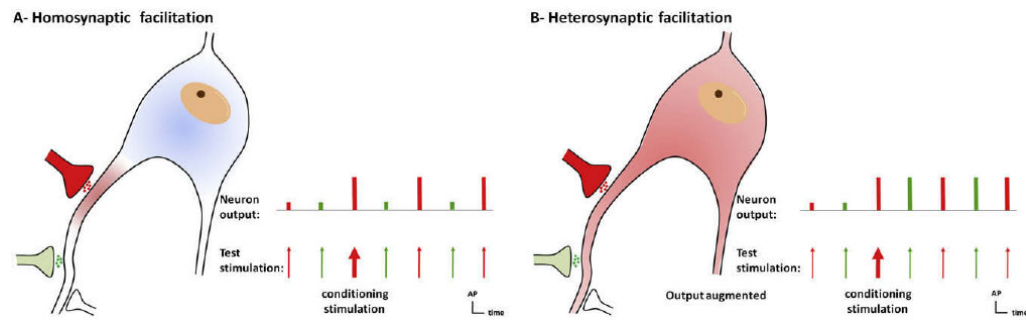


Figure 10.

Homo synaptic and heterosynaptic facilitation. **(A)**, Homosynaptic potentiation is a form of use-dependent facilitation of a synapse evoked by activation of that same synapse (in red). A nonconditioned synapse (green) is not potentiated. This type of potentiation is commonly called long-term potentiation (LTP). LTP-like homosynaptic potentiation can contribute to primary hyperalgesia. **(B)**, Heterosynaptic facilitation represents a form of activity-dependent facilitation in which activity in 1 set of synapses (conditioning input, red) augments subsequent activity in other, nonactivated groups of synapses (test input, green). Heterosynaptic potentiation is responsible for the major sensory manifestations of use-dependent central sensitization: pain in response to low-threshold afferents (allodynia) and spread of pain sensitivity to noninjured areas (secondary hyperalgesia).

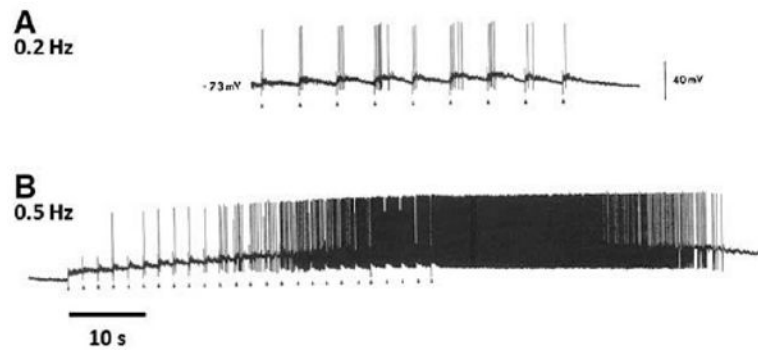


Figure 11.

Action potential windup. Windup is the consequence of a cumulative membrane depolarization resulting from the temporal summation of slow synaptic potentials. Under normal conditions, low-frequency stimulations of C-fibers (0.2 Hz) cause steady neuronal discharges (**A**) as the membrane potential has sufficient time to return to resting potential between stimuli. At a frequency of 0.5 Hz or higher, activation of NK1 and CGRP1 receptors by release of substance P and CGRP from peptidergic nociceptors produces a cumulative increase in membrane depolarization. This then enables activation of NMDAR by removal of the voltage-dependent Mg^{2+} block, further boosting the responses in a non-linear fashion (**B**). Windup disappears within tens of seconds of the end of the stimulus train as the membrane potential returns to its normal resting level (**B**). Modified from ^{Reference 323}.

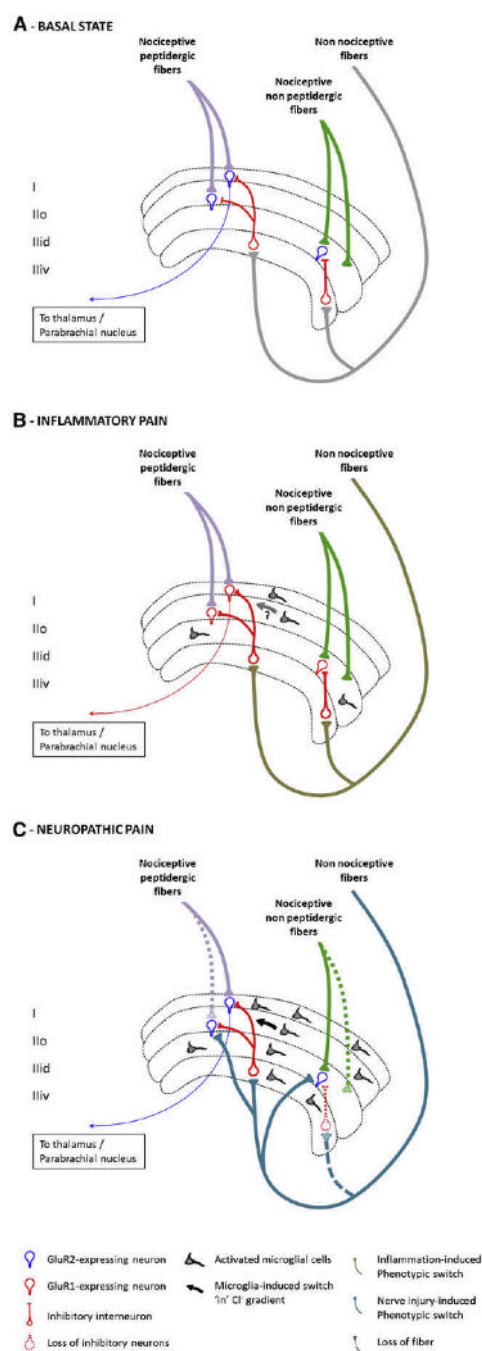


Figure 12.

Central sensitization in pathological settings. **A**, Representation of the superficial lamina of the dorsal horn of the spinal cord. Nociceptive peptidergic fibers contact lamina I and II outer (I and IIo) neurons that express GluR2-containing AMPAR (in blue). Some of these neurons project to the thalamus, parabrachial nucleus, and PAG. Nociceptive nonpeptidergic fibers contact neurons in dorsal lamina II inner (IIid), which also express GluR2-containing AMPAR. Non-nociceptive large fibers contact deeper laminae but also send collaterals to inhibitory interneurons in the ventral part of lamina II (IIiv).²²² **B**, Alterations in the superficial lamina in inflammatory pain. Because large DRG neurons begin to express SP and BDNF, stimulation of these afferents acquires the capacity to generate central sensitization. Neurons now express

GluR1-containing AMPAR at their synapse (in red), resulting in an increase of Ca^{2+} influx on their activation. Some spinal cord microglial cells are activated and release factors that contribute to the development of central sensitization by enhancing excitatory and reducing inhibitory currents. **C**, Representation of superficial lamina in neuropathic pain states. After peripheral nerve injury, there is a loss of C-fiber central terminals and the sprouting of myelinated A- β fibers from deep to superficial lamina. Injured sensory neurons in the dorsal root ganglion exhibit a change in transcription that alters their membrane properties, growth, and transmission. Large fibers begin to express substance P, BDNF, and the synthetic enzymes for tetrahydrobiopterin, an essential cofactor for NOS and can drive central sensitization. Recruitment and activation of microglial cells is an essential step in the development of pain after nerve injury and to trigger central sensitization by releasing proinflammatory cytokines that increase neuronal excitability and BDNF that induces a switch in Cl^- gradients, resulting in a loss of inhibition. Loss of inhibition is also caused by excitotoxic apoptosis of inhibitory interneurons.

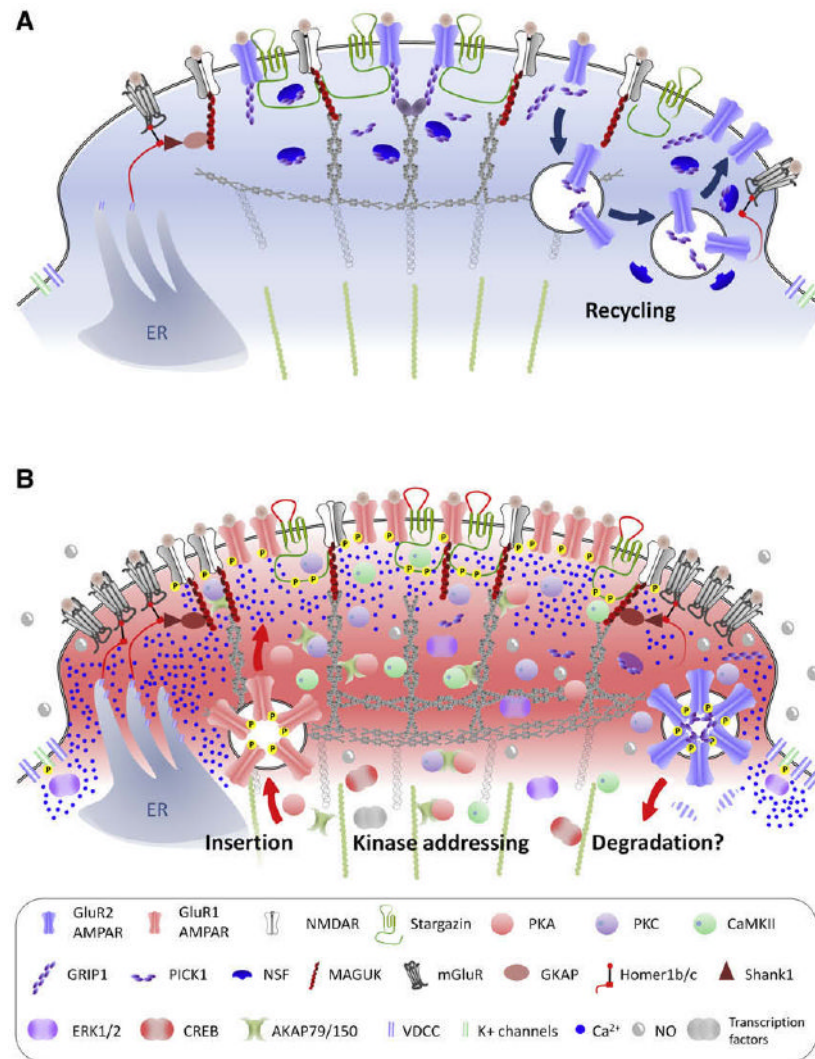


Figure 13.

Role of scaffolding proteins in central sensitization. Representation of the post synaptic density (PSD) region of a synapse of a nociceptive neuron in the superficial lamina of the spinal cord under basal conditions (**A**) and during inflammatory pain (**B**). The proteins that make up the PSD can drive a major functional reorganization of synapses, modifying postsynaptic efficacy by altering receptor density at the membrane and producing switches from Ca^{2+} -impermeable to Ca^{2+} -permeable AMPARs. In addition, scaffolding proteins mediate the “addressing” of kinases to the receptors, thereby increasing their activity. Under normal conditions (**A**), neurons mostly express GluR2-containing AMPAR that are anchored to the PSD via GRIP-1. NMDAR are recycled in an activity-dependent manner via PICK-1 and NSF. During inflammation (**B**), there is an endocytosis of GluR2-containing AMPAR (initiated by PKC phosphorylation of GluR2) along with a loss of NSF that prevents their reinsertion into the synapse. GluR1-containing AMPAR are expressed at the membrane, where their activity is increased by phosphorylation with PKA as well as by the ectodomain of stargazin, whose C-terminus segment is phosphorylated by PKC and CaMKII. MAGUK can participate to increase glutamate receptors density at the synapse, and phosphorylated stargazin promotes AMPAR and NMDAR clustering. Scaffolding proteins such as AKAP79/150 and the MAGUKs also “address” kinases to specific synaptic position at the right time to phosphorylate AMPAR and

NMDAR subunits, thereby increasing their activity. The Homer-Shank1-GKAP-MAGUK complex couples mGluR and NMDAR to intrasomal Ca^{2+} stores, so that activation of glutamate receptors leads to high levels of Ca^{2+} in the PSD that activate PKC and CaMKII, also recruited to the PSD by scaffolding proteins. PKC, CaMKII, and other kinases converge to activate ERK, which reduce K^{+} channels activity and promote transcriptional activity.

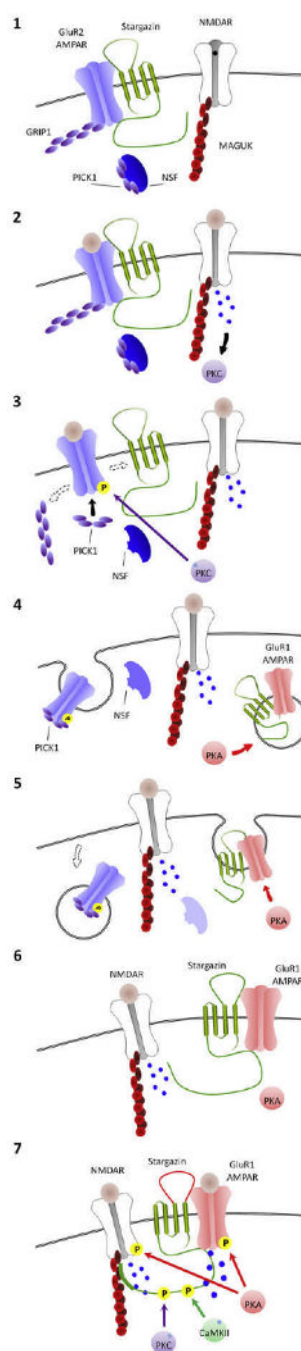


Figure 14.

Synaptic scaffolding proteins and the switch to GluR1-containing AMPAR after peripheral inflammation. Under basal conditions, GluR2-containing AMPARs are associated in the synapse with stargazin and GRIP-1. The C-terminus of stargazin binds to PSD-95 (1). Peripheral inflammation leads to glutamate release from nociceptors and AMPA and NMDA receptor activation. NMDAR activation leads to a Ca^{2+} influx that activates PKC (2). Activated PKC phosphorylates GluR2-containing AMPARs at ser880, which leads to a loss of affinity for stargazin and GRIP-1, allowing PICK-1 to bind to the receptor (3). The GluR2-containing AMPAR and PICK-1 complex undergo endocytosis, whereas NSF is downregulated, preventing reinsertion of GluR2-containing AMPAR into the synapse. Meanwhile, peripheral

inflammation also leads to activation of PKA that promotes GluR1-containing AMPAR to be inserted into the membrane (4). Because GluR2-containing AMPARs are internalized and the Ca^{2+} -permeable GluR1-containing AMPAR are inserted into the membrane, there is a switch from GluR2- to GluR1-containing AMPAR (5 and 6). Increased $[\text{Ca}^{2+}]_i$ causes an activation of PKC and CaMKII, which phosphorylate the C-terminus of stargazin, increasing its affinity for PSD-95 (CaMKII) and modifying its ectodomain (PKC), which causes increased GluR1 affinity for glutamate and single-channel conductance and a higher channel opening probability, further increasing Ca^{2+} influx (7).